

Human review packet: PRPKG24AccuracyDiversity

Generated from byte-pinned source excerpts, the expanded PaperInterface specifications, and hash-bound raw-source-to-Spec screening records.

Paper: PRPKG24AccuracyDiversity

Paper id: PRPKG24AccuracyDiversity

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Source record: audit/paper_statement_map.json

Rows: 25

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Source URL: [official source](#)

How to use this packet

For each source claim, compare the exact source excerpts with the expanded PaperInterface specification. Mark up this PDF or fill the blank reviewer box in the TeX source; return the annotated file for reconciliation into the paper-local human-review log.

Open the interactive dashboard

The interactive dashboard is an optional alternative to using this PDF; it presents the same review surface rather than an additional review requirement. After forking and cloning (or pulling) AppliedModelingLib, run the following from the repository root. The cache command is needed only the first time in a new clone.

```
cd <your-repository-checkout>
lake exe cache get
python3 scripts/review_dashboard.py --paper PRPKG24AccuracyDiversity --serve
```

Open <http://127.0.0.1:8765/> in a browser and keep that terminal running while reviewing. The dashboard records saved annotations in this paper's local reviewer-owned review record.

Contents

- [Governing Lean declarations \(151; supporting context\)](#)
- [Paper-specific semantic prerequisites \(7; not paper claims\)](#)
 - [PRPKG24AccuracyDiversity.AllocationSequence.ConvergesToProfile](#)
 - [PRPKG24AccuracyDiversity.CountAllocation.representation](#)
 - [PRPKG24AccuracyDiversity.GammaHomogeneityProfile.Exact](#)
 - [PRPKG24AccuracyDiversity.InfiniteGammaHomogeneityProfile.Exact](#)
 - [PRPKG24AccuracyDiversity.definition3IidOrderStatisticMean](#)
 - [PRPKG24AccuracyDiversity.gammaLikelihoodProfile](#)
 - [PRPKG24AccuracyDiversity.infiniteLikelihoodProfile](#)
- [Source claims \(25\)](#)
 - [theorem1_i_formulaSpec](#)
 - [theorem1_ii_formulaSpec](#)
 - [theorem1_iii_formulaSpec](#)
 - [theorem1_iv_formulaSpec](#)
 - [theorem1_v_selected_maximizer_conclusionSpec](#)
 - [corollary1Spec](#)
 - [proposition2_corrected_finite_and_sequence_sourceSpec](#)
 - [corollary3_iid_bernoulli_conclusionSpec](#)
 - [theorem2_i_corrected_pmf_source_model_endpointSpec](#)
 - [theorem2_ii_corrected_pmf_source_model_endpointSpec](#)
 - [theorem2_iii_corrected_pmf_source_model_endpointSpec](#)
 - [theorem2_iv_corrected_pmf_source_model_endpointSpec](#)
 - [theorem3_log_share_conclusionSpec](#)
 - [theorem3_all_consumed_argmax_conclusionSpec](#)
 - [lemma1_bounded_topk_loss_asymptotic_of_pdf_sourceSpec](#)
 - [proposition4Spec](#)
 - [lemmaD1_i_corrected_uniform_optimizer_shares_sourceSpec](#)

- lemmaD1_ii_corrected_powerTail_optimizer_shares_sourceSpec
- lemmaD1_iii_corrected_probability_optimizer_shares_sourceSpec
- lemmaD1_iv_corrected_powerTail_optimizer_shares_sourceSpec
- lemmaD2_bounded_fixed_rank_integral_asymptotic_of_pdf_sourceSpec
- lemmaD3_corrected_exponential_fixed_rank_sourceSpec
- lemmaD4_corrected_pareto_fixed_rank_sourceSpec
- lemmaD5Spec
- proposition5_iid_topk_identitySpec

Governing Lean declarations

These exact graph-bound declaration bodies are retained by the displayed specifications or reviewed prerequisites. They are supporting context and do not add source-result rows or semantic judgments.

AppliedModelingLib.Allocation

Declaration location: reusable library

Lean declaration body

field count:
{κ : Type u_1} → AppliedModelingLib.Allocation κ → κ → ℕ

AppliedModelingLib.Allocation.allOnTypeAllocation

Declaration location: reusable library

Lean declaration body

type:
{κ : Type u_1} → [DecidableEq κ] → ℕ → κ → AppliedModelingLib.Allocation κ
value:
fun {κ : Type u_1} [DecidableEq κ] (N : ℕ) (best : κ) => { count := fun (k : κ) => if k = best then N else 0 }

Direct retained dependencies

- [AppliedModelingLib.Allocation](#)

AppliedModelingLib.Allocation.objective

Declaration location: reusable library

Lean declaration body

type:
{κ : Type u_1} → [Fintype κ] → AppliedModelingLib.Allocation κ → (κ → ℝ) → (κ → ℕ → ℝ) → ℝ
value:
fun {κ : Type u_1} [Fintype κ] (a : AppliedModelingLib.Allocation κ) (weight : κ → ℝ) (valueOfCount : κ → ℕ → ℝ) =>
Σ k : κ, weight k * valueOfCount k (a.count k)

Direct retained dependencies

- [AppliedModelingLib.Allocation](#)

AppliedModelingLib.Allocation.total

Declaration location: reusable library

Lean declaration body

type:
{κ : Type u_1} → [Fintype κ] → AppliedModelingLib.Allocation κ → ℕ
value:
fun {κ : Type u_1} [Fintype κ] (a : AppliedModelingLib.Allocation κ) => Σ k : κ, a.count k

Direct retained dependencies

- [AppliedModelingLib.Allocation](#)

AppliedModelingLib.Allocation.HasTotal

Declaration location: reusable library

Lean declaration body

```
type:
{κ : Type u_1} → [Fintype κ] → AppliedModelingLib.Allocation κ → ℕ → Prop

value:
fun {κ : Type u_1} [Fintype κ] (a : AppliedModelingLib.Allocation κ) (N : ℕ) => a.total = N
```

Direct retained dependencies

- [AppliedModelingLib.Allocation](#)
- [AppliedModelingLib.Allocation.total](#)

AppliedModelingLib.Allocation.IsOptimalAtTotal

Declaration location: reusable library

Lean declaration body

```
type:
{κ : Type u_1} → [Fintype κ] → (κ → ℝ) → (κ → ℕ → ℝ) → ℕ → AppliedModelingLib.Allocation κ → Prop

value:
fun {κ : Type u_1} [Fintype κ] (weight : κ → ℝ) (valueOfCount : κ → ℕ → ℝ) (N : ℕ)
(a : AppliedModelingLib.Allocation κ) =>
a.HasTotal N ∧
∀ (b : AppliedModelingLib.Allocation κ),
b.HasTotal N → b.objective weight valueOfCount ≤ a.objective weight valueOfCount
```

Direct retained dependencies

- [AppliedModelingLib.Allocation](#)
- [AppliedModelingLib.Allocation.HasTotal](#)
- [AppliedModelingLib.Allocation.objective](#)

AppliedModelingLib.Allocation.OptimalSequence

Declaration location: reusable library

Lean declaration body

```
field allocation:
{κ : Type u_1} →
[inst : Fintype κ] →
{objectiveWeightSeq : ℕ → κ → ℝ} →
{valueOfCountSeq : ℕ → κ → ℕ → ℝ} →
AppliedModelingLib.Allocation.OptimalSequence objectiveWeightSeq valueOfCountSeq →
ℕ → AppliedModelingLib.Allocation κ

field optimal:
∀ {κ : Type u_1} [inst : Fintype κ] {objectiveWeightSeq : ℕ → κ → ℝ} {valueOfCountSeq : ℕ → κ → ℕ → ℝ}
(self : AppliedModelingLib.Allocation.OptimalSequence objectiveWeightSeq valueOfCountSeq) (N : ℕ),
AppliedModelingLib.Allocation.IsOptimalAtTotal (objectiveWeightSeq N) (valueOfCountSeq N) N (self.allocation N)
```

Direct retained dependencies

- [AppliedModelingLib.Allocation](#)
- [AppliedModelingLib.Allocation.IsOptimalAtTotal](#)

AppliedModelingLib.Allocation.Sequence

Declaration location: reusable library

Lean declaration body

```
field allocation:
{κ : Type u_2} → [inst : Fintype κ] → AppliedModelingLib.Allocation.Sequence κ → ℕ → AppliedModelingLib.Allocation κ

field feasible:
∀ {κ : Type u_2} [inst : Fintype κ] (self : AppliedModelingLib.Allocation.Sequence κ) (N : ℕ),
(self.allocation N).HasTotal N
```

Direct retained dependencies

- [AppliedModelingLib.Allocation](#)
- [AppliedModelingLib.Allocation.HasTotal](#)

AppliedModelingLib.Allocation.OptimalSequence.toSequence

Declaration location: reusable library

Lean declaration body

```
type:
{κ : Type u_1} →
[inst : Fintype κ] →
{objectiveWeightSeq : ℕ → κ → ℝ} →
{valueOfCountSeq : ℕ → κ → ℕ → ℝ} →
AppliedModelingLib.Allocation.OptimalSequence objectiveWeightSeq valueOfCountSeq →
AppliedModelingLib.Allocation.Sequence κ

value:
fun {κ : Type u_1} [Fintype κ] {objectiveWeightSeq : ℕ → κ → ℝ} {valueOfCountSeq : ℕ → κ → ℕ → ℝ}
(seq : AppliedModelingLib.Allocation.OptimalSequence objectiveWeightSeq valueOfCountSeq) =>
{ allocation := seq.allocation, feasible := AppliedModelingLib.Allocation.OptimalSequence.toSequence._proof_1 seq }
```

Direct retained dependencies

- [AppliedModelingLib.Allocation.HasTotal](#)
- [AppliedModelingLib.Allocation.OptimalSequence](#)
- [AppliedModelingLib.Allocation.Sequence](#)

AppliedModelingLib.Allocation.share

Declaration location: reusable library

Lean declaration body

```
type:
{κ : Type u_1} → [Fintype κ] → AppliedModelingLib.Allocation κ → κ → ℝ

value:
fun {κ : Type u_1} [Fintype κ] (a : AppliedModelingLib.Allocation κ) (k : κ) =>
if h : a.total = 0 then 0 else ↑(a.count k) / ↑a.total
```

Direct retained dependencies

- [AppliedModelingLib.Allocation](#)
- [AppliedModelingLib.Allocation.total](#)

AppliedModelingLib.Allocation.Sequence.share

Declaration location: reusable library

Lean declaration body

```
type:
{κ : Type u_1} → [inst : Fintype κ] → AppliedModelingLib.Allocation.Sequence κ → ℕ → κ → ℝ

value:
fun {κ : Type u_1} [Fintype κ] (seq : AppliedModelingLib.Allocation.Sequence κ) (N : ℕ) (k : κ) =>
(seq.allocation N).share k
```

Direct retained dependencies

- [AppliedModelingLib.Allocation.Sequence](#)
- [AppliedModelingLib.Allocation.share](#)

AppliedModelingLib.Allocation.Sequence.ConvergesToProfile

Declaration location: reusable library

Lean declaration body

```
type:
{κ : Type u_1} → [inst : Fintype κ] → AppliedModelingLib.Allocation.Sequence κ → (κ → ℝ) → Prop

value:
fun {κ : Type u_1} [Fintype κ] (seq : AppliedModelingLib.Allocation.Sequence κ) (target : κ → ℝ) =>
  ∀ (k : κ), Filter.Tendsto (fun (N : ℕ) => seq.share N k) Filter.atTop (nhds (target k))
```

Direct retained dependencies

- [AppliedModelingLib.Allocation.Sequence](#)
- [AppliedModelingLib.Allocation.Sequence.share](#)

AppliedModelingLib.Math.AsymptoticEquivalent

Declaration location: reusable library

Lean declaration body

```
type:
(ℕ → ℝ) → (ℕ → ℝ) → Prop

value:
fun (x y : ℕ → ℝ) => Filter.Tendsto (fun (n : ℕ) => x n / y n) Filter.atTop (nhds 1)
```

AppliedModelingLib.Math.TendsToZero

Declaration location: reusable library

Lean declaration body

```
type:
(ℕ → ℝ) → Prop

value:
fun (ε : ℕ → ℝ) => Filter.Tendsto ε Filter.atTop (nhds 0)
```

AppliedModelingLib.Probability.Exponential.Model

Declaration location: reusable library

Lean declaration body

```
field rate:
AppliedModelingLib.Probability.Exponential.Model → ℝ

field rate_pos:
∀ (self : AppliedModelingLib.Probability.Exponential.Model), 0 < self.rate
```

AppliedModelingLib.Probability.Exponential.Model.measure

Declaration location: reusable library

Lean declaration body

```
type:
AppliedModelingLib.Probability.Exponential.Model → MeasureTheory.Measure ℝ

value:
fun (M : AppliedModelingLib.Probability.Exponential.Model) => ProbabilityTheory.expMeasure M.rate
```

Direct retained dependencies

- [AppliedModelingLib.Probability.Exponential.Model](#)

AppliedModelingLib.Probability.Exponential.Model.iidProductMeasure

Declaration location: reusable library

Lean declaration body

```
type:
AppliedModelingLib.Probability.Exponential.Model → (q : ℕ) → MeasureTheory.Measure (Fin q → ℝ)

value:
fun (M : AppliedModelingLib.Probability.Exponential.Model) (q : ℕ) =>
  MeasureTheory.Measure.pi fun (x : Fin q) => M.measure
```

Direct retained dependencies

- [AppliedModelingLib.Probability.Exponential.Model](#)
- [AppliedModelingLib.Probability.Exponential.Model.measure](#)

AppliedModelingLib.Probability.Exponential.harmonicReal

Declaration location: reusable library

Lean declaration body

```
type:
ℕ → ℝ

value:
fun (q : ℕ) => ∑ j ∈ Finset.range q, 1 / ↑(j + 1)
```

AppliedModelingLib.Probability.Pareto.iidProductMeasure

Declaration location: reusable library

Lean declaration body

```
type:
ℝ → ℝ → (q : ℕ) → MeasureTheory.Measure (Fin q → ℝ)

value:
fun (t r : ℝ) (q : ℕ) => MeasureTheory.Measure.pi fun (x : Fin q) => ProbabilityTheory.paretoMeasure t r
```

AppliedModelingLib.Probability.TopKExpectationOracle

Declaration location: reusable library

Lean declaration body

```
field expectedTopSum:
{τ : Type u_1} → AppliedModelingLib.Probability.TopKExpectationOracle τ → ℕ → τ → ℕ → ℝ
```

AppliedModelingLib.Probability.TopKExpectationOracle.common

Declaration location: reusable library

Lean declaration body

```
type:
(τ : Type u_2) → (ℕ → ℝ) → AppliedModelingLib.Probability.TopKExpectationOracle τ

value:
fun (τ : Type u_2) (h : ℕ → ℝ) => { expectedTopSum := fun (_k : ℕ) (_t : τ) (q : ℕ) => h q }
```

Direct retained dependencies

- [AppliedModelingLib.Probability.TopKExpectationOracle](#)

AppliedModelingLib.Probability.ascendingOrderStatistic

Declaration location: reusable library

Lean declaration body

```
type:
{n : ℕ} → (Fin n → ℝ) → Fin n → ℝ

value:
fun {n : ℕ} (sample : Fin n → ℝ) (rank : Fin n) => sample ((Tuple.sort sample) rank)
```

AppliedModelingLib.Probability.binaryRatingScore

Declaration location: reusable library

Lean declaration body

```
type:
  Bool → ℝ

value:
  fun (b : Bool) => if b = true then 1 else 0
```

AppliedModelingLib.Probability.orderStatisticTopKSumFromMean

Declaration location: reusable library

Lean declaration body

```
type:
  (ℕ → ℕ → ℝ) → ℕ → ℕ → ℝ

value:
  fun (μ : ℕ → ℕ → ℝ) (k a : ℕ) => ∑ i ∈ Finset.range (min k a), μ (a - i) a
```

AppliedModelingLib.Probability.TopKExpectationOracle.orderStatisticTopKExpectationOracle

Declaration location: reusable library

Lean declaration body

```
type:
  (τ : Type u_2) → (ℕ → ℕ → ℝ) → AppliedModelingLib.Probability.TopKExpectationOracle τ

value:
  fun (τ : Type u_2) (μ : ℕ → ℕ → ℝ) =>
  {
    expectedTopSum := fun (k : ℕ) (t : τ) (q : ℕ) =>
      AppliedModelingLib.Probability.orderStatisticTopKSumFromMean μ k q }
  }
```

Direct retained dependencies

- [AppliedModelingLib.Probability.TopKExpectationOracle](#)
- [AppliedModelingLib.Probability.orderStatisticTopKSumFromMean](#)

AppliedModelingLib.Probability.realBernoulliPMF

Declaration location: reusable library

Lean declaration body

```
type:
  (p : ℝ) → 0 ≤ p → p ≤ 1 → PMF Bool

value:
  fun (p : ℝ) (hp0 : 0 ≤ p) (hp1 : p ≤ 1) =>
    PMF.bernoulli (p, hp0) (AppliedModelingLib.Probability.realBernoulliPMF_proof_1 p hp0 hp1)
```

AppliedModelingLib.Probability.reflectedCDFMass

Declaration location: reusable library

Lean declaration body

```
type:
  MeasureTheory.Measure ℝ → ℝ → ℝ → ℝ

value:
  fun (μ : MeasureTheory.Measure ℝ) (M x : ℝ) => μ.real {y : ℝ | M ≤ x + y}
```

AppliedModelingLib.Probability.sampleOrderStatisticValue

Declaration location: reusable library

Lean declaration body

```
type:
{a : ℕ} → (Fin a → ℝ) → ℕ → ℝ

value:
fun {a : ℕ} (sample : Fin a → ℝ) (rank : ℕ) =>
  if hrank : 0 < rank ∧ rank ≤ a then
    AppliedModelingLib.Probability.ascendingOrderStatistic sample
      (rank - 1, AppliedModelingLib.Probability.sampleOrderStatisticValue._proof_2 rank hrank)
  else 0
```

Direct retained dependencies

- [AppliedModelingLib.Probability.ascendingOrderStatistic](#)

AppliedModelingLib.Probability.expectedSampleOrderStatisticMean

Declaration location: reusable library

Lean declaration body

```
type:
{a : ℕ} → MeasureTheory.Measure (Fin a → ℝ) → ℕ → ℕ → ℝ

value:
fun {a : ℕ} (μ : MeasureTheory.Measure (Fin a → ℝ)) (rank sampleSize : ℕ) =>
  if sampleSize = a then ∫ (sample : Fin a → ℝ), AppliedModelingLib.Probability.sampleOrderStatisticValue sample rank μ
  else 0
```

Direct retained dependencies

- [AppliedModelingLib.Probability.sampleOrderStatisticValue](#)

AppliedModelingLib.Probability.expectedOrderStatisticMeanSeq

Declaration location: reusable library

Lean declaration body

```
type:
((a : ℕ) → MeasureTheory.Measure (Fin a → ℝ)) → ℕ → ℕ → ℝ

value:
fun (sampleMeasure : (a : ℕ) → MeasureTheory.Measure (Fin a → ℝ)) (rank sampleSize : ℕ) =>
  AppliedModelingLib.Probability.expectedSampleOrderStatisticMean (sampleMeasure sampleSize) rank sampleSize
```

Direct retained dependencies

- [AppliedModelingLib.Probability.expectedSampleOrderStatisticMean](#)

AppliedModelingLib.Probability.topKRankEmbedding

Declaration location: reusable library

Lean declaration body

```
type:
(k n : ℕ) → Fin (min k n) → Fin n

value:
fun (k n : ℕ) (i : Fin (min k n)) => (↑i, AppliedModelingLib.Probability.topKRankEmbedding._proof_1 k n i)
```

AppliedModelingLib.Probability.upperOrderStatistic

Declaration location: reusable library

Lean declaration body

```
type:
{n : ℕ} → (Fin n → ℝ) → Fin n → ℝ

value:
fun {n : ℕ} (sample : Fin n → ℝ) (rankFromTop : Fin n) =>
  AppliedModelingLib.Probability.ascendingOrderStatistic sample rankFromTop.rev
```

Direct retained dependencies

- [AppliedModelingLib.Probability.ascendingOrderStatistic](#)

AppliedModelingLib.Probability.sampleTopKSum

Declaration location: reusable library

Lean declaration body

```
type:
{n : ℕ} → (Fin n → ℝ) → ℕ → ℝ

value:
fun {n : ℕ} (sample : Fin n → ℝ) (k : ℕ) =>
  ∑ i : Fin (min k n),
    AppliedModelingLib.Probability.upperOrderStatistic sample (AppliedModelingLib.Probability.topKRankEmbedding k n i)
```

Direct retained dependencies

- [AppliedModelingLib.Probability.topKRankEmbedding](#)
- [AppliedModelingLib.Probability.upperOrderStatistic](#)

AppliedModelingLib.Probability.expectedSampleTopKSum

Declaration location: reusable library

Lean declaration body

```
type:
{n : ℕ} → MeasureTheory.Measure (Fin n → ℝ) → ℕ → ℝ

value:
fun {n : ℕ} (μ : MeasureTheory.Measure (Fin n → ℝ)) (k : ℕ) =>
  ∫ (sample : Fin n → ℝ), AppliedModelingLib.Probability.sampleTopKSum sample k ∂μ
```

Direct retained dependencies

- [AppliedModelingLib.Probability.sampleTopKSum](#)

AppliedModelingLib.finiteMin

Declaration location: reusable library

Lean declaration body

```
type:
{α : Type u_1} → [Fintype α] → [Nonempty α] → (α → ℝ) → ℝ

value:
fun {α : Type u_1} [Fintype α] [Nonempty α] (f : α → ℝ) => Finset.univ.inf' Finset.univ_nonempty f
```

AppliedModelingLib.pmfExp

Declaration location: reusable library

Lean declaration body

```
type:
{α : Type u_1} → [Fintype α] → PMF α → (α → ℝ) → ℝ

value:
fun {α : Type u_1} [Fintype α] (μ : PMF α) (f : α → ℝ) => ∑ a : α, (μ a).toReal * f a
```

AppliedModelingLib.pmfPi

Declaration location: reusable library

Lean declaration body

```
type:
{ι : Type u_1} →
[Fintype ι] →
[DecidableEq ι] → {α : ι → Type u_2} → [(i : ι) → Fintype (α i)] → ((i : ι) → PMF (α i)) → PMF ((i : ι) → α i)

value:
fun {ι : Type u_1} [Fintype ι] [DecidableEq ι] {α : ι → Type u_2} [(i : ι) → Fintype (α i)] (μ : (i : ι) → PMF (α i)) =>
  PMF.ofFintype (fun (f : (i : ι) → α i) => ∏ i : ι, (μ i) (f i)) (AppliedModelingLib.pmfPi._proof_1 μ)
```

AppliedModelingLib.pmfProb

Declaration location: reusable library

Lean declaration body

```
type:
{α : Type u_1} → [Fintype α] → PMF α → (p : α → Prop) → [DecidablePred p] → ℝ

value:
fun {α : Type u_1} [Fintype α] (μ : PMF α) (p : α → Prop) [DecidablePred p] =>
  AppliedModelingLib.pmfExp μ fun (a : α) => if p a then 1 else 0
```

Direct retained dependencies

- [AppliedModelingLib.pmfExp](#)

AppliedModelingLib.pmfProduct

Declaration location: reusable library

Lean declaration body

```
type:
(ι : Type u_1) → (α : Type u_2) → [Fintype ι] → [DecidableEq ι] → [Fintype α] → PMF α → PMF (ι → α)

value:
fun (ι : Type u_1) (α : Type u_2) [Fintype ι] [DecidableEq ι] [Fintype α] (μ : PMF α) =>
  PMF.ofFintype (fun (f : ι → α) => ∏ i : ι, μ (f i)) (AppliedModelingLib.pmfProduct._proof_1 ι α μ)
```

PRPKG24AccuracyDiversity.AppendixD1GenericI.saturationGap

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → (ℕ → ℝ) → ℕ → ℝ

value:
fun (A : ℝ) (h : ℕ → ℝ) (a : ℕ) => A - h a
```

PRPKG24AccuracyDiversity.AppendixD1GenericII.saturationGap

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → (ℕ → ℝ) → ℕ → ℝ

value:
fun (A : ℝ) (h : ℕ → ℝ) (a : ℕ) => PRPKG24AccuracyDiversity.AppendixD1GenericI.saturationGap A h a
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.AppendixD1GenericI.saturationGap](#)

PRPKG24AccuracyDiversity.AppendixD1GenericIII.forwardIncrement

Declaration location: paper-local

Lean declaration body

```
type:
( $\mathbb{N} \rightarrow \mathbb{R}$ )  $\rightarrow \mathbb{N} \rightarrow \mathbb{R}$ 

value:
fun (h :  $\mathbb{N} \rightarrow \mathbb{R}$ ) (a :  $\mathbb{N}$ ) => h (a + 1) - h a
```

PRPKG24AccuracyDiversity.AppendixD1GenericIII.StrictDiscreteConcave

Declaration location: paper-local

Lean declaration body

```
type:
( $\mathbb{N} \rightarrow \mathbb{R}$ )  $\rightarrow$  Prop

value:
fun (h :  $\mathbb{N} \rightarrow \mathbb{R}$ ) => StrictAnti (PRPKG24AccuracyDiversity.AppendixD1GenericIII.forwardIncrement h)
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.AppendixD1GenericIII.forwardIncrement](#)

PRPKG24AccuracyDiversity.AppendixD1GenericIII.logRemainder

Declaration location: paper-local

Lean declaration body

```
type:
( $\mathbb{N} \rightarrow \mathbb{R}$ )  $\rightarrow \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{N} \rightarrow \mathbb{R}$ 

value:
fun (h :  $\mathbb{N} \rightarrow \mathbb{R}$ ) (B C :  $\mathbb{R}$ ) (a :  $\mathbb{N}$ ) => h a - B * Real.log ↑a - C
```

PRPKG24AccuracyDiversity.AppendixD1GenericIV.forwardIncrement

Declaration location: paper-local

Lean declaration body

```
type:
( $\mathbb{N} \rightarrow \mathbb{R}$ )  $\rightarrow \mathbb{N} \rightarrow \mathbb{R}$ 

value:
fun (h :  $\mathbb{N} \rightarrow \mathbb{R}$ ) (a :  $\mathbb{N}$ ) => h (a + 1) - h a
```

PRPKG24AccuracyDiversity.AppendixD1GenericIV.StrictDiscreteConcave

Declaration location: paper-local

Lean declaration body

```
type:
( $\mathbb{N} \rightarrow \mathbb{R}$ )  $\rightarrow$  Prop

value:
fun (h :  $\mathbb{N} \rightarrow \mathbb{R}$ ) => StrictAnti (PRPKG24AccuracyDiversity.AppendixD1GenericIV.forwardIncrement h)
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.AppendixD1GenericIV.forwardIncrement](#)

PRPKG24AccuracyDiversity.AppendixD1GenericIV.powerTailQuotient

Declaration location: paper-local

Lean declaration body

```
type:
( $\mathbb{N} \rightarrow \mathbb{R}$ )  $\rightarrow \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{N} \rightarrow \mathbb{R}$ 

value:
fun (h :  $\mathbb{N} \rightarrow \mathbb{R}$ ) (B sigma :  $\mathbb{R}$ ) (a :  $\mathbb{N}$ ) => h a / (B * ↑a ^ sigma)
```

PRPKG24AccuracyDiversity.AppendixD1GenericPower.sourcePowerWeight

Declaration location: paper-local

Lean declaration body

```
type:
{κ : Type u_1} → (κ → ℝ) → ℝ → κ → ℝ

value:
fun {κ : Type u_1} (p : κ → ℝ) (sigma : ℝ) (i : κ) => p i ^ (1 / (1 - sigma))
```

PRPKG24AccuracyDiversity.AppendixD1GenericIV.targetWeight

Declaration location: paper-local

Lean declaration body

```
type:
{m : ℕ} → (Fin m → ℝ) → ℝ → Fin m → ℝ

value:
fun {m : ℕ} (p : Fin m → ℝ) (sigma : ℝ) (i : Fin m) =>
  PRPKG24AccuracyDiversity.AppendixD1GenericPower.sourcePowerWeight p sigma i
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.AppendixD1GenericPower.sourcePowerWeight](#)

PRPKG24AccuracyDiversity.AppendixD1GenericPower.weightSum

Declaration location: paper-local

Lean declaration body

```
type:
{κ : Type u_1} → [Fintype κ] → (κ → ℝ) → ℝ

value:
fun {κ : Type u_1} [Fintype κ] (w : κ → ℝ) => ∑ i : κ, w i
```

PRPKG24AccuracyDiversity.AppendixD1GenericPower.targetShare

Declaration location: paper-local

Lean declaration body

```
type:
{κ : Type u_1} → [Fintype κ] → (κ → ℝ) → κ → ℝ

value:
fun {κ : Type u_1} [Fintype κ] (w : κ → ℝ) (i : κ) => w i / PRPKG24AccuracyDiversity.AppendixD1GenericPower.weightSum w
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.AppendixD1GenericPower.weightSum](#)

PRPKG24AccuracyDiversity.AppendixD1GenericIV.targetShare

Declaration location: paper-local

Lean declaration body

```
type:
{m : ℕ} → [NeZero m] → (Fin m → ℝ) → ℝ → Fin m → ℝ

value:
fun {m : ℕ} [NeZero m] (p : Fin m → ℝ) (sigma : ℝ) (i : Fin m) =>
  PRPKG24AccuracyDiversity.AppendixD1GenericPower.targetShare
    (PRPKG24AccuracyDiversity.AppendixD1GenericIV.targetWeight p sigma) i
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.AppendixD1GenericIV.targetWeight](#)
- [PRPKG24AccuracyDiversity.AppendixD1GenericPower.targetShare](#)

PRPKG24AccuracyDiversity.BoundedLemmaD2Index

Declaration location: paper-local

Lean declaration body

```
type:
  ℕ → Type

value:
  fun (k : ℕ) => (i : Fin k) × Fin (↑i + 1)
```

PRPKG24AccuracyDiversity.GeneralRounding.objective

Declaration location: paper-local

Lean declaration body

```
type:
  {κ : Type u_1} → [Fintype κ] → (κ → ℝ → ℝ) → (κ → ℝ) → ℝ

value:
  fun {κ : Type u_1} [Fintype κ] (g : κ → ℝ → ℝ) (x : κ → ℝ) => ∑ i : κ, g i (x i)
```

PRPKG24AccuracyDiversity.ItemType

Declaration location: paper-local

Lean declaration body

```
type:
  ℕ → Type

value:
  fun (T : ℕ) => Fin T
```

PRPKG24AccuracyDiversity.AppendixD1GenericIIFull.targetWeight

Declaration location: paper-local

Lean declaration body

```
type:
  {m : ℕ} → (PRPKG24AccuracyDiversity.ItemType m → ℝ) → ℝ → PRPKG24AccuracyDiversity.ItemType m → ℝ

value:
  fun {m : ℕ} (p : PRPKG24AccuracyDiversity.ItemType m → ℝ) (sigma : ℝ) (i : PRPKG24AccuracyDiversity.ItemType m) =>
    PRPKG24AccuracyDiversity.AppendixD1GenericPower.sourcePowerWeight p sigma i
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.AppendixD1GenericPower.sourcePowerWeight](#)
- [PRPKG24AccuracyDiversity.ItemType](#)

PRPKG24AccuracyDiversity.AppendixD1GenericIII.normalizedWeight

Declaration location: paper-local

Lean declaration body

```
type:
  {m : ℕ} → (PRPKG24AccuracyDiversity.ItemType m → ℝ) → PRPKG24AccuracyDiversity.ItemType m → ℝ

value:
  fun {m : ℕ} (p : PRPKG24AccuracyDiversity.ItemType m → ℝ) (i : PRPKG24AccuracyDiversity.ItemType m) =>
    p i / ∑ j : PRPKG24AccuracyDiversity.ItemType m, p j
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.ItemType](#)

PRPKG24AccuracyDiversity.AppendixD1GenericIIFull.targetShare

Declaration location: paper-local

Lean declaration body

```
type:
{m : ℕ} → (PRPKG24AccuracyDiversity.ItemType m → ℝ) → ℝ → PRPKG24AccuracyDiversity.ItemType m → ℝ

value:
fun {m : ℕ} (p : PRPKG24AccuracyDiversity.ItemType m → ℝ) (sigma : ℝ) (i : PRPKG24AccuracyDiversity.ItemType m) =>
  PRPKG24AccuracyDiversity.AppendixD1GenericIII.normalizedWeight
  (PRPKG24AccuracyDiversity.AppendixD1GenericIIFull.targetWeight p sigma) i
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.AppendixD1GenericIIFull.targetWeight](#)
- [PRPKG24AccuracyDiversity.AppendixD1GenericIII.normalizedWeight](#)
- [PRPKG24AccuracyDiversity.ItemType](#)

PRPKG24AccuracyDiversity.BernoulliSatisfactionModel

Declaration location: paper-local

Lean declaration body

```
field likelihood:
{T : ℕ} → PRPKG24AccuracyDiversity.BernoulliSatisfactionModel T → PRPKG24AccuracyDiversity.ItemType T → ℝ

field successProb:
{T : ℕ} → PRPKG24AccuracyDiversity.BernoulliSatisfactionModel T → PRPKG24AccuracyDiversity.ItemType T → ℝ
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.ItemType](#)

PRPKG24AccuracyDiversity.ConsumptionModel

Declaration location: paper-local

Lean declaration body

```
field likelihood:
{T : ℕ} → PRPKG24AccuracyDiversity.ConsumptionModel T → PRPKG24AccuracyDiversity.ItemType T → ℝ

field valueOfCount:
{T : ℕ} → PRPKG24AccuracyDiversity.ConsumptionModel T → PRPKG24AccuracyDiversity.ItemType T → ℕ → ℝ
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.ItemType](#)

PRPKG24AccuracyDiversity.ConsumptionModel.linearValueOfCount

Declaration location: paper-local

Lean declaration body

```
type:
{T : ℕ} → (PRPKG24AccuracyDiversity.ItemType T → ℝ) → PRPKG24AccuracyDiversity.ItemType T → ℕ → ℝ

value:
fun {T : ℕ} (perItemValue : PRPKG24AccuracyDiversity.ItemType T → ℝ) (a : PRPKG24AccuracyDiversity.ItemType T)
  (a_1 : ℕ) =>
  ↑a_1 * perItemValue a
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.ItemType](#)

PRPKG24AccuracyDiversity.ConsumptionModel.linearized

Declaration location: paper-local

Lean declaration body

```
type:
{T : ℕ} →
(PRPKG24AccuracyDiversity.ItemType T → ℝ) →
(PRPKG24AccuracyDiversity.ItemType T → ℝ) → PRPKG24AccuracyDiversity.ConsumptionModel T

value:
fun {T : ℕ} (likelihood perItemValue : PRPKG24AccuracyDiversity.ItemType T → ℝ) =>
{ likelihood := likelihood,
  valueOfCount := PRPKG24AccuracyDiversity.ConsumptionModel.linearValueOfCount perItemValue }
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.ConsumptionModel](#)
- [PRPKG24AccuracyDiversity.ConsumptionModel.linearValueOfCount](#)
- [PRPKG24AccuracyDiversity.ItemType](#)

PRPKG24AccuracyDiversity.CountAllocation

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → Type

value:
fun (T : ℕ) => AppliedModelingLib.Allocation (PRPKG24AccuracyDiversity.ItemType T)
```

Direct retained dependencies

- [AppliedModelingLib.Allocation](#)
- [PRPKG24AccuracyDiversity.ItemType](#)

PRPKG24AccuracyDiversity.ConsumptionModel.FeasibleAtTotal

Declaration location: paper-local

Lean declaration body

```
type:
{T : ℕ} → ℕ → PRPKG24AccuracyDiversity.CountAllocation T → Prop

value:
fun {T : ℕ} (N : ℕ) (a : PRPKG24AccuracyDiversity.CountAllocation T) => AppliedModelingLib.Allocation.HasTotal a N
```

Direct retained dependencies

- [AppliedModelingLib.Allocation.HasTotal](#)
- [PRPKG24AccuracyDiversity.CountAllocation](#)
- [PRPKG24AccuracyDiversity.ItemType](#)

PRPKG24AccuracyDiversity.AllocationSequence

Declaration location: paper-local

Lean declaration body

field allocation:

$\{T : \mathbb{N}\} \rightarrow \text{PRPKG24AccuracyDiversity.AllocationSequence } T \rightarrow \mathbb{N} \rightarrow \text{PRPKG24AccuracyDiversity.CountAllocation } T$

field feasible:

$\forall \{T : \mathbb{N}\} \text{ (self : PRPKG24AccuracyDiversity.AllocationSequence } T) \text{ (N : } \mathbb{N}),$
 $\text{PRPKG24AccuracyDiversity.ConsumptionModel.FeasibleAtTotal } N \text{ (self.allocation } N)$

Direct retained dependencies

- [PRPKG24AccuracyDiversity.ConsumptionModel.FeasibleAtTotal](#)
- [PRPKG24AccuracyDiversity.CountAllocation](#)

PRPKG24AccuracyDiversity.ConsumptionModel.objective

Declaration location: paper-local

Lean declaration body

type:

$\{T : \mathbb{N}\} \rightarrow \text{PRPKG24AccuracyDiversity.ConsumptionModel } T \rightarrow \text{PRPKG24AccuracyDiversity.CountAllocation } T \rightarrow \mathbb{R}$

value:

$\text{fun } \{T : \mathbb{N}\} \text{ (M : PRPKG24AccuracyDiversity.ConsumptionModel } T) \text{ (a : PRPKG24AccuracyDiversity.CountAllocation } T) \Rightarrow$
 $\text{AppliedModelingLib.Allocation.objective } a \text{ M.likelihood } M.\text{valueOfCount}$

Direct retained dependencies

- [AppliedModelingLib.Allocation.objective](#)
- [PRPKG24AccuracyDiversity.ConsumptionModel](#)
- [PRPKG24AccuracyDiversity.CountAllocation](#)
- [PRPKG24AccuracyDiversity.ItemType](#)

PRPKG24AccuracyDiversity.ConsumptionModel.IsOptimalAtTotal

Declaration location: paper-local

Lean declaration body

type:

$\{T : \mathbb{N}\} \rightarrow \text{PRPKG24AccuracyDiversity.ConsumptionModel } T \rightarrow \mathbb{N} \rightarrow \text{PRPKG24AccuracyDiversity.CountAllocation } T \rightarrow \text{Prop}$

value:

$\text{fun } \{T : \mathbb{N}\} \text{ (M : PRPKG24AccuracyDiversity.ConsumptionModel } T) \text{ (N : } \mathbb{N})$
 $\text{ (a : PRPKG24AccuracyDiversity.CountAllocation } T) \Rightarrow$
 $\text{PRPKG24AccuracyDiversity.ConsumptionModel.FeasibleAtTotal } N \text{ a } \wedge$
 $\forall \text{ (b : PRPKG24AccuracyDiversity.CountAllocation } T),$
 $\text{PRPKG24AccuracyDiversity.ConsumptionModel.FeasibleAtTotal } N \text{ b} \rightarrow M.\text{objective } b \leq M.\text{objective } a$

Direct retained dependencies

- [PRPKG24AccuracyDiversity.ConsumptionModel](#)
- [PRPKG24AccuracyDiversity.ConsumptionModel.FeasibleAtTotal](#)
- [PRPKG24AccuracyDiversity.ConsumptionModel.objective](#)
- [PRPKG24AccuracyDiversity.CountAllocation](#)

PRPKG24AccuracyDiversity.AllocationSequence.representation

Declaration location: paper-local

Lean declaration body

type:

$\{T : \mathbb{N}\} \rightarrow \text{PRPKG24AccuracyDiversity.AllocationSequence } T \rightarrow \mathbb{N} \rightarrow \text{PRPKG24AccuracyDiversity.ItemType } T \rightarrow \mathbb{R}$

value:

$\text{fun } \{T : \mathbb{N}\} \text{ (seq : PRPKG24AccuracyDiversity.AllocationSequence } T) \text{ (N : } \mathbb{N}) \text{ (t : PRPKG24AccuracyDiversity.ItemType } T) \Rightarrow$
 $\text{(seq.allocation } N).\text{representation } t$

Direct retained dependencies

- [PRPKG24AccuracyDiversity.AllocationSequence](#)
- [PRPKG24AccuracyDiversity.CountAllocation.representation](#)
- [PRPKG24AccuracyDiversity.ItemType](#)

PRPKG24AccuracyDiversity.CountAllocation.HasApproxRepresentation

Declaration location: paper-local

Lean declaration body

```
type:
{T : ℕ} → PRPKG24AccuracyDiversity.CountAllocation T → (PRPKG24AccuracyDiversity.ItemType T → ℝ) → ℝ → Prop

value:
fun {T : ℕ} (a : PRPKG24AccuracyDiversity.CountAllocation T) (target : PRPKG24AccuracyDiversity.ItemType T → ℝ)
  (ε : ℝ) =>
  ∀ (t : PRPKG24AccuracyDiversity.ItemType T), |a.representation t - target t| ≤ ε
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.CountAllocation](#)
- [PRPKG24AccuracyDiversity.CountAllocation.representation](#)
- [PRPKG24AccuracyDiversity.ItemType](#)

PRPKG24AccuracyDiversity.CountAllocation.HasExactRepresentation

Declaration location: paper-local

Lean declaration body

```
type:
{T : ℕ} → PRPKG24AccuracyDiversity.CountAllocation T → (PRPKG24AccuracyDiversity.ItemType T → ℝ) → Prop

value:
fun {T : ℕ} (a : PRPKG24AccuracyDiversity.CountAllocation T) (target : PRPKG24AccuracyDiversity.ItemType T → ℝ) =>
  ∀ (t : PRPKG24AccuracyDiversity.ItemType T), a.representation t = target t
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.CountAllocation](#)
- [PRPKG24AccuracyDiversity.CountAllocation.representation](#)
- [PRPKG24AccuracyDiversity.ItemType](#)

PRPKG24AccuracyDiversity.GammaHomogeneityProfile

Declaration location: paper-local

Lean declaration body

```
field gamma:
{T : ℕ} → PRPKG24AccuracyDiversity.GammaHomogeneityProfile T → ℝ

field targetWeight:
{T : ℕ} → PRPKG24AccuracyDiversity.GammaHomogeneityProfile T → PRPKG24AccuracyDiversity.ItemType T → ℝ
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.ItemType](#)

PRPKG24AccuracyDiversity.GammaHomogeneityProfile.normalizer

Declaration location: paper-local

Lean declaration body

```
type:
{T : ℕ} → PRPKG24AccuracyDiversity.GammaHomogeneityProfile T → ℝ

value:
fun {T : ℕ} (G : PRPKG24AccuracyDiversity.GammaHomogeneityProfile T) =>
  ∑ t : PRPKG24AccuracyDiversity.ItemType T, G.targetWeight t
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.GammaHomogeneityProfile](#)
- [PRPKG24AccuracyDiversity.ItemType](#)

PRPKG24AccuracyDiversity.GammaHomogeneityProfile.targetShare

Declaration location: paper-local

Lean declaration body

```
type:
{T : ℕ} → PRPKG24AccuracyDiversity.GammaHomogeneityProfile T → PRPKG24AccuracyDiversity.ItemType T → ℝ

value:
fun {T : ℕ} (G : PRPKG24AccuracyDiversity.GammaHomogeneityProfile T) (t : PRPKG24AccuracyDiversity.ItemType T) =>
  if h : G.normalizer = 0 then 0 else G.targetWeight t / G.normalizer
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.GammaHomogeneityProfile](#)
- [PRPKG24AccuracyDiversity.GammaHomogeneityProfile.normalizer](#)
- [PRPKG24AccuracyDiversity.ItemType](#)

PRPKG24AccuracyDiversity.GammaHomogeneityProfile.Approx

Declaration location: paper-local

Lean declaration body

```
type:
{T : ℕ} → PRPKG24AccuracyDiversity.GammaHomogeneityProfile T → PRPKG24AccuracyDiversity.CountAllocation T → ℝ → Prop

value:
fun {T : ℕ} (G : PRPKG24AccuracyDiversity.GammaHomogeneityProfile T) (a : PRPKG24AccuracyDiversity.CountAllocation T)
  (ε : ℝ) =>
  a.HasApproxRepresentation G.targetShare ε
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.CountAllocation](#)
- [PRPKG24AccuracyDiversity.CountAllocation.HasApproxRepresentation](#)
- [PRPKG24AccuracyDiversity.GammaHomogeneityProfile](#)
- [PRPKG24AccuracyDiversity.GammaHomogeneityProfile.targetShare](#)

PRPKG24AccuracyDiversity.ConsumptionModel.AsymptoticHomogeneityTarget

Declaration location: paper-local

Lean declaration body

```
type:
{T : ℕ} →
  (N → PRPKG24AccuracyDiversity.ConsumptionModel T) →
  PRPKG24AccuracyDiversity.GammaHomogeneityProfile T → ((N → ℝ) → Prop) → Prop

value:
fun {T : ℕ} (Mseq : N → PRPKG24AccuracyDiversity.ConsumptionModel T)
  (G : PRPKG24AccuracyDiversity.GammaHomogeneityProfile T) (TendsToZero : (N → ℝ) → Prop) =>
  ∃ (ε : N → ℝ),
    TendsToZero ε ∧
    ∀ (N : ℕ) (a : PRPKG24AccuracyDiversity.CountAllocation T),
      0 < N → (Mseq N).IsOptimalAtTotal N a → G.Approx a (ε N)
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.ConsumptionModel](#)
- [PRPKG24AccuracyDiversity.ConsumptionModel.IsOptimalAtTotal](#)
- [PRPKG24AccuracyDiversity.CountAllocation](#)
- [PRPKG24AccuracyDiversity.GammaHomogeneityProfile](#)
- [PRPKG24AccuracyDiversity.GammaHomogeneityProfile.Approx](#)

PRPKG24AccuracyDiversity.ConsumptionModel.AsymptoticHomogeneity

Declaration location: paper-local

Lean declaration body

```
type:
{T : ℕ} → (ℕ → PRPKG24AccuracyDiversity.ConsumptionModel T) → PRPKG24AccuracyDiversity.GammaHomogeneityProfile T → Prop

value:
fun {T : ℕ} (Mseq : ℕ → PRPKG24AccuracyDiversity.ConsumptionModel T)
  (G : PRPKG24AccuracyDiversity.GammaHomogeneityProfile T) =>
  PRPKG24AccuracyDiversity.ConsumptionModel.AsymptoticHomogeneityTarget Mseq G AppliedModelingLib.Math.TendsToZero
```

Direct retained dependencies

- [AppliedModelingLib.Math.TendsToZero](#)
- [PRPKG24AccuracyDiversity.ConsumptionModel](#)
- [PRPKG24AccuracyDiversity.ConsumptionModel.AsymptoticHomogeneityTarget](#)
- [PRPKG24AccuracyDiversity.GammaHomogeneityProfile](#)

PRPKG24AccuracyDiversity.InfiniteGammaHomogeneityProfile

Declaration location: paper-local

Lean declaration body

```
field likelihood:
{T : ℕ} → PRPKG24AccuracyDiversity.InfiniteGammaHomogeneityProfile T → PRPKG24AccuracyDiversity.ItemType T → ℝ

field best:
{T : ℕ} → PRPKG24AccuracyDiversity.InfiniteGammaHomogeneityProfile T → PRPKG24AccuracyDiversity.ItemType T

field isArgmax:
∀ {T : ℕ} (self : PRPKG24AccuracyDiversity.InfiniteGammaHomogeneityProfile T) (t : PRPKG24AccuracyDiversity.ItemType T),
  self.likelihood t ≤ self.likelihood self.best
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.ItemType](#)

PRPKG24AccuracyDiversity.InfiniteGammaHomogeneityProfile.targetShare

Declaration location: paper-local

Lean declaration body

```
type:
{T : ℕ} → PRPKG24AccuracyDiversity.InfiniteGammaHomogeneityProfile T → PRPKG24AccuracyDiversity.ItemType T → ℝ

value:
fun {T : ℕ} (G : PRPKG24AccuracyDiversity.InfiniteGammaHomogeneityProfile T)
  (t : PRPKG24AccuracyDiversity.ItemType T) =>
  if t = G.best then 1 else 0
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.InfiniteGammaHomogeneityProfile](#)
- [PRPKG24AccuracyDiversity.ItemType](#)

PRPKG24AccuracyDiversity.OptimalAllocationSequence

Declaration location: paper-local

Lean declaration body

field allocation:

```
{T : ℕ} →  
{Mseq : ℕ → PRPKG24AccuracyDiversity.ConsumptionModel T} →  
  PRPKG24AccuracyDiversity.OptimalAllocationSequence Mseq → ℕ → PRPKG24AccuracyDiversity.CountAllocation T
```

field optimal:

```
∀ {T : ℕ} {Mseq : ℕ → PRPKG24AccuracyDiversity.ConsumptionModel T}  
(self : PRPKG24AccuracyDiversity.OptimalAllocationSequence Mseq) (N : ℕ),  
(Mseq N).IsOptimalAtTotal N (self.allocation N)
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.ConsumptionModel](#)
- [PRPKG24AccuracyDiversity.ConsumptionModel.IsOptimalAtTotal](#)
- [PRPKG24AccuracyDiversity.CountAllocation](#)

PRPKG24AccuracyDiversity.OptimalAllocationSequence.toAllocationSequence

Declaration location: paper-local

Lean declaration body

type:

```
{T : ℕ} →  
{Mseq : ℕ → PRPKG24AccuracyDiversity.ConsumptionModel T} →  
  PRPKG24AccuracyDiversity.OptimalAllocationSequence Mseq → PRPKG24AccuracyDiversity.AllocationSequence T
```

value:

```
fun {T : ℕ} {Mseq : ℕ → PRPKG24AccuracyDiversity.ConsumptionModel T}  
(seq : PRPKG24AccuracyDiversity.OptimalAllocationSequence Mseq) =>  
{ allocation := seq.allocation,  
  feasible := PRPKG24AccuracyDiversity.OptimalAllocationSequence.toAllocationSequence._proof_1 seq }
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.AllocationSequence](#)
- [PRPKG24AccuracyDiversity.ConsumptionModel](#)
- [PRPKG24AccuracyDiversity.ConsumptionModel.FeasibleAtTotal](#)
- [PRPKG24AccuracyDiversity.OptimalAllocationSequence](#)

PRPKG24AccuracyDiversity.Proposition4Sphere.UnitSphere

Declaration location: paper-local

Lean declaration body

type:

```
(E : Type u_1) → [NormedAddCommGroup E] → Type (max 0 u_1)
```

value:

```
fun (E : Type u_1) [NormedAddCommGroup E] => { x : E // x ∈ Metric.sphere 0 1 }
```

PRPKG24AccuracyDiversity.Proposition4Sphere.logRadialDistanceKernel

Declaration location: paper-local

Lean declaration body

type:

```
{E : Type u_1} →  
[inst : NormedAddCommGroup E] →  
(ℝ → ℝ) →  
  PRPKG24AccuracyDiversity.Proposition4Sphere.UnitSphere E →  
  PRPKG24AccuracyDiversity.Proposition4Sphere.UnitSphere E → ℝ
```

value:

```
fun {E : Type u_1} [NormedAddCommGroup E] (p : ℝ → ℝ)  
(x u : PRPKG24AccuracyDiversity.Proposition4Sphere.UnitSphere E) =>  
  Real.log (p (dist ↑x ↑u))
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.Proposition4Sphere.UnitSphere](#)

PRPKG24AccuracyDiversity.Proposition4Sphere.logRadialDistanceProfilePayoff

Declaration location: paper-local

Lean declaration body

```
type:
{E : Type u_1} →
[inst : NormedAddCommGroup E] →
[inst_1 : MeasurableSpace E] →
( $\mathbb{R} \rightarrow \mathbb{R}$ ) →
MeasureTheory.ProbabilityMeasure (PRPKG24AccuracyDiversity.Proposition4Sphere.UnitSphere E) →
PRPKG24AccuracyDiversity.Proposition4Sphere.logRadialDistanceProfilePayoff E →  $\mathbb{R}$ 

value:
fun {E : Type u_1} [NormedAddCommGroup E] [MeasurableSpace E] (p :  $\mathbb{R} \rightarrow \mathbb{R}$ )
(alpha : MeasureTheory.ProbabilityMeasure (PRPKG24AccuracyDiversity.Proposition4Sphere.UnitSphere E))
(u : PRPKG24AccuracyDiversity.Proposition4Sphere.UnitSphere E) =>
f (x : PRPKG24AccuracyDiversity.Proposition4Sphere.UnitSphere E),
PRPKG24AccuracyDiversity.Proposition4Sphere.logRadialDistanceKernel p x u  $\uparrow$  alpha
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.Proposition4Sphere.UnitSphere](#)
- [PRPKG24AccuracyDiversity.Proposition4Sphere.logRadialDistanceKernel](#)

PRPKG24AccuracyDiversity.Proposition4Sphere.logRadialDistanceProfileSupValue

Declaration location: paper-local

Lean declaration body

```
type:
{E : Type u_1} →
[inst : NormedAddCommGroup E] →
[inst_1 : MeasurableSpace E] →
( $\mathbb{R} \rightarrow \mathbb{R}$ ) → MeasureTheory.ProbabilityMeasure (PRPKG24AccuracyDiversity.Proposition4Sphere.UnitSphere E) →  $\mathbb{R}$ 

value:
fun {E : Type u_1} [NormedAddCommGroup E] [MeasurableSpace E] (p :  $\mathbb{R} \rightarrow \mathbb{R}$ )
(alpha : MeasureTheory.ProbabilityMeasure (PRPKG24AccuracyDiversity.Proposition4Sphere.UnitSphere E)) =>
sSup (Set.range (PRPKG24AccuracyDiversity.Proposition4Sphere.logRadialDistanceProfilePayoff p alpha))
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.Proposition4Sphere.UnitSphere](#)
- [PRPKG24AccuracyDiversity.Proposition4Sphere.logRadialDistanceProfilePayoff](#)

PRPKG24AccuracyDiversity.Proposition4Sphere.sphereSurfaceMeasure

Declaration location: paper-local

Lean declaration body

```
type:
{E : Type u_1} →
[inst : NormedAddCommGroup E] →
[InnerProductSpace  $\mathbb{R}$  E] →
[inst_2 : MeasurableSpace E] →
MeasureTheory.Measure E → MeasureTheory.Measure (PRPKG24AccuracyDiversity.Proposition4Sphere.UnitSphere E)

value:
fun {E : Type u_1} [NormedAddCommGroup E] [InnerProductSpace  $\mathbb{R}$  E] [MeasurableSpace E] ( $\mu$  : MeasureTheory.Measure E) =>
 $\mu$ .toSphere
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.Proposition4Sphere.UnitSphere](#)

PRPKG24AccuracyDiversity.Proposition4Sphere.sphereUniformMeasure

Declaration location: paper-local

Lean declaration body

```
type:
{E : Type u_1} →
[inst : NormedAddCommGroup E] →
[InnerProductSpace ℝ E] →
[inst_2 : MeasurableSpace E] →
MeasureTheory.Measure E → MeasureTheory.Measure (PRPKG24AccuracyDiversity.Proposition4Sphere.UnitSphere E)

value:
fun {E : Type u_1} [NormedAddCommGroup E] [InnerProductSpace ℝ E] [MeasurableSpace E] (μ : MeasureTheory.Measure E) =>
((PRPKG24AccuracyDiversity.Proposition4Sphere.sphereSurfaceMeasure μ) Set.univ)-1 •
PRPKG24AccuracyDiversity.Proposition4Sphere.sphereSurfaceMeasure μ
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.Proposition4Sphere.UnitSphere](#)
- [PRPKG24AccuracyDiversity.Proposition4Sphere.sphereSurfaceMeasure](#)

PRPKG24AccuracyDiversity.Proposition4Sphere.sphereVolumeUniformProbabilityMeasure

Declaration location: paper-local

Lean declaration body

```
type:
{E : Type u_1} →
[inst : NormedAddCommGroup E] →
[inst_1 : InnerProductSpace ℝ E] →
[inst_2 : MeasurableSpace E] →
[BorelSpace E] →
[FiniteDimensional ℝ E] →
[Nontrivial E] → MeasureTheory.ProbabilityMeasure (PRPKG24AccuracyDiversity.Proposition4Sphere.UnitSphere E)

value:
fun {E : Type u_1} [NormedAddCommGroup E] [InnerProductSpace ℝ E] [MeasurableSpace E] [BorelSpace E]
[FiniteDimensional ℝ E] [Nontrivial E] =>
(PRPKG24AccuracyDiversity.Proposition4Sphere.sphereUniformMeasure MeasureTheory.volume,
PRPKG24AccuracyDiversity.Proposition4Sphere.sphereVolumeUniformProbabilityMeasure._proof_1)
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.Proposition4Sphere.UnitSphere](#)
- [PRPKG24AccuracyDiversity.Proposition4Sphere.sphereUniformMeasure](#)

PRPKG24AccuracyDiversity.SourcePreferenceLaw

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → Type

value:
fun (T : ℕ) => PMF (PRPKG24AccuracyDiversity.ItemType T)
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.ItemType](#)

PRPKG24AccuracyDiversity.TopKValueOracle

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → Type

value:
fun (T : ℕ) => AppliedModelingLib.Probability.TopKExpectationOracle (PRPKG24AccuracyDiversity.ItemType T)
```

Direct retained dependencies

- [AppliedModelingLib.Probability.TopKExpectationOracle](#)
- [PRPKG24AccuracyDiversity.ItemType](#)

PRPKG24AccuracyDiversity.TopKValueOracle.common

Declaration location: paper-local

Lean declaration body

```
type:
(T : ℕ) → (ℕ → ℝ) → PRPKG24AccuracyDiversity.TopKValueOracle T

value:
fun (T : ℕ) (h : ℕ → ℝ) =>
  AppliedModelingLib.Probability.TopKExpectationOracle.common (PRPKG24AccuracyDiversity.ItemType T) h
```

Direct retained dependencies

- [AppliedModelingLib.Probability.TopKExpectationOracle.common](#)
- [PRPKG24AccuracyDiversity.ItemType](#)
- [PRPKG24AccuracyDiversity.TopKValueOracle](#)

PRPKG24AccuracyDiversity.TopKValueOracle.ofOrderStatisticMean

Declaration location: paper-local

Lean declaration body

```
type:
(T : ℕ) → (ℕ → ℕ → ℝ) → PRPKG24AccuracyDiversity.TopKValueOracle T

value:
fun (T : ℕ) (μ : ℕ → ℕ → ℝ) =>
  AppliedModelingLib.Probability.TopKExpectationOracle.orderStatisticTopKExpectationOracle
    (PRPKG24AccuracyDiversity.ItemType T) μ
```

Direct retained dependencies

- [AppliedModelingLib.Probability.TopKExpectationOracle.orderStatisticTopKExpectationOracle](#)
- [PRPKG24AccuracyDiversity.ItemType](#)
- [PRPKG24AccuracyDiversity.TopKValueOracle](#)

PRPKG24AccuracyDiversity.TopKValueOracle.toConsumptionModel

Declaration location: paper-local

Lean declaration body

```
type:
{T : ℕ} →
  PRPKG24AccuracyDiversity.TopKValueOracle T →
  (PRPKG24AccuracyDiversity.ItemType T → ℝ) → ℕ → PRPKG24AccuracyDiversity.ConsumptionModel T

value:
fun {T : ℕ} (O : PRPKG24AccuracyDiversity.TopKValueOracle T) (likelihood : PRPKG24AccuracyDiversity.ItemType T → ℝ)
  (ell : ℕ) =>
  { likelihood := likelihood,
    valueOfCount := fun (t : PRPKG24AccuracyDiversity.ItemType T) (q : ℕ) => O.expectedTopSum ell t q }
```

Direct retained dependencies

- [AppliedModelingLib.Probability.TopKExpectationOracle](#)
- [PRPKG24AccuracyDiversity.ConsumptionModel](#)
- [PRPKG24AccuracyDiversity.ItemType](#)
- [PRPKG24AccuracyDiversity.TopKValueOracle](#)

PRPKG24AccuracyDiversity.allOnTypeAllocation

Declaration location: paper-local

Lean declaration body

```
type:
{T : ℕ} → ℕ → PRPKG24AccuracyDiversity.ItemType T → PRPKG24AccuracyDiversity.CountAllocation T

value:
fun {T : ℕ} (N : ℕ) (best : PRPKG24AccuracyDiversity.ItemType T) =>
  AppliedModelingLib.Allocation.allOnTypeAllocation N best
```

Direct retained dependencies

- [AppliedModelingLib.Allocation.allOnTypeAllocation](#)
- [PRPKG24AccuracyDiversity.CountAllocation](#)
- [PRPKG24AccuracyDiversity.ItemType](#)

PRPKG24AccuracyDiversity.assumption_bernoulli_success_probability_range

Declaration location: paper-local

Lean declaration body

```
type:
{T : ℕ} → PRPKG24AccuracyDiversity.BernoulliSatisfactionModel T → Prop

value:
fun {T : ℕ} (B : PRPKG24AccuracyDiversity.BernoulliSatisfactionModel T) =>
  ∀ (t : PRPKG24AccuracyDiversity.ItemType T), 0 ≤ B.successProb t ∧ B.successProb t ≤ 1
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.BernoulliSatisfactionModel](#)
- [PRPKG24AccuracyDiversity.ItemType](#)

PRPKG24AccuracyDiversity.bernoulliAllConsumedModel

Declaration location: paper-local

Lean declaration body

```
type:
{T : ℕ} → PRPKG24AccuracyDiversity.BernoulliSatisfactionModel T → PRPKG24AccuracyDiversity.ConsumptionModel T

value:
fun {T : ℕ} (B : PRPKG24AccuracyDiversity.BernoulliSatisfactionModel T) =>
  PRPKG24AccuracyDiversity.ConsumptionModel.linearized B.likelihood B.successProb
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.BernoulliSatisfactionModel](#)
- [PRPKG24AccuracyDiversity.ConsumptionModel](#)
- [PRPKG24AccuracyDiversity.ConsumptionModel.linearized](#)

PRPKG24AccuracyDiversity.bernoulliAtLeastOneValue

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ℕ → ℝ

value:
fun (p : ℝ) (q : ℕ) => 1 - (1 - p) ^ q
```

PRPKG24AccuracyDiversity.BernoulliSatisfactionModel.toConsumptionModel

Declaration location: paper-local

Lean declaration body

```
type:
{T : ℕ} → PRPKG24AccuracyDiversity.BernoulliSatisfactionModel T → PRPKG24AccuracyDiversity.ConsumptionModel T

value:
fun {T : ℕ} (B : PRPKG24AccuracyDiversity.BernoulliSatisfactionModel T) =>
{ likelihood := B.likelihood,
  valueOfCount := fun (t : PRPKG24AccuracyDiversity.ItemType T) (q : ℕ) =>
    PRPKG24AccuracyDiversity.bernoulliAtLeastOneValue (B.successProb t) q }
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.BernoulliSatisfactionModel](#)
- [PRPKG24AccuracyDiversity.ConsumptionModel](#)
- [PRPKG24AccuracyDiversity.ItemType](#)
- [PRPKG24AccuracyDiversity.bernoulliAtLeastOneValue](#)

PRPKG24AccuracyDiversity.boundedLemmaD2IntegralKernel

Declaration location: paper-local

Lean declaration body

```
type:
(ℝ → ℝ) → ℕ → ℕ → ℝ → ℝ

value:
fun (G : ℝ → ℝ) (j a : ℕ) (x : ℝ) => ↑(a.choose j) * G x ^ j * (1 - G x) ^ (a - j)
```

PRPKG24AccuracyDiversity.boundedLemmaD2IntegralTerm

Declaration location: paper-local

Lean declaration body

```
type:
(ℝ → ℝ) → ℕ → ℕ → ℝ

value:
fun (G : ℝ → ℝ) (j a : ℕ) => ∫ (x : ℝ) in Set.Ioi 0, PRPKG24AccuracyDiversity.boundedLemmaD2IntegralKernel G j a x
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.boundedLemmaD2IntegralKernel](#)

PRPKG24AccuracyDiversity.boundedLemmaD2LimitKernel

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ℝ → ℕ → ℝ → ℝ

value:
fun (beta c : ℝ) (j : ℕ) (y : ℝ) =>
(c / beta) ^ j / ↑j.factorial * (y ^ (beta * ↑j) * Real.exp (-(c / beta) * y ^ beta))
```

PRPKG24AccuracyDiversity.boundedLemmaD2LimitCoeff

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ℝ → ℕ → ℝ

value:
fun (beta c : ℝ) (j : ℕ) => ∫ (y : ℝ) in Set.Ioi 0, PRPKG24AccuracyDiversity.boundedLemmaD2LimitKernel beta c j y
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.boundedLemmaD2LimitKernel](#)

PRPKG24AccuracyDiversity.boundedReflectedPowerSourceMap

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ 

value:
fun (beta u :  $\mathbb{R}$ ) => 1 - u ^ (1 / beta)
```

PRPKG24AccuracyDiversity.boundedTailScale

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{N} \rightarrow \mathbb{R}$ 

value:
fun (beta :  $\mathbb{R}$ ) (a :  $\mathbb{N}$ ) => ↑a ^ (-{1 / beta})
```

PRPKG24AccuracyDiversity.decayingBernoulliSuccess

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{N} \rightarrow \mathbb{R}$ 

value:
fun (c d  $\alpha$  :  $\mathbb{R}$ ) (i :  $\mathbb{N}$ ) => c * (↑(i + 1) + d) ^ (- $\alpha$ )
```

PRPKG24AccuracyDiversity.definition3lidSampleMeasure

Declaration location: paper-local

Lean declaration body

```
type:
MeasureTheory.Measure  $\mathbb{R} \rightarrow (a : \mathbb{N}) \rightarrow$  MeasureTheory.Measure (Fin a  $\rightarrow \mathbb{R}$ )

value:
fun (D : MeasureTheory.Measure  $\mathbb{R}$ ) (a :  $\mathbb{N}$ ) => MeasureTheory.Measure.pi fun (x : Fin a) => D
```

PRPKG24AccuracyDiversity.expectedOrderStatisticMeanSeq

Declaration location: paper-local

Lean declaration body

```
type:
((a :  $\mathbb{N}$ )  $\rightarrow$  MeasureTheory.Measure (Fin a  $\rightarrow \mathbb{R}$ ))  $\rightarrow \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{R}$ 

value:
fun (sampleMeasure : (a :  $\mathbb{N}$ )  $\rightarrow$  MeasureTheory.Measure (Fin a  $\rightarrow \mathbb{R}$ )) (rank sampleSize :  $\mathbb{N}$ ) =>
  AppliedModelingLib.Probability.expectedOrderStatisticMeanSeq sampleMeasure rank sampleSize
```

Direct retained dependencies

- [AppliedModelingLib.Probability.expectedOrderStatisticMeanSeq](#)

PRPKG24AccuracyDiversity.boundedlidOrderStatisticMeanSeq

Declaration location: paper-local

Lean declaration body

```
type:
MeasureTheory.Measure  $\mathbb{R} \rightarrow \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{R}$ 

value:
fun (baseMeasure : MeasureTheory.Measure  $\mathbb{R}$ ) (a a_1 :  $\mathbb{N}$ ) =>
  PRPKG24AccuracyDiversity.expectedOrderStatisticMeanSeq
    (fun (a :  $\mathbb{N}$ ) => MeasureTheory.Measure.pi fun (x : Fin a) => baseMeasure) a a_1
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.expectedOrderStatisticMeanSeq](#)

PRPKG24AccuracyDiversity.boundedlidOrderStatisticOracle

Declaration location: paper-local

Lean declaration body

```
type:
( $T : \mathbb{N}$ )  $\rightarrow$  MeasureTheory.Measure  $\mathbb{R} \rightarrow$  PRPKG24AccuracyDiversity.TopKValueOracle T

value:
fun (T :  $\mathbb{N}$ ) (baseMeasure : MeasureTheory.Measure  $\mathbb{R}$ ) =>
  PRPKG24AccuracyDiversity.TopKValueOracle.ofOrderStatisticMean T
    (PRPKG24AccuracyDiversity.boundedlidOrderStatisticMeanSeq baseMeasure)
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.TopKValueOracle](#)
- [PRPKG24AccuracyDiversity.TopKValueOracle.ofOrderStatisticMean](#)
- [PRPKG24AccuracyDiversity.boundedlidOrderStatisticMeanSeq](#)

PRPKG24AccuracyDiversity.boundedlidOrderStatisticConsumptionModel

Declaration location: paper-local

Lean declaration body

```
type:
{ $T : \mathbb{N}$ }  $\rightarrow$ 
  (PRPKG24AccuracyDiversity.ItemType T  $\rightarrow \mathbb{R}$ )  $\rightarrow \mathbb{N} \rightarrow$  MeasureTheory.Measure  $\mathbb{R} \rightarrow$  PRPKG24AccuracyDiversity.ConsumptionModel T

value:
fun { $T : \mathbb{N}$ } (likelihood : PRPKG24AccuracyDiversity.ItemType T  $\rightarrow \mathbb{R}$ ) (k :  $\mathbb{N}$ ) (baseMeasure : MeasureTheory.Measure  $\mathbb{R}$ ) =>
  (PRPKG24AccuracyDiversity.boundedlidOrderStatisticOracle T baseMeasure).toConsumptionModel likelihood k
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.ConsumptionModel](#)
- [PRPKG24AccuracyDiversity.ItemType](#)
- [PRPKG24AccuracyDiversity.TopKValueOracle.toConsumptionModel](#)
- [PRPKG24AccuracyDiversity.boundedlidOrderStatisticOracle](#)

PRPKG24AccuracyDiversity.exponentialDistributionModel

Declaration location: paper-local

Lean declaration body

```
type:
( $\lambda : \mathbb{R}$ )  $\rightarrow 0 < \lambda \rightarrow$  AppliedModelingLib.Probability.Exponential.Model

value:
fun ( $\lambda : \mathbb{R}$ ) (hlambda_pos :  $0 < \lambda$ ) => { rate :=  $\lambda$ , rate_pos := hlambda_pos }
```

Direct retained dependencies

- [AppliedModelingLib.Probability.Exponential.Model](#)

PRPKG24AccuracyDiversity.exponentialTopKOrderStatisticValue

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{R}$ 

value:
fun (lambda :  $\mathbb{R}$ ) (k q :  $\mathbb{N}$ ) =>  $\sum j \in \text{Finset.range } q, 1 / \text{lambda} * (\uparrow(\min k (j + 1)) / \uparrow(j + 1))$ 
```

PRPKG24AccuracyDiversity.exponentialTopKOrderStatisticOracle

Declaration location: paper-local

Lean declaration body

```
type:
(T :  $\mathbb{N}$ )  $\rightarrow \mathbb{R} \rightarrow \mathbb{N} \rightarrow \text{PRPKG24AccuracyDiversity.TopKValueOracle } T$ 

value:
fun (T :  $\mathbb{N}$ ) (lambda :  $\mathbb{R}$ ) (k :  $\mathbb{N}$ ) =>
  PRPKG24AccuracyDiversity.TopKValueOracle.common T
  (PRPKG24AccuracyDiversity.exponentialTopKOrderStatisticValue lambda k)
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.TopKValueOracle](#)
- [PRPKG24AccuracyDiversity.TopKValueOracle.common](#)
- [PRPKG24AccuracyDiversity.exponentialTopKOrderStatisticValue](#)

PRPKG24AccuracyDiversity.harmonicReal

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{N} \rightarrow \mathbb{R}$ 

value:
fun (q :  $\mathbb{N}$ ) => AppliedModelingLib.Probability.Exponential.harmonicReal q
```

Direct retained dependencies

- [AppliedModelingLib.Probability.Exponential.harmonicReal](#)

PRPKG24AccuracyDiversity.iidAllConsumedSampleLaw

Declaration location: paper-local

Lean declaration body

```
type:
MeasureTheory.Measure  $\mathbb{R} \rightarrow (q : \mathbb{N}) \rightarrow \text{MeasureTheory.Measure } (\text{Fin } q \rightarrow \mathbb{R})$ 

value:
fun (D : MeasureTheory.Measure  $\mathbb{R}$ ) (q :  $\mathbb{N}$ ) => MeasureTheory.Measure.pi fun (x : Fin q) => D
```

PRPKG24AccuracyDiversity.iidAllConsumedSampleValue

Declaration location: paper-local

Lean declaration body

```
type:
{q :  $\mathbb{N}$ }  $\rightarrow (\text{Fin } q \rightarrow \mathbb{R}) \rightarrow \mathbb{R}$ 

value:
fun {q :  $\mathbb{N}$ } (sample : Fin q  $\rightarrow \mathbb{R}$ ) =>  $\sum i : \text{Fin } q, \text{sample } i$ 
```

PRPKG24AccuracyDiversity.iidAllConsumedSourceModel

Declaration location: paper-local

Lean declaration body

```
type:
  {T : ℕ} →
    PRPKG24AccuracyDiversity.SourcePreferenceLaw T → MeasureTheory.Measure ℝ → PRPKG24AccuracyDiversity.ConsumptionModel T

value:
  fun {T : ℕ} (preferenceLaw : PRPKG24AccuracyDiversity.SourcePreferenceLaw T) (D : MeasureTheory.Measure ℝ) =>
    { likelihood := fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal,
      valueOfCount := fun (x : PRPKG24AccuracyDiversity.ItemType T) (q : ℕ) =>
        ∫ (sample : Fin q → ℝ),
          PRPKG24AccuracyDiversity.iidAllConsumedSampleValue
            sample ∂PRPKG24AccuracyDiversity.iidAllConsumedSampleLaw D q }
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.ConsumptionModel](#)
- [PRPKG24AccuracyDiversity.ItemType](#)
- [PRPKG24AccuracyDiversity.SourcePreferenceLaw](#)
- [PRPKG24AccuracyDiversity.iidAllConsumedSampleLaw](#)
- [PRPKG24AccuracyDiversity.iidAllConsumedSampleValue](#)

PRPKG24AccuracyDiversity.orderStatisticTopKSumFromMean

Declaration location: paper-local

Lean declaration body

```
type:
  (ℕ → ℕ → ℝ) → ℕ → ℕ → ℝ

value:
  fun (a : ℕ → ℕ → ℝ) (a_1 a_2 : ℕ) => AppliedModelingLib.Probability.orderStatisticTopKSumFromMean a a_1 a_2
```

Direct retained dependencies

- [AppliedModelingLib.Probability.orderStatisticTopKSumFromMean](#)

PRPKG24AccuracyDiversity.paretolidSampleMeasure

Declaration location: paper-local

Lean declaration body

```
type:
  ℝ → (q : ℕ) → MeasureTheory.Measure (Fin q → ℝ)

value:
  fun (α : ℝ) (q : ℕ) => AppliedModelingLib.Probability.Pareto.iidProductMeasure 1 α q
```

Direct retained dependencies

- [AppliedModelingLib.Probability.Pareto.iidProductMeasure](#)

PRPKG24AccuracyDiversity.paretolidOrderStatisticMeanSeq

Declaration location: paper-local

Lean declaration body

```
type:
  ℝ → ℕ → ℕ → ℝ

value:
  fun (α : ℝ) (a a_1 : ℕ) =>
    PRPKG24AccuracyDiversity.expectedOrderStatisticMeanSeq (PRPKG24AccuracyDiversity.paretolidSampleMeasure α) a a_1
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.expectedOrderStatisticMeanSeq](#)
- [PRPKG24AccuracyDiversity.paretoIidSampleMeasure](#)

PRPKG24AccuracyDiversity.paretoIidOrderStatisticOracle

Declaration location: paper-local

Lean declaration body

```
type:
(T : ℕ) → ℝ → PRPKG24AccuracyDiversity.TopKValueOracle T

value:
fun (T : ℕ) (α : ℝ) =>
  PRPKG24AccuracyDiversity.TopKValueOracle.ofOrderStatisticMean T
  (PRPKG24AccuracyDiversity.paretoIidOrderStatisticMeanSeq α)
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.TopKValueOracle](#)
- [PRPKG24AccuracyDiversity.TopKValueOracle.ofOrderStatisticMean](#)
- [PRPKG24AccuracyDiversity.paretoIidOrderStatisticMeanSeq](#)

PRPKG24AccuracyDiversity.paretoIidOrderStatisticConsumptionModel

Declaration location: paper-local

Lean declaration body

```
type:
{T : ℕ} → (PRPKG24AccuracyDiversity.ItemType T → ℝ) → ℕ → ℝ → PRPKG24AccuracyDiversity.ConsumptionModel T

value:
fun {T : ℕ} (likelihood : PRPKG24AccuracyDiversity.ItemType T → ℝ) (k : ℕ) (α : ℝ) =>
  (PRPKG24AccuracyDiversity.paretoIidOrderStatisticOracle T α).toConsumptionModel likelihood k
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.ConsumptionModel](#)
- [PRPKG24AccuracyDiversity.ItemType](#)
- [PRPKG24AccuracyDiversity.TopKValueOracle.toConsumptionModel](#)
- [PRPKG24AccuracyDiversity.paretoIidOrderStatisticOracle](#)

PRPKG24AccuracyDiversity.paretoRankValueCoeff

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ℕ → ℝ

value:
fun (α : ℝ) (r : ℕ) => Real.Gamma (↑r + 1 - 1 / α) / Real.Gamma (↑r + 1)
```

PRPKG24AccuracyDiversity.rankBernoulliAllConsumedValue

Declaration location: paper-local

Lean declaration body

```
type:
(ℕ → ℝ) → ℕ → ℝ

value:
fun (success : ℕ → ℝ) (q : ℕ) => ∑ i ∈ Finset.range q, success i
```

PRPKG24AccuracyDiversity.rankBernoulliAllConsumedConsumptionModel

Declaration location: paper-local

Lean declaration body

```
type:
{T : ℕ} → (PRPKG24AccuracyDiversity.ItemType T → ℝ) → (ℕ → ℝ) → PRPKG24AccuracyDiversity.ConsumptionModel T

value:
fun {T : ℕ} (likelihood : PRPKG24AccuracyDiversity.ItemType T → ℝ) (success : ℕ → ℝ) =>
{ likelihood := likelihood,
  valueOfCount := fun (x : PRPKG24AccuracyDiversity.ItemType T) (q : ℕ) =>
    PRPKG24AccuracyDiversity.rankBernoulliAllConsumedValue success q }
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.ConsumptionModel](#)
- [PRPKG24AccuracyDiversity.ItemType](#)
- [PRPKG24AccuracyDiversity.rankBernoulliAllConsumedValue](#)

PRPKG24AccuracyDiversity.decayingBernoulliAllConsumedConsumptionModel

Declaration location: paper-local

Lean declaration body

```
type:
{T : ℕ} → (PRPKG24AccuracyDiversity.ItemType T → ℝ) → ℝ → ℝ → ℝ → PRPKG24AccuracyDiversity.ConsumptionModel T

value:
fun {T : ℕ} (likelihood : PRPKG24AccuracyDiversity.ItemType T → ℝ) (c d α : ℝ) =>
  PRPKG24AccuracyDiversity.rankBernoulliAllConsumedConsumptionModel likelihood
  (PRPKG24AccuracyDiversity.decayingBernoulliSuccess c d α)
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.ConsumptionModel](#)
- [PRPKG24AccuracyDiversity.ItemType](#)
- [PRPKG24AccuracyDiversity.decayingBernoulliSuccess](#)
- [PRPKG24AccuracyDiversity.rankBernoulliAllConsumedConsumptionModel](#)

PRPKG24AccuracyDiversity.rankBernoulliFiniteAllConsumedSampleValue

Declaration location: paper-local

Lean declaration body

```
type:
{q : ℕ} → (Fin q → Bool) → ℝ

value:
fun {q : ℕ} (sample : Fin q → Bool) => ∑ i : Fin q, AppliedModelingLib.Probability.binaryRatingScore (sample i)
```

Direct retained dependencies

- [AppliedModelingLib.Probability.binaryRatingScore](#)

PRPKG24AccuracyDiversity.rankBernoulliFiniteLaw

Declaration location: paper-local

Lean declaration body

```
type:
(success : ℕ → ℝ) → (q : ℕ) → (∀ (i : Fin q), 0 ≤ success ↑i ∧ success ↑i ≤ 1) → PMF (Fin q → Bool)

value:
fun (success : ℕ → ℝ) (q : ℕ) (hvalid : ∀ (i : Fin q), 0 ≤ success ↑i ∧ success ↑i ≤ 1) =>
  AppliedModelingLib.pmfPi fun (i : Fin q) =>
    AppliedModelingLib.Probability.realBernoulliPMF (success ↑i)
    (PRPKG24AccuracyDiversity.rankBernoulliFiniteLaw._proof_1 success q hvalid i)
    (PRPKG24AccuracyDiversity.rankBernoulliFiniteLaw._proof_2 success q hvalid i)
```


Direct retained dependencies

- [AppliedModelingLib.Probability.realBernoulliPMF](#)
- [AppliedModelingLib.pmfPi](#)

PRPKG24AccuracyDiversity.decayingBernoulliFiniteLaw

Declaration location: paper-local

Lean declaration body

```
type:
(c d alpha : ℝ) →
(q : ℕ) →
0 ≤ c → 0 ≤ d → 0 ≤ alpha → PRPKG24AccuracyDiversity.decayingBernoulliSuccess c d alpha 0 ≤ 1 → PMF (Fin q → Bool)

value:
fun (c d alpha : ℝ) (q : ℕ) (hc_nonneg : 0 ≤ c) (hd_nonneg : 0 ≤ d) (halpha_nonneg : 0 ≤ alpha)
(hfirst_le_one : PRPKG24AccuracyDiversity.decayingBernoulliSuccess c d alpha 0 ≤ 1) =>
PRPKG24AccuracyDiversity.rankBernoulliFiniteLaw (PRPKG24AccuracyDiversity.decayingBernoulliSuccess c d alpha) q
(PRPKG24AccuracyDiversity.decayingBernoulliSuccess_finite_valid c d alpha q hc_nonneg hd_nonneg halpha_nonneg
hfirst_le_one)
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.decayingBernoulliSuccess](#)
- [PRPKG24AccuracyDiversity.rankBernoulliFiniteLaw](#)

PRPKG24AccuracyDiversity.rankBernoulliFiniteTopOneSampleValue

Declaration location: paper-local

Lean declaration body

```
type:
{q : ℕ} → (Fin q → Bool) → ℝ

value:
fun {q : ℕ} (sample : Fin q → Bool) => if ∀ (i : Fin q), sample i = false then 0 else 1
```

PRPKG24AccuracyDiversity.rankBernoulliTopOneValue

Declaration location: paper-local

Lean declaration body

```
type:
(ℕ → ℝ) → ℕ → ℝ

value:
fun (success : ℕ → ℝ) (q : ℕ) => 1 - ∏ i ∈ Finset.range q, (1 - success i)
```

PRPKG24AccuracyDiversity.rankBernoulliTopOneConsumptionModel

Declaration location: paper-local

Lean declaration body

```
type:
{T : ℕ} → (PRPKG24AccuracyDiversity.ItemType T → ℝ) → (ℕ → ℝ) → PRPKG24AccuracyDiversity.ConsumptionModel T

value:
fun {T : ℕ} (likelihood : PRPKG24AccuracyDiversity.ItemType T → ℝ) (success : ℕ → ℝ) =>
{ likelihood := likelihood,
  valueOfCount := fun (x : PRPKG24AccuracyDiversity.ItemType T) (q : ℕ) =>
    PRPKG24AccuracyDiversity.rankBernoulliTopOneValue success q }
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.ConsumptionModel](#)
- [PRPKG24AccuracyDiversity.ItemType](#)
- [PRPKG24AccuracyDiversity.rankBernoulliTopOneValue](#)

PRPKG24AccuracyDiversity.decayingBernoulliTopOneConsumptionModel

Declaration location: paper-local

Lean declaration body

```
type:
{T : ℕ} → (PRPKG24AccuracyDiversity.ItemType T → ℝ) → ℝ → ℝ → ℝ → PRPKG24AccuracyDiversity.ConsumptionModel T
value:
fun {T : ℕ} (likelihood : PRPKG24AccuracyDiversity.ItemType T → ℝ) (c d α : ℝ) =>
  PRPKG24AccuracyDiversity.rankBernoulliTopOneConsumptionModel likelihood
  (PRPKG24AccuracyDiversity.decayingBernoulliSuccess c d α)
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.ConsumptionModel](#)
- [PRPKG24AccuracyDiversity.ItemType](#)
- [PRPKG24AccuracyDiversity.decayingBernoulliSuccess](#)
- [PRPKG24AccuracyDiversity.rankBernoulliTopOneConsumptionModel](#)

PRPKG24AccuracyDiversity.sqrtLikelihoodProfile

Declaration location: paper-local

Lean declaration body

```
type:
{T : ℕ} → (PRPKG24AccuracyDiversity.ItemType T → ℝ) → PRPKG24AccuracyDiversity.GammaHomogeneityProfile T
value:
fun {T : ℕ} (likelihood : PRPKG24AccuracyDiversity.ItemType T → ℝ) =>
  { gamma := 1 / 2, targetWeight := fun (t : PRPKG24AccuracyDiversity.ItemType T) => √(likelihood t) }
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.GammaHomogeneityProfile](#)
- [PRPKG24AccuracyDiversity.ItemType](#)

PRPKG24AccuracyDiversity.PaperInterface.proposition2SqrtProfile

Declaration location: paper-local

Lean declaration body

```
type:
{T : ℕ} → (PRPKG24AccuracyDiversity.ItemType T → ℝ) → PRPKG24AccuracyDiversity.GammaHomogeneityProfile T
value:
fun {T : ℕ} (likelihood : PRPKG24AccuracyDiversity.ItemType T → ℝ) =>
  PRPKG24AccuracyDiversity.sqrtLikelihoodProfile likelihood
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.GammaHomogeneityProfile](#)
- [PRPKG24AccuracyDiversity.ItemType](#)
- [PRPKG24AccuracyDiversity.sqrtLikelihoodProfile](#)

PRPKG24AccuracyDiversity.theorem3LogShareWeight

Declaration location: paper-local

Lean declaration body

```
type:
{T : ℕ} → PRPKG24AccuracyDiversity.BernoulliSatisfactionModel T → PRPKG24AccuracyDiversity.ItemType T → ℝ
value:
fun {T : ℕ} (B : PRPKG24AccuracyDiversity.BernoulliSatisfactionModel T) (t : PRPKG24AccuracyDiversity.ItemType T) =>
  (Real.log (1 - B.successProb t))-1
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.BernoulliSatisfactionModel](#)
- [PRPKG24AccuracyDiversity.ItemType](#)

PRPKG24AccuracyDiversity.topKCandidateSets

Declaration location: paper-local

Lean declaration body

```
type:
  (ι : Type u_1) → [Fintype ι] → [DecidableEq ι] → ℕ → Finset (Finset ι)

value:
  fun (ι : Type u_1) [Fintype ι] [DecidableEq ι] (k : ℕ) => {s ∈ Finset.univ.powerset | s.card ≤ k}
```

PRPKG24AccuracyDiversity.topKSumOn

Declaration location: paper-local

Lean declaration body

```
type:
  {ι : Type u_1} → [Fintype ι] → [DecidableEq ι] → ℕ → (ι → ℝ) → ℝ

value:
  fun {ι : Type u_1} [Fintype ι] [DecidableEq ι] (k : ℕ) (v : ι → ℝ) =>
    (PRPKG24AccuracyDiversity.topKCandidateSets ι k).sup' (PRPKG24AccuracyDiversity.topKCandidateSets_nonempty ι k)
    fun (s : Finset ι) => ∑ i ∈ s, v i
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.topKCandidateSets](#)

PRPKG24AccuracyDiversity.iidTopKExpectedOn

Declaration location: paper-local

Lean declaration body

```
type:
  (ι : Type u_1) →
  (Ω : Type u_2) → [Fintype ι] → [DecidableEq ι] → [Fintype Ω] → [DecidableEq Ω] → PMF Ω → ℕ → (Ω → ℝ) → ℝ

value:
  fun (ι : Type u_1) (Ω : Type u_2) [Fintype ι] [DecidableEq ι] [Fintype Ω] [DecidableEq Ω] (itemLaw : PMF Ω) (k : ℕ)
    (value : Ω → ℝ) =>
    AppliedModelingLib.pmfExp (AppliedModelingLib.pmfProduct ι Ω itemLaw) fun (sample : ι → Ω) =>
      PRPKG24AccuracyDiversity.topKSumOn k fun (i : ι) => value (sample i)
```

Direct retained dependencies

- [AppliedModelingLib.pmfExp](#)
- [AppliedModelingLib.pmfProduct](#)
- [PRPKG24AccuracyDiversity.topKSumOn](#)

PRPKG24AccuracyDiversity.finiteDiscretelidTopKExpected

Declaration location: paper-local

Lean declaration body

```
type:
  (Ω : Type u_1) → [Fintype Ω] → [DecidableEq Ω] → PMF Ω → ℕ → (Ω → ℝ) → ℕ → ℝ

value:
  fun (Ω : Type u_1) [Fintype Ω] [DecidableEq Ω] (itemLaw : PMF Ω) (k : ℕ) (value : Ω → ℝ) (a : ℕ) =>
    PRPKG24AccuracyDiversity.iidTopKExpectedOn (Fin a) Ω itemLaw k value
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.iidTopKExpectedOn](#)

PRPKG24AccuracyDiversity.corollary1FiniteTwoPointBernoulliSourceIdModel

Declaration location: paper-local

Lean declaration body

```

type:
{T : ℕ} →
(PRPKG24AccuracyDiversity.ItemType T → ℝ) → ℕ → (q : ℝ) → 0 ≤ q → q ≤ 1 → PRPKG24AccuracyDiversity.ConsumptionModel T

value:
fun {T : ℕ} (likelihood : PRPKG24AccuracyDiversity.ItemType T → ℝ) (k : ℕ) (q : ℝ) (hq_nonneg : 0 ≤ q)
  (hq_le_one : q ≤ 1) =>
(PRPKG24AccuracyDiversity.TopKValueOracle.common T
  (PRPKG24AccuracyDiversity.finiteDiscreteIdTopKExpected Bool
    (AppliedModelingLib.Probability.realBernoulliPMF q hq_nonneg hq_le_one) k
    AppliedModelingLib.Probability.binaryRatingScore)).toConsumptionModel
likelihood k

```

Direct retained dependencies

- [AppliedModelingLib.Probability.binaryRatingScore](#)
- [AppliedModelingLib.Probability.realBernoulliPMF](#)
- [PRPKG24AccuracyDiversity.ConsumptionModel](#)
- [PRPKG24AccuracyDiversity.ItemType](#)
- [PRPKG24AccuracyDiversity.TopKValueOracle.common](#)
- [PRPKG24AccuracyDiversity.TopKValueOracle.toConsumptionModel](#)
- [PRPKG24AccuracyDiversity.finiteDiscreteIdTopKExpected](#)

PRPKG24AccuracyDiversity.uniform01Measure

Declaration location: paper-local

Lean declaration body

```

type:
MeasureTheory.Measure ℝ

value:
MeasureTheory.volume[|Set.Icc 0 1]

```

PRPKG24AccuracyDiversity.boundedReflectedPowerSourceMeasure

Declaration location: paper-local

Lean declaration body

```

type:
ℝ → MeasureTheory.Measure ℝ

value:
fun (beta : ℝ) =>
  MeasureTheory.Measure.map (PRPKG24AccuracyDiversity.boundedReflectedPowerSourceMap beta)
    PRPKG24AccuracyDiversity.uniform01Measure

```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.boundedReflectedPowerSourceMap](#)
- [PRPKG24AccuracyDiversity.uniform01Measure](#)

PRPKG24AccuracyDiversity.boundedReflectedPowerSourceIdOrderStatisticMeanSeq

Declaration location: paper-local

Lean declaration body

```

type:
ℝ → ℕ → ℕ → ℝ

value:
fun (beta : ℝ) (rank sampleSize : ℕ) =>
  AppliedModelingLib.Probability.expectedOrderStatisticMeanSeq
    (fun (a : ℕ) =>
      MeasureTheory.Measure.pi fun (x : Fin a) => PRPKG24AccuracyDiversity.boundedReflectedPowerSourceMeasure beta)
    rank sampleSize

```

Direct retained dependencies

- [AppliedModelingLib.Probability.expectedOrderStatisticMeanSeq](#)
- [PRPKG24AccuracyDiversity.boundedReflectedPowerSourceMeasure](#)

PRPKG24AccuracyDiversity.boundedReflectedPowerSourceIdOrderStatisticOracle

Declaration location: paper-local

Lean declaration body

```
type:
(T : ℕ) → ℝ → PRPKG24AccuracyDiversity.TopKValueOracle T

value:
fun (T : ℕ) (beta : ℝ) =>
  PRPKG24AccuracyDiversity.TopKValueOracle.ofOrderStatisticMean T
  (PRPKG24AccuracyDiversity.boundedReflectedPowerSourceIdOrderStatisticMeanSeq beta)
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.TopKValueOracle](#)
- [PRPKG24AccuracyDiversity.TopKValueOracle.ofOrderStatisticMean](#)
- [PRPKG24AccuracyDiversity.boundedReflectedPowerSourceIdOrderStatisticMeanSeq](#)

PRPKG24AccuracyDiversity.boundedReflectedPowerSourceIdOrderStatisticConsumptionModel

Declaration location: paper-local

Lean declaration body

```
type:
{T : ℕ} → (PRPKG24AccuracyDiversity.ItemType T → ℝ) → ℕ → ℝ → PRPKG24AccuracyDiversity.ConsumptionModel T

value:
fun {T : ℕ} (likelihood : PRPKG24AccuracyDiversity.ItemType T → ℝ) (k : ℕ) (beta : ℝ) =>
  (PRPKG24AccuracyDiversity.boundedReflectedPowerSourceIdOrderStatisticOracle T beta).toConsumptionModel likelihood k
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.ConsumptionModel](#)
- [PRPKG24AccuracyDiversity.ItemType](#)
- [PRPKG24AccuracyDiversity.TopKValueOracle.toConsumptionModel](#)
- [PRPKG24AccuracyDiversity.boundedReflectedPowerSourceIdOrderStatisticOracle](#)

PRPKG24AccuracyDiversity.Corollary1SourceIdFamily

Declaration location: paper-local

Lean declaration body

```
type:
{T : ℕ} → (PRPKG24AccuracyDiversity.ItemType T → ℝ) → ℝ → ℕ → PRPKG24AccuracyDiversity.ConsumptionModel T → Prop

constructor PRPKG24AccuracyDiversity.Corollary1SourceIdFamily.finiteTwoPointBernoulli:
∀ {T : ℕ} {likelihood : PRPKG24AccuracyDiversity.ItemType T → ℝ} {gamma : ℝ} {k : ℕ} (q : ℝ) (hq_pos : 0 < q)
  (hq_lt_one : q < 1),
  gamma = 0 →
  PRPKG24AccuracyDiversity.Corollary1SourceIdFamily likelihood gamma k
  (PRPKG24AccuracyDiversity.corollary1FiniteTwoPointBernoulliSourceIdModel likelihood k q (LT.lt.le hq_pos)
    (LT.lt.le hq_lt_one))

constructor PRPKG24AccuracyDiversity.Corollary1SourceIdFamily.boundedReflectedPowerOrderStatistic:
∀ {T : ℕ} {likelihood : PRPKG24AccuracyDiversity.ItemType T → ℝ} {gamma : ℝ} {k : ℕ} (beta : ℝ),
  0 < beta →
  0 < k →
  beta / (beta + 1) = gamma →
  PRPKG24AccuracyDiversity.Corollary1SourceIdFamily likelihood gamma k
  (PRPKG24AccuracyDiversity.boundedReflectedPowerSourceIdOrderStatisticConsumptionModel likelihood k beta)

constructor PRPKG24AccuracyDiversity.Corollary1SourceIdFamily.exponentialOrderStatistic:
∀ {T : ℕ} {likelihood : PRPKG24AccuracyDiversity.ItemType T → ℝ} {gamma : ℝ} {k : ℕ} (lambda : ℝ),
  0 < lambda →
```

```

0 < k →
  gamma = 1 →
    PRPKG24AccuracyDiversity.Corollary1SourceIdFamily likelihood gamma k
    ((PRPKG24AccuracyDiversity.exponentialTopKOrderStatisticOracle T lambda k).toConsumptionModel likelihood k)

constructor PRPKG24AccuracyDiversity.Corollary1SourceIdFamily.paretoOrderStatistic:
  ∀ {T : ℕ} {likelihood : PRPKG24AccuracyDiversity.ItemType T → ℝ} {gamma : ℝ} {k : ℕ} (alpha : ℝ),
  1 < alpha →
  0 < k →
    alpha / (alpha - 1) = gamma →
      PRPKG24AccuracyDiversity.Corollary1SourceIdFamily likelihood gamma k
      (PRPKG24AccuracyDiversity.paretoIdOrderStatisticConsumptionModel likelihood k alpha)

```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.ConsumptionModel](#)
- [PRPKG24AccuracyDiversity.ItemType](#)
- [PRPKG24AccuracyDiversity.TopKValueOracle.toConsumptionModel](#)
- [PRPKG24AccuracyDiversity.boundedReflectedPowerSourceIdOrderStatisticConsumptionModel](#)
- [PRPKG24AccuracyDiversity.corollary1FiniteTwoPointBernoulliSourceIdModel](#)
- [PRPKG24AccuracyDiversity.exponentialTopKOrderStatisticOracle](#)
- [PRPKG24AccuracyDiversity.paretoIdOrderStatisticConsumptionModel](#)

PRPKG24AccuracyDiversity.uniformProfile

Declaration location: paper-local

Lean declaration body

```

type:
(T : ℕ) → PRPKG24AccuracyDiversity.GammaHomogeneityProfile T

value:
fun (T : ℕ) => { gamma := 0, targetWeight := fun (x : PRPKG24AccuracyDiversity.ItemType T) => 1 }

```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.GammaHomogeneityProfile](#)
- [PRPKG24AccuracyDiversity.ItemType](#)

PRPKG24AccuracyDiversity.uniformSqrtMinShare

Declaration location: paper-local

Lean declaration body

```

type:
{T : ℕ} → [NeZero T] → (PRPKG24AccuracyDiversity.ItemType T → ℝ) → ℝ

value:
fun {T : ℕ} [NeZero T] (likelihood : PRPKG24AccuracyDiversity.ItemType T → ℝ) =>
  AppliedModelingLib.finiteMin fun (t : PRPKG24AccuracyDiversity.ItemType T) =>
    √(likelihood t) / ∑ i : PRPKG24AccuracyDiversity.ItemType T, √(likelihood i)

```

Direct retained dependencies

- [AppliedModelingLib.finiteMin](#)
- [PRPKG24AccuracyDiversity.ItemType](#)

PRPKG24AccuracyDiversity.uniformTopKFactor

Declaration location: paper-local

Lean declaration body

```

type:
ℕ → ℝ

value:
fun (k : ℕ) => ↑k * (↑k + 1) / 2

```

PRPKG24AccuracyDiversity.uniformTopKValue

Declaration location: paper-local

Lean declaration body

type:
 $\mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{R}$

value:
fun (k q : $\mathbb{N}$) => if q $\leq$ k then $\uparrow q / 2$ else $\uparrow k - \text{PRPKG24AccuracyDiversity.uniformTopKFactor } k / (\uparrow q + 1)$

Direct retained dependencies

- [PRPKG24AccuracyDiversity.uniformTopKFactor](#)

PRPKG24AccuracyDiversity.uniformTopKConsumptionModel

Declaration location: paper-local

Lean declaration body

type:
 $\{T : \mathbb{N}\} \rightarrow (\text{PRPKG24AccuracyDiversity.ItemType } T \rightarrow \mathbb{R}) \rightarrow \mathbb{N} \rightarrow \text{PRPKG24AccuracyDiversity.ConsumptionModel } T$

value:
fun {T : $\mathbb{N}$ } (likelihood : PRPKG24AccuracyDiversity.ItemType T $\rightarrow \mathbb{R}$) (k : $\mathbb{N}$) =>
{ likelihood := likelihood,
valueOfCount := fun (x : PRPKG24AccuracyDiversity.ItemType T) (q : $\mathbb{N}$) =>
PRPKG24AccuracyDiversity.uniformTopKValue k q }

Direct retained dependencies

- [PRPKG24AccuracyDiversity.ConsumptionModel](#)
- [PRPKG24AccuracyDiversity.ItemType](#)
- [PRPKG24AccuracyDiversity.uniformTopKValue](#)

Paper-specific semantic prerequisites

PRPKG24AccuracyDiversity.AllocationSequence.ConvergesToProfile

Paper-local declaration:

PRPKG24AccuracyDiversity.AllocationSequence.ConvergesToProfile

Source locator: cited publication, lines 301-338

Verbatim paper-source connection

Definition 2 (γ -homogeneity for set sequences). A sequence of sets $\{S_n\}_{n=1}^\infty$

$n=1$ is γ -homogeneous if for all

$t \in [m]$,

p_t

(6)

$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n p_i(t) = p_t$.

$i=1$ p_i

One perhaps surprising aspect of our results is that γ -homogeneity is sufficient to characterize diversity in a large class of settings, as opposed to requiring more complicated functions of proportions p_t .

3

Main results

We now state our main results, which consider several settings reflecting different assumptions about the

(t)

(t) iid

conditional item values X_i . In Section 3.1, we assume that $X_i \sim D$ are drawn from a shared distribution, which implies that the recommender has little information about the value of specific items. In Section 3.2,

(t)

X_i are Bernoulli random variables with success probabilities differing depending on i and t , meaning that the recommender has information about which items are more likely to satisfy a user.

In each setting, we analyze the diversity of optimal sets $S_{n,k}$ for when k is fixed (i.e., accounting for consumption constraints) and $k = n$ (i.e., not accounting for consumption constraints). We sketch our proof strategy in Section 4 and defer full proofs to Appendix D.

3.1

i.i.d. conditional item values

(t) iid

Consider the setting in which $X_i \sim D$, i.e., conditional item values are drawn from a shared distribution.

This implies that the values of items behave similarly across types and within types, and the platform cannot easily distinguish between the items in a type. From the perspective of a hiring platform, there may be many candidates with similar backgrounds (e.g., education or work history), none of whom can be distinguished from another by the platform. Conditional on a recruiter preferring this background, candidate values can be modeled as coming from a shared distribution.

[Next verbatim source excerpt]

X_i as a conditional item value (the value of an item conditional on the user preferring the item's type).

Let $S_{n,k}$ be the set of n items that maximizes the expected total value of the k items with the highest realized values. We call $S_{n,k}$ the optimal set of n recommendations with respect to this objective. (We omit the dependency of $S_{n,k}$ on the other model parameters, as these will be clear from context.)

Lean-expanded paper semantic target

type:

$\{T : \mathbb{N}\} \rightarrow \text{PRPKG24AccuracyDiversity.AllocationSequence } T \rightarrow \text{PRPKG24AccuracyDiversity.GammaHomogeneityProfile } T \rightarrow \text{Prop}$

value:

$\text{fun } \{T : \mathbb{N}\} \text{ (seq : PRPKG24AccuracyDiversity.AllocationSequence } T)$

$(G : \text{PRPKG24AccuracyDiversity.GammaHomogeneityProfile } T) \Rightarrow$

$\forall (t : \text{PRPKG24AccuracyDiversity.ItemType } T),$

$\text{Filter.Tendsto (fun (N : } \mathbb{N}) \Rightarrow \text{seq.representation } N \ t) \text{ Filter.atTop (nhds (G.targetShare } t))}$

Direct retained dependencies

- [PRPKG24AccuracyDiversity.AllocationSequence](#)
- [PRPKG24AccuracyDiversity.AllocationSequence.representation](#)
- [PRPKG24AccuracyDiversity.GammaHomogeneityProfile](#)
- [PRPKG24AccuracyDiversity.GammaHomogeneityProfile.targetShare](#)
- [PRPKG24AccuracyDiversity.ItemType](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The source's Definition 2 requires every type's representation along the set sequence to converge to its normalized p_t^γ share; the expanded target quantifies every $\text{Fin } T$ type and applies Tendsto atTop to the allocation share with the profile's normalized target share, so the source case is preserved.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

PRPKG24AccuracyDiversity.CountAllocation.representation

Paper-local declaration:

PRPKG24AccuracyDiversity.CountAllocation.representation

Source locator: cited publication, lines 258-272

Verbatim paper-source connection

Evaluating diversity

We now formalize our approach to evaluating diversity. For a set S of items, we define

$$rt(S) :=$$

of items in S of type t

$|S|$

(4)

the representation of type t in S . Intuitively, a set of recommendations is diverse if all types are well represented. We now define an interpretable family of representations that interpolates between maximum diversity and maximum homogeneity, and which arises naturally in many of our results.

Lean-expanded paper semantic target

type:

$\{T : \mathbb{N}\} \rightarrow \text{PRPKG24AccuracyDiversity.CountAllocation } T \rightarrow \text{PRPKG24AccuracyDiversity.ItemType } T \rightarrow \mathbb{R}$

value:

fun $\{T : \mathbb{N}\}$ (a : PRPKG24AccuracyDiversity.CountAllocation T) (t : PRPKG24AccuracyDiversity.ItemType T) =>
AppliedModelingLib.Allocation.share a t

Direct retained dependencies

- [AppliedModelingLib.Allocation.share](#)
- [PRPKG24AccuracyDiversity.CountAllocation](#)
- [PRPKG24AccuracyDiversity.ItemType](#)

Recorded source-to-declaration screening Verdict: matches

Reason: Equation (4) defines representation as the number of type- t items divided by total set size; the expanded dependencies compute total as the sum of counts and representation as count t divided by that total, with only a harmless totalization to zero for the empty allocation outside the source's positive-size uses.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

PRPKG24AccuracyDiversity.GammaHomogeneityProfile.Exact

Paper-local declaration:

PRPKG24AccuracyDiversity.GammaHomogeneityProfile.Exact

Source locator: cited publication, lines 273-300

Verbatim paper-source connection

Definition 1 (γ -homogeneity). A set S is γ -homogeneous if for all $t \in [m]$,

$$\frac{p_t}{r_t(S)} = \frac{p_m}{r_m(S)}$$

$$\gamma = \frac{1}{m} \sum_{i=1}^m p_i$$

(5)

γ -homogeneity captures several intuitive notions of diversity, using $p_1, \dots, p_m$ as a benchmark:

1 The mathematical result in this example also appears in the context of resource allocation (e.g., [31]) and betting (e.g., §22.2 in [32]) in the presence of logarithmic utility.

4

[form-feed in source extraction]

1

- When $\gamma = 0$, $r_t(S) = m$. There is “equal representation.”

- When $\gamma = 1$, $r_t(S) = p_t$. There is “proportional representation,” where an item type is represented in proportion to its likelihood.

- When $\gamma = \infty$, $r_t(S) = 1$ for $t = \arg \max_{i \in [m]} p_i$ and $r_t(S) = 0$ otherwise. There is “complete homogeneity,” where only the highest-likelihood item type is represented.

A smaller γ corresponds to more diversity, with $\gamma \leq 1$ indicating at least proportional representation. In practice, it is challenging to show that individual sets are γ -homogeneous; for one, since sets have an integer number of items from each type, it is typically impossible to obtain the exact ratios in (5). Instead, we will give primarily asymptotic results, showing that as n grows large, the optimal set $S_{n,k}$ approaches γ -homogeneity. Formally, we define γ -homogeneity over sequences of sets:

Lean-expanded paper semantic target

type:

$\{T : \mathbb{N}\} \rightarrow \text{PRPKG24AccuracyDiversity.GammaHomogeneityProfile } T \rightarrow \text{PRPKG24AccuracyDiversity.CountAllocation } T \rightarrow \text{Prop}$

value:

fun $\{T : \mathbb{N}\}$ (G : PRPKG24AccuracyDiversity.GammaHomogeneityProfile T) (a : PRPKG24AccuracyDiversity.CountAllocation T) => a.HasExactRepresentation G.targetShare

Direct retained dependencies

- [PRPKG24AccuracyDiversity.CountAllocation](#)
- [PRPKG24AccuracyDiversity.CountAllocation.HasExactRepresentation](#)
- [PRPKG24AccuracyDiversity.GammaHomogeneityProfile](#)
- [PRPKG24AccuracyDiversity.GammaHomogeneityProfile.targetShare](#)

Recorded source-to-declaration screening Verdict: matches

Reason: Definition 1 requires exact equality of every type share to its normalized gamma-profile weight; the expanded target is precisely the pointwise HasExactRepresentation predicate against targetWeight divided by its finite-type normalizer.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

PRPKG24AccuracyDiversity.InfiniteGammaHomogeneityProfile.Exact

Paper-local declaration:

PRPKG24AccuracyDiversity.InfiniteGammaHomogeneityProfile.Exact

Source locator: cited publication, lines 273-300

Verbatim paper-source connection

Definition 1 (γ -homogeneity). A set S is γ -homogeneous if for all $t \in [m]$,

$$\frac{p_t}{r_t(S)} = \frac{p_m}{r_t(S)}$$

$$\gamma = \frac{1}{m} \sum_{i=1}^m p_i$$

(5)

γ -homogeneity captures several intuitive notions of diversity, using $p_1, \dots, p_m$ as a benchmark:

1 The mathematical result in this example also appears in the context of resource allocation (e.g., [31]) and betting (e.g., §22.2 in [32]) in the presence of logarithmic utility.

4

[form-feed in source extraction]

1

- When $\gamma = 0$, $r_t(S) = m$. There is “equal representation.”

- When $\gamma = 1$, $r_t(S) = p_t$. There is “proportional representation,” where an item type is represented in proportion to its likelihood.

- When $\gamma = \infty$, $r_t(S) = 1$ for $t = \arg \max_{i \in [m]} p_i$ and $r_t(S) = 0$ otherwise. There is “complete homogeneity,” where only the highest-likelihood item type is represented.

A smaller γ corresponds to more diversity, with $\gamma \leq 1$ indicating at least proportional representation. In practice, it is challenging to show that individual sets are γ -homogeneous; for one, since sets have an integer number of items from each type, it is typically impossible to obtain the exact ratios in (5). Instead, we will give primarily asymptotic results, showing that as n grows large, the optimal set $S_{n,k}$ approaches γ -homogeneity. Formally, we define γ -homogeneity over sequences of sets:

Lean-expanded paper semantic target

type:

{T : ℕ} → PRPKG24AccuracyDiversity.InfiniteGammaHomogeneityProfile T → PRPKG24AccuracyDiversity.CountAllocation T → Prop

value:

fun {T : ℕ} (G : PRPKG24AccuracyDiversity.InfiniteGammaHomogeneityProfile T)

(a : PRPKG24AccuracyDiversity.CountAllocation T) =>

a.HasExactRepresentation G.targetShare

Direct retained dependencies

- [PRPKG24AccuracyDiversity.CountAllocation](#)
- [PRPKG24AccuracyDiversity.CountAllocation.HasExactRepresentation](#)
- [PRPKG24AccuracyDiversity.InfiniteGammaHomogeneityProfile](#)
- [PRPKG24AccuracyDiversity.InfiniteGammaHomogeneityProfile.targetShare](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The source’s $\gamma = \infty$ case assigns share one to a highest-likelihood type and zero to every other type; the expanded profile records a proved maximizer, defines exactly that indicator target share, and Exact requires pointwise equality to it.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

PRPKG24AccuracyDiversity.definition3lidOrderStatisticMean

Paper-local declaration:

PRPKG24AccuracyDiversity.definition3lidOrderStatisticMean

Source locator: cited publication, lines 663-677

Verbatim paper-source connection

Definition 3. Define $\mu_D(i, a)$ to be the expected value of the i -th order statistic² of a random variables drawn i.i.d. from D . (Thus, $\mu_D(1, a)$ is the expected minimum of a i.i.d. draws from D and $\mu_D(a, a)$ is the expected maximum.) Then conditioned on the user preferring type t , the expected sum of the k highest value of items in S is $\sum_{i=1}^k \mu_D(a_t - i + 1, a_t)$, which follows from the linearity of expectation. Therefore, the expected sum of the k highest value items in S is $\sum_{i=1}^k \mu_D(a_t - i + 1, a_t)$. Define A_n to be the set of tuples of non-negative integers whose entries sum to n . Then $(a_1, a_2, \dots, a_m) = \arg \max_{(a_1, \dots, a_m) \in A_n} \sum_{t=1}^m \mu_D(a_t, a_t)$. We can then determine the limit in (10) given asymptotic information about h . In Lemma D.1 in the appendix, we develop general technical machinery for this task. Below, we state Lemma D.1(ii), which can be used to prove Theorem 1(ii).

[Next verbatim source excerpt]

Theorem 1. Suppose X_i

$\sim D$ where D has finite mean. Then the following statements hold.

[Next verbatim source excerpt]

We now sketch the proof of Theorem 1(ii). (The general strategy applies to the rest of Theorem 1, as well as Theorem 2 and Theorem 3.) First, recall the general setup in Theorem 1, where the conditional item values X_i are drawn i.i.d. from a shared distribution D . In other words, a user prefers type $t \in [m]$ with probability p_t such that the user prefers exactly one type of item, and conditioned on the user preferring type t

Lean-expanded paper semantic target

```
type:
  MeasureTheory.Measure ℝ → ℕ → ℕ → ℝ

value:
  fun (D : MeasureTheory.Measure ℝ) (a a_1 : ℕ) =>
    AppliedModelingLib.Probability.expectedOrderStatisticMeanSeq (PRPKG24AccuracyDiversity.definition3lidSampleMeasure D)
      a a_1
```

Direct retained dependencies

- [AppliedModelingLib.Probability.expectedOrderStatisticMeanSeq](#)
- [PRPKG24AccuracyDiversity.definition3lidSampleMeasure](#)

Recorded source-to-declaration screening Verdict: matches

Reason: Definition 3 takes rank i and sample size a and returns the expected ascending i -th order statistic of a iid draws from D ; the expanded target uses the iid product measure on $\text{Fin } a$, validates one-based ranks, sorts ascending, and integrates that statistic, matching the source's minimum-at-rank-1 convention.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

PRPKG24AccuracyDiversity.gammaLikelihoodProfile

Paper-local declaration:

PRPKG24AccuracyDiversity.gammaLikelihoodProfile

Source locator: cited publication, lines 273-300

Verbatim paper-source connection

Definition 1 (γ -homogeneity). A set S is γ -homogeneous if for all $t \in [m]$,

$$\frac{p_t}{r_t(S)} = \frac{p_m}{r_m(S)}$$

$$\frac{p_t}{r_t(S)} = \frac{p_m}{r_m(S)}$$

(5)

γ -homogeneity captures several intuitive notions of diversity, using $p_1, \dots, p_m$ as a benchmark:

1 The mathematical result in this example also appears in the context of resource allocation (e.g., [31]) and betting (e.g., §22.2 in [32]) in the presence of logarithmic utility.

4

[form-feed in source extraction]

1

- When $\gamma = 0$, $r_t(S) = m$. There is “equal representation.”

- When $\gamma = 1$, $r_t(S) = p_t$. There is “proportional representation,” where an item type is represented in proportion to its likelihood.

- When $\gamma = \infty$, $r_t(S) = 1$ for $t = \arg \max_{i \in [m]} p_i$ and $r_t(S) = 0$ otherwise. There is “complete homogeneity,” where only the highest-likelihood item type is represented.

A smaller γ corresponds to more diversity, with $\gamma \leq 1$ indicating at least proportional representation. In practice, it is challenging to show that individual sets are γ -homogeneous; for one, since sets have an integer number of items from each type, it is typically impossible to obtain the exact ratios in (5). Instead, we will give primarily asymptotic results, showing that as n grows large, the optimal set $S_{n,k}$ approaches γ -homogeneity. Formally, we define γ -homogeneity over sequences of sets:

Lean-expanded paper semantic target

type:

$\{T : \mathbb{N}\} \rightarrow (\text{PRPKG24AccuracyDiversity.ItemType } T \rightarrow \mathbb{R}) \rightarrow \mathbb{R} \rightarrow \text{PRPKG24AccuracyDiversity.GammaHomogeneityProfile } T$

value:

$\text{fun } \{T : \mathbb{N}\} \text{ (likelihood : PRPKG24AccuracyDiversity.ItemType } T \rightarrow \mathbb{R}) \text{ (gamma : } \mathbb{R}) \Rightarrow$

$\{ \text{gamma} := \text{gamma}, \text{targetWeight} := \text{fun } (t : \text{PRPKG24AccuracyDiversity.ItemType } T) \Rightarrow \text{likelihood } t ^ \text{gamma} \}$

Direct retained dependencies

- [PRPKG24AccuracyDiversity.GammaHomogeneityProfile](#)
- [PRPKG24AccuracyDiversity.ItemType](#)

Recorded source-to-declaration screening Verdict: matches

Reason: Equation (5) uses p_t^γ as the unnormalized weight; the expanded constructor stores likelihood t raised to the real exponent gamma as targetWeight , which the displayed profile normalizer and targetShare convert to the stated ratio.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

PRPKG24AccuracyDiversity.infiniteLikelihoodProfile

Paper-local declaration:

PRPKG24AccuracyDiversity.infiniteLikelihoodProfile

Source locator: cited publication, lines 273-300

Verbatim paper-source connection

Definition 1 (γ -homogeneity). A set S is γ -homogeneous if for all $t \in [m]$,

$$\frac{p_t}{r_t(S)} = \frac{p_{m+1}}{r_{m+1}(S)}$$

$$\gamma = \frac{1}{m} \sum_{i=1}^m p_i$$

(5)

γ -homogeneity captures several intuitive notions of diversity, using $p_1, \dots, p_m$ as a benchmark:

1 The mathematical result in this example also appears in the context of resource allocation (e.g., [31]) and betting (e.g., §2.2 in [32]) in the presence of logarithmic utility.

4

[form-feed in source extraction]

1

- When $\gamma = 0$, $r_t(S) = m$. There is “equal representation.”

- When $\gamma = 1$, $r_t(S) = p_t$. There is “proportional representation,” where an item type is represented in proportion to its likelihood.

- When $\gamma = \infty$, $r_t(S) = 1$ for $t = \arg \max_{i \in [m]} p_i$ and $r_t(S) = 0$ otherwise. There is “complete homogeneity,” where only the highest-likelihood item type is represented.

A smaller γ corresponds to more diversity, with $\gamma \leq 1$ indicating at least proportional representation. In practice, it is challenging to show that individual sets are γ -homogeneous; for one, since sets have an integer number of items from each type, it is typically impossible to obtain the exact ratios in (5). Instead, we will give primarily asymptotic results, showing that as n grows large, the optimal set $S_{n,k}$ approaches γ -homogeneity. Formally, we define γ -homogeneity over sequences of sets:

Lean-expanded paper semantic target

type:

```
{T : ℕ} →  
(likelihood : PRPKG24AccuracyDiversity.ItemType T → ℝ) →  
(best : PRPKG24AccuracyDiversity.ItemType T →  
(∀ (t : PRPKG24AccuracyDiversity.ItemType T), likelihood t ≤ likelihood best) →  
PRPKG24AccuracyDiversity.InfiniteGammaHomogeneityProfile T
```

value:

```
fun {T : ℕ} (likelihood : PRPKG24AccuracyDiversity.ItemType T → ℝ) (best : PRPKG24AccuracyDiversity.ItemType T →  
(hbest : ∀ (t : PRPKG24AccuracyDiversity.ItemType T), likelihood t ≤ likelihood best) =>  
{ likelihood := likelihood, best := best, isArgmax := hbest }
```

Direct retained dependencies

- [PRPKG24AccuracyDiversity.InfiniteGammaHomogeneityProfile](#)
- [PRPKG24AccuracyDiversity.ItemType](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The source’s $\gamma = \infty$ reading concentrates representation on an argmax of the likelihoods; the expanded constructor takes a type with a proof that it weakly maximizes likelihood and records exactly those data for the indicator target share.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Source claims

Review row 1

Source locator: cited publication, lines 349-350

Verbatim source input

(i) [Finite Discrete] If D is a finite discrete distribution, $\{S_{n,k}\}_{n=1}^{\infty}$ is 0-homogeneous.

[Next verbatim source excerpt]

There are m types of items indexed by $[m] = \{1, 2, \dots, m\}$. A user prefers exactly one type of item, preferring type $t \in [m]$ with probability p_t . The value of the i -th item of type t is a random variable X_i if the user prefers type t and 0 otherwise (so its expected value is $p_t E[X_i]$). The recommender knows only how X_i are distributed, not their realizations. We refer to p_t as the likelihood of a type and the random variable X_i as a conditional item value (the value of an item conditional on the user preferring the item's type). Let $S_{n,k}$ be the set of n items that maximizes the expected total value of the k items with the highest realized values. We call $S_{n,k}$ the optimal set of n recommendations with respect to this objective. (We omit the dependency of $S_{n,k}$ on the other model parameters, as these will be clear from context.) $S_{n,k}$ arises naturally from an assumption that the user can only use k items, so that they derive value from the k highest value recommended items. So $S_{n,1}$ results from an assumption that the user uses only one item, while $S_{n,n}$ results from an assumption that the user uses every recommended item. $S_{n,n}$ maximizes the expected total value of all of the recommended items, so by linearity of expectation, $S_{n,n}$ contains the n items with the highest individual expected values.

[Next verbatim source excerpt]

Consider the setting in which $X_i \sim D$, i.e., conditional item values are drawn from a shared distribution. This implies that the values of items behave similarly across types and within types, and the platform cannot easily distinguish between the items in a type. From the perspective of a hiring platform, there may be many candidates with similar backgrounds (e.g., education or work history), none of whom can be distinguished from another by the platform. Conditional on a recruiter preferring this background, candidate values can be modeled as coming from a shared distribution. We show that for a fixed k , as n grows large, the diversity of $S_{n,k}$ theoretically varies between equal representation, proportional representation, and near-complete homogeneity depending on the tail-behavior of D . In particular, distributions that are bounded or have exponential tails induce at least proportional representation. Meanwhile, $S_{n,n}$ is completely homogeneous.

[Next verbatim source excerpt]

Theorem 1. Suppose X_i

$\sim D$ where D has finite mean. Then the following statements hold.

Formalized review target The archival source above is not asserted equivalent to this formalized target.

For a nonempty finite type space with a preferred-type PMF having every atom $p_t > 0$, a fixed positive k , and a shared iid finite discrete nonnegative

- conditional-value law having a positive top value, a strictly smaller nonnegative second bound, and positive law mass both at the top and below it, the
- source exact $\min(k, a)$ top- k objective is the stated two-stage PMF experiment and every fixed-total optimal allocation sequence has limiting uniform
- type shares.

Archival source anchor: cited publication, lines 349-350

Expanded PaperInterface specification

```
∀ {T : ℕ} {NeZero T} {Omega : Type u_1} [inst : Fintype Omega] [inst_1 : DecidableEq Omega]
  (preferenceLaw : PRPKG24AccuracyDiversity.SourcePreferenceLaw T) (itemLaw : PMF Omega) (value : Omega → ℝ)
  {xTop xSecond : ℝ} (k : ℕ)
  (seq :
    PRPKG24AccuracyDiversity.OptimalAllocationSequence fun (x : ℕ) =>
      (PRPKG24AccuracyDiversity.TopKValueOracle.common T
        (PRPKG24AccuracyDiversity.finiteDiscretelidTopKExpected Omega itemLaw k value)).toConsumptionModel
        (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) k),
  0 < k →
  0 < xTop →
  0 ≤ xSecond →
  xSecond ≤ xTop →
  xSecond < xTop →
  (∀ (omega : Omega), 0 ≤ value omega) →
  (∀ (omega : Omega), value omega ≤ xTop) →
  (∀ (omega : Omega), value omega = xTop ∨ value omega ≤ xSecond) →
  (0 < AppliedModelingLib.pmfProb itemLaw fun (omega : Omega) => value omega = xTop) →
  (0 < AppliedModelingLib.pmfProb itemLaw fun (omega : Omega) => ¬value omega = xTop) →
  (∀ (t : PRPKG24AccuracyDiversity.ItemType T), 0 < (preferenceLaw t).toReal) →
  (∀ (a : PRPKG24AccuracyDiversity.CountAllocation T),
    ((PRPKG24AccuracyDiversity.TopKValueOracle.common T
      (PRPKG24AccuracyDiversity.finiteDiscretelidTopKExpected Omega itemLaw k
        value)).toConsumptionModel
        (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal)
        k).objective
      a =
    AppliedModelingLib.pmfExp preferenceLaw fun (t : PRPKG24AccuracyDiversity.ItemType T) =>
      AppliedModelingLib.pmfExp
```



```

(AppliedModelingLib.pmfProduct (Fin (a.count t)) Omega itemLaw)
fun (sample : Fin (a.count t) → Omega) =>
  AppliedModelingLib.Probability.sampleTopKSum
    (fun (i : Fin (a.count t)) => value (sample i)) k) ∧
  ∀ (t : PRPKG24AccuracyDiversity.ItemType T),
    Filter.Tendsto (fun (N : ℕ) => (seq.allocation N).representation t) Filter.atTop
      (nhds (1 / ↑T))

```

Retained governing declarations

- [AppliedModelingLib.Allocation](#)
- [AppliedModelingLib.Probability.sampleTopKSum](#)
- [AppliedModelingLib.pmfExp](#)
- [AppliedModelingLib.pmfProb](#)
- [AppliedModelingLib.pmfProduct](#)
- [PRPKG24AccuracyDiversity.ConsumptionModel.objective](#)
- [PRPKG24AccuracyDiversity.CountAllocation](#)
- [PRPKG24AccuracyDiversity.CountAllocation.representation](#)
- [PRPKG24AccuracyDiversity.ItemType](#)
- [PRPKG24AccuracyDiversity.OptimalAllocationSequence](#)
- [PRPKG24AccuracyDiversity.SourcePreferenceLaw](#)
- [PRPKG24AccuracyDiversity.TopKValueOracle.common](#)
- [PRPKG24AccuracyDiversity.TopKValueOracle.toConsumptionModel](#)
- [PRPKG24AccuracyDiversity.finiteDiscreteIdTopKExpected](#)

Lean-elaborated claim roles

```

1. parameter: ℕ
2. assumption: NeZero T
3. parameter: Type u_1
4. parameter: Fintype Omega
5. parameter: DecidableEq Omega
6. parameter: PRPKG24AccuracyDiversity.SourcePreferenceLaw T
7. parameter: PMF Omega
8. parameter: Omega → ℝ
9. parameter: ℝ
10. parameter: ℝ
11. parameter: ℕ
12. parameter: PRPKG24AccuracyDiversity.OptimalAllocationSequence fun (x : ℕ) =>
  (PRPKG24AccuracyDiversity.TopKValueOracle.common T
    (PRPKG24AccuracyDiversity.finiteDiscreteIdTopKExpected Omega itemLaw k value)).toConsumptionModel
    (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) k
13. assumption: 0 < k
14. assumption: 0 < xTop
15. assumption: 0 ≤ xSecond
16. assumption: xSecond ≤ xTop
17. assumption: xSecond < xTop
18. assumption: ∀ (omega : Omega), 0 ≤ value omega
19. assumption: ∀ (omega : Omega), value omega ≤ xTop
20. assumption: ∀ (omega : Omega), value omega = xTop ∨ value omega ≤ xSecond
21. assumption: 0 < AppliedModelingLib.pmfProb itemLaw fun (omega : Omega) => value omega = xTop
22. assumption: 0 < AppliedModelingLib.pmfProb itemLaw fun (omega : Omega) => ¬value omega = xTop
23. assumption: ∀ (t : PRPKG24AccuracyDiversity.ItemType T), 0 < (preferenceLaw t).toReal
24. conclusion: (∀ (a : PRPKG24AccuracyDiversity.CountAllocation T),
  ((PRPKG24AccuracyDiversity.TopKValueOracle.common T
    (PRPKG24AccuracyDiversity.finiteDiscreteIdTopKExpected Omega itemLaw k value)).toConsumptionModel
    (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) k).objective
    a =
  AppliedModelingLib.pmfExp preferenceLaw fun (t : PRPKG24AccuracyDiversity.ItemType T) =>
    AppliedModelingLib.pmfExp (AppliedModelingLib.pmfProduct (Fin (a.count t)) Omega itemLaw)
      fun (sample : Fin (a.count t) → Omega) =>
        AppliedModelingLib.Probability.sampleTopKSum (fun (i : Fin (a.count t)) => value (sample i)) k) ∧
    ∀ (t : PRPKG24AccuracyDiversity.ItemType T),
      Filter.Tendsto (fun (N : ℕ) => (seq.allocation N).representation t) Filter.atTop (nhds (1 / ↑T))

```

Lean proof endpoint (built separately)

PRPKG24AccuracyDiversity.PaperInterface.theorem1_i_formulaSpec_proof

Recorded source-to-Spec screening **Verdict:** matches approved corrected target

Reason: Approved corrected target: Lean exposes the selected-type PMF and finite iid top-k experiment, positive support, and the approved nondegenerate finite top/support split before concluding uniform shares.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 2

Source locator: cited publication, lines 351-365

Verbatim source input

(ii) [Bounded] If D has support bounded from above by M with pdf f_D satisfying
 $f_D(x)$
 $\lim_{x \rightarrow M} (M - x)^{\beta-1} = c$
for some $\beta, c > 0$, then $\{S_{n,k}\}_{n=1}^{\infty}$ is $\beta+1$ -homogeneous.
(This pdf class contains beta distributions, including the uniform distribution.)

5

(7)

[form-feed in source extraction]
(t) iid

[Next verbatim source excerpt]

There are m types of items indexed by $[m] = \{1, 2, \dots, m\}$. A user prefers exactly one type of item, preferring
(t)
type $t \in [m]$ with probability p_t . The value of the i -th item of type t is a random variable X_i if the user
(t)
(t)
prefers type t and 0 otherwise (so its expected value is $p_t E[X_i]$). The recommender knows only how X_i are distributed, not their realizations. We refer to p_t as the likelihood of a type and the random variable
(t)
 X_i as a conditional item value (the value of an item conditional on the user preferring the item's type). Let $S_{n,k}$ be the set of n items that maximizes the expected total value of the k items with the highest realized values. We call $S_{n,k}$ the optimal set of n recommendations with respect to this objective. (We omit the dependency of $S_{n,k}$ on the other model parameters, as these will be clear from context.) $S_{n,k}$ arises naturally from an assumption that the user can only use k items, so that they derive value from the k highest value recommended items. So $S_{n,1}$ results from an assumption that the user uses only one item, while $S_{n,n}$ results from an assumption that the user uses every recommended item. $S_{n,n}$ maximizes the expected total value of all of the recommended items, so by linearity of expectation, $S_{n,n}$ contains the n items with the highest individual expected values.

[Next verbatim source excerpt]

Consider the setting in which $X_i \sim D$, i.e., conditional item values are drawn from a shared distribution. This implies that the values of items behave similarly across types and within types, and the platform cannot easily distinguish between the items in a type. From the perspective of a hiring platform, there may be many candidates with similar backgrounds (e.g., education or work history), none of whom can be distinguished from another by the platform. Conditional on a recruiter preferring this background, candidate values can be modeled as coming from a shared distribution. We show that for a fixed k , as n grows large, the diversity of $S_{n,k}$ theoretically varies between equal representation, proportional representation, and near-complete homogeneity depending on the tail-behavior of D . In particular, distributions that are bounded or have exponential tails induce at least proportional representation. Meanwhile, $S_{n,n}$ is completely homogeneous.

[Next verbatim source excerpt]

Theorem 1. Suppose X_i

$\sim D$ where D has finite mean. Then the following statements hold.

Settled source reading Theorem 1(ii): an upper-bounded conditional value law $\rightarrow$ an upper-bounded law with non-negative values almost surely. The current allocation proof does not establish the translated negative-value domain.

Settled source reading All-coordinate share limits: possibly zero type masses $\rightarrow$ strictly positive mass for every named type.

Expanded PaperInterface specification

$\forall \{T : \mathbb{N}\} \{NeZero T\} \{\text{beta } c \text{ M } L : \mathbb{R}\} \{k : \mathbb{N}\} \text{ (preferenceLaw : PRPKG24AccuracyDiversity.SourcePreferenceLaw } T) \text{ (baseMeasure : MeasureTheory.Measure } \mathbb{R}) \text{ [MeasureTheory.IsProbabilityMeasure baseMeasure], MeasureTheory.Integrable (fun (x : } \mathbb{R}) => x) \text{ baseMeasure } \rightarrow$
 $(\forall^m (y : \mathbb{R}) \partial \text{baseMeasure, } L \leq y \wedge y \leq M) \rightarrow$
 $(\forall^m (y : \mathbb{R}) \partial \text{baseMeasure, } 0 \leq y) \rightarrow$
 $0 < M \rightarrow$
 $\forall (f : \mathbb{R} \rightarrow \mathbb{R}),$
 $(\text{baseMeasure} = \text{MeasureTheory.volume.withDensity fun (y : } \mathbb{R}) => \text{ENNReal.ofReal (f y)}) \rightarrow$
 $(\forall (y : \mathbb{R}), 0 \leq f y) \rightarrow$
 $\text{Measurable } f \rightarrow$
 $0 < \text{beta} \rightarrow$
 $0 < c \rightarrow$
 $\text{Filter.Tendsto (fun (x : } \mathbb{R}) => f x / (M - x) ^ (\text{beta} - 1)) (\text{nhdsWithin } M (\text{Set.lim } M)) (\text{nhds } c) \rightarrow$
 $(\forall (t : \text{PRPKG24AccuracyDiversity.ItemType } T), 0 < (\text{preferenceLaw } t).toReal) \rightarrow$
 $0 < k \rightarrow$
 $0 < M - L \rightarrow$
 $\forall$
(seq :

```

PRPKG24AccuracyDiversity.OptimalAllocationSequence fun (x : ℕ) =>
  PRPKG24AccuracyDiversity.boundedlidOrderStatisticConsumptionModel
    (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) k
    baseMeasure),
(∀ (a : PRPKG24AccuracyDiversity.CountAllocation T),
  (PRPKG24AccuracyDiversity.boundedlidOrderStatisticConsumptionModel
    (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal)
    k baseMeasure).objective
    a =
    AppliedModelingLib.pmfExp preferenceLaw
    fun (t : PRPKG24AccuracyDiversity.ItemType T) =>
      AppliedModelingLib.Probability.expectedSampleTopKSum
        (PRPKG24AccuracyDiversity.definition3lidSampleMeasure baseMeasure (a.count t))
    k) ∧
  ∀ (t : PRPKG24AccuracyDiversity.ItemType T),
  Filter.Tendsto (fun (N : ℕ) => (seq.allocation N).representation t) Filter.atTop
    (nhds
      ((preferenceLaw t).toReal ^ (beta / (beta + 1))) /
      (Σ i : PRPKG24AccuracyDiversity.ItemType T,
        (preferenceLaw i).toReal ^ (beta / (beta + 1)))))

```

Retained governing declarations

- [AppliedModelingLib.Allocation](#)
- [AppliedModelingLib.Probability.expectedSampleTopKSum](#)
- [AppliedModelingLib.pmfExp](#)
- [PRPKG24AccuracyDiversity.ConsumptionModel.objective](#)
- [PRPKG24AccuracyDiversity.CountAllocation](#)
- [PRPKG24AccuracyDiversity.CountAllocation.representation](#)
- [PRPKG24AccuracyDiversity.ItemType](#)
- [PRPKG24AccuracyDiversity.OptimalAllocationSequence](#)
- [PRPKG24AccuracyDiversity.SourcePreferenceLaw](#)
- [PRPKG24AccuracyDiversity.boundedlidOrderStatisticConsumptionModel](#)
- [PRPKG24AccuracyDiversity.definition3lidSampleMeasure](#)

Lean-elaborated claim roles

1. parameter: $\mathbb{N}$
2. assumption: NeZero T
3. parameter: $\mathbb{R}$
4. parameter: $\mathbb{R}$
5. parameter: $\mathbb{R}$
6. parameter: $\mathbb{R}$
7. parameter: $\mathbb{N}$
8. parameter: PRPKG24AccuracyDiversity.SourcePreferenceLaw T
9. parameter: MeasureTheory.Measure $\mathbb{R}$
10. assumption: MeasureTheory.IsProbabilityMeasure baseMeasure
11. assumption: MeasureTheory.Integrable (fun (x : $\mathbb{R}$) => x) baseMeasure
12. assumption: $\forall^m (y : \mathbb{R}) \partial \text{baseMeasure}, L \leq y \wedge y \leq M$
13. assumption: $\forall^m (y : \mathbb{R}) \partial \text{baseMeasure}, 0 \leq y$
14. assumption: $0 < M$
15. parameter: $\mathbb{R} \rightarrow \mathbb{R}$
16. assumption: baseMeasure = MeasureTheory.volume.withDensity fun (y : $\mathbb{R}$) => ENNReal.ofReal (f y)
17. assumption: $\forall (y : \mathbb{R}), 0 \leq f y$
18. assumption: Measurable f
19. assumption: $0 < \text{beta}$
20. assumption: $0 < c$
21. assumption: Filter.Tendsto (fun (x : $\mathbb{R}$) => f x / (M - x) ^ (beta - 1)) (nhdsWithin M (Set.lim M)) (nhds c)
22. assumption: $\forall (t : \text{PRPKG24AccuracyDiversity.ItemType } T), 0 < (\text{preferenceLaw } t).\text{toReal}$
23. assumption: $0 < k$
24. assumption: $0 < M - L$
25. parameter: PRPKG24AccuracyDiversity.OptimalAllocationSequence fun (x : $\mathbb{N}$) =>
 PRPKG24AccuracyDiversity.boundedlidOrderStatisticConsumptionModel
 (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) k baseMeasure
26. conclusion: (∀ (a : PRPKG24AccuracyDiversity.CountAllocation T),
 (PRPKG24AccuracyDiversity.boundedlidOrderStatisticConsumptionModel
 (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) k baseMeasure).objective
 a =
 AppliedModelingLib.pmfExp preferenceLaw fun (t : PRPKG24AccuracyDiversity.ItemType T) =>
 AppliedModelingLib.Probability.expectedSampleTopKSum
 (PRPKG24AccuracyDiversity.definition3lidSampleMeasure baseMeasure (a.count t)) k) ∧
 ∀ (t : PRPKG24AccuracyDiversity.ItemType T),
 Filter.Tendsto (fun (N : $\mathbb{N}$) => (seq.allocation N).representation t) Filter.atTop
 (nhds
 ((preferenceLaw t).toReal ^ (beta / (beta + 1))) /
 (Σ i : PRPKG24AccuracyDiversity.ItemType T, (preferenceLaw i).toReal ^ (beta / (beta + 1)))))

Lean proof endpoint (built separately)

PRPKG24AccuracyDiversity.PaperInterface.theorem1_ii_formulaSpec_proof

Recorded source-to-Spec screening Verdict: matches

Reason: Matches: Lean gives the source bounded-PDF endpoint, fixed positive k , two-stage iid expectation bridge, and normalized $p^{(\beta/(\beta+1))}$ shares under its visible positive-support convention.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

--

Review row 3

Source locator: cited publication, lines 431-433

Verbatim source input

(iii) [Exponential tail] If $D = \text{Exp}(\lambda)$ for $\lambda > 0$, then $\{S_{n,k}\}_{n=1}^\infty$ is 1-homogeneous.
 α

[Next verbatim source excerpt]

There are m types of items indexed by $[m] = \{1, 2, \dots, m\}$. A user prefers exactly one type of item, preferring (t)
type $t \in [m]$ with probability p_t . The value of the i -th item of type t is a random variable X_i if the user (t)
(t)
prefers type t and 0 otherwise (so its expected value is $p_t E[X_i]$). The recommender knows only how X_i are distributed, not their realizations. We refer to p_t as the likelihood of a type and the random variable (t)
 X_i as a conditional item value (the value of an item conditional on the user preferring the item's type). Let $S_{n,k}$ be the set of n items that maximizes the expected total value of the k items with the highest realized values. We call $S_{n,k}$ the optimal set of n recommendations with respect to this objective. (We omit the dependency of $S_{n,k}$ on the other model parameters, as these will be clear from context.)
 $S_{n,k}$ arises naturally from an assumption that the user can only use k items, so that they derive value from the k highest value recommended items. So $S_{n,1}$ results from an assumption that the user uses only one item, while $S_{n,n}$ results from an assumption that the user uses every recommended item.
 $S_{n,n}$ maximizes the expected total value of all of the recommended items, so by linearity of expectation, $S_{n,n}$ contains the n items with the highest individual expected values.

[Next verbatim source excerpt]

Consider the setting in which $X_i \sim D$, i.e., conditional item values are drawn from a shared distribution. This implies that the values of items behave similarly across types and within types, and the platform cannot easily distinguish between the items in a type. From the perspective of a hiring platform, there may be many candidates with similar backgrounds (e.g., education or work history), none of whom can be distinguished from another by the platform. Conditional on a recruiter preferring this background, candidate values can be modeled as coming from a shared distribution.
We show that for a fixed k , as n grows large, the diversity of $S_{n,k}$ theoretically varies between equal representation, proportional representation, and near-complete homogeneity depending on the tail-behavior of D . In particular, distributions that are bounded or have exponential tails induce at least proportional representation. Meanwhile, $S_{n,n}$ is completely homogeneous.

[Next verbatim source excerpt]

Theorem 1. Suppose X_i

$\sim D$ where D has finite mean. Then the following statements hold.

Settled source reading All-coordinate share limits: possibly zero type masses $\rightarrow$ strictly positive mass for every named type.

Expanded PaperInterface specification

```
∀ {T : ℕ} [NeZero T] (preferenceLaw : PRPKG24AccuracyDiversity.SourcePreferenceLaw T) (lambda : ℝ) (k : ℕ)
(hlambda_pos : 0 < lambda),
0 < k →
(∀ (t : PRPKG24AccuracyDiversity.ItemType T), 0 < (preferenceLaw t).toReal) →
  ∀
    (seq :
      PRPKG24AccuracyDiversity.OptimalAllocationSequence fun (x : ℕ) =>
        (PRPKG24AccuracyDiversity.exponentialTopKOrderStatisticOracle T lambda k).toConsumptionModel
          (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) k),
    (∀ (a : PRPKG24AccuracyDiversity.CountAllocation T),
      ((PRPKG24AccuracyDiversity.exponentialTopKOrderStatisticOracle T lambda k).toConsumptionModel
        (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) k).objective
        a =
        AppliedModelingLib.pmfExp preferenceLaw fun (t : PRPKG24AccuracyDiversity.ItemType T) =>
          AppliedModelingLib.Probability.expectedSampleTopKSum
            ((PRPKG24AccuracyDiversity.exponentialDistributionModel lambda hlambda_pos).iidProductMeasure
              (a.count t))
          k) ∧
    ∀ (t : PRPKG24AccuracyDiversity.ItemType T),
      Filter.Tendsto (fun (N : ℕ) => (seq.allocation N).representation t) Filter.atTop
        (nhds
          ((preferenceLaw t).toReal ^ 1 /
            ∑ i : PRPKG24AccuracyDiversity.ItemType T, (preferenceLaw i).toReal ^ 1))
```

Retained governing declarations

- [AppliedModelingLib.Allocation](#)
- [AppliedModelingLib.Probability.Exponential.Model.iidProductMeasure](#)
- [AppliedModelingLib.Probability.expectedSampleTopKSum](#)
- [AppliedModelingLib.pmfExp](#)

- [PRPKG24AccuracyDiversity.ConsumptionModel.objective](#)
- [PRPKG24AccuracyDiversity.CountAllocation](#)
- [PRPKG24AccuracyDiversity.CountAllocation.representation](#)
- [PRPKG24AccuracyDiversity.ItemType](#)
- [PRPKG24AccuracyDiversity.OptimalAllocationSequence](#)
- [PRPKG24AccuracyDiversity.SourcePreferenceLaw](#)
- [PRPKG24AccuracyDiversity.TopKValueOracle.toConsumptionModel](#)
- [PRPKG24AccuracyDiversity.exponentialDistributionModel](#)
- [PRPKG24AccuracyDiversity.exponentialTopKOrderStatisticOracle](#)

Lean-elaborated claim roles

```

1. parameter:  $\mathbb{N}$ 
2. assumption: NeZero T
3. parameter: PRPKG24AccuracyDiversity.SourcePreferenceLaw T
4. parameter:  $\mathbb{R}$ 
5. parameter:  $\mathbb{N}$ 
6. assumption:  $0 < \text{lambda}$ 
7. assumption:  $0 < k$ 
8. assumption:  $\forall (t : \text{PRPKG24AccuracyDiversity.ItemType } T), 0 < (\text{preferenceLaw } t).\text{toReal}$ 
9. parameter: PRPKG24AccuracyDiversity.OptimalAllocationSequence fun (x :  $\mathbb{N}$ ) =>
  (PRPKG24AccuracyDiversity.exponentialTopKOrderStatisticOracle T lambda k).toConsumptionModel
  (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) k
10. conclusion: ( $\forall (a : \text{PRPKG24AccuracyDiversity.CountAllocation } T),$ 
  ((PRPKG24AccuracyDiversity.exponentialTopKOrderStatisticOracle T lambda k).toConsumptionModel
    (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) k).objective
    a =
    AppliedModelingLib.pmfExp preferenceLaw fun (t : PRPKG24AccuracyDiversity.ItemType T) =>
    AppliedModelingLib.Probability.expectedSampleTopKSum
      ((PRPKG24AccuracyDiversity.exponentialDistributionModel lambda hlambda_pos).iidProductMeasure (a.count t)
        k)  $\wedge$ 
 $\forall (t : \text{PRPKG24AccuracyDiversity.ItemType } T),$ 
    Filter.Tendsto (fun (N :  $\mathbb{N}$ ) => (seq.allocation N).representation t) Filter.atTop
      (nhds ((preferenceLaw t).toReal ^ 1 /  $\sum i : \text{PRPKG24AccuracyDiversity.ItemType } T, (\text{preferenceLaw } i).\text{toReal} ^ 1$ ))

```

Lean proof endpoint (built separately)

PRPKG24AccuracyDiversity.PaperInterface.theorem1_iii_formulaSpec_proof

Recorded source-to-Spec screening Verdict: matches

Reason: Matches: the exponential $\text{lambda} > 0$, fixed-positive-k source model and its exact selected-type iid expectation bridge yield the stated proportional limiting shares.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 4

Source locator: cited publication, lines 434-437

Verbatim source input

(iv) [Heavy tail] If $D = \text{Pareto}(\alpha)$ for $\alpha > 1$, then $\{S_{n,k}\}_{n=1}^{\infty}$ is $\alpha-1$ -homogeneous.

Additionally,

[Next verbatim source excerpt]

There are m types of items indexed by $[m] = \{1, 2, \dots, m\}$. A user prefers exactly one type of item, preferring

(t)
type $t \in [m]$ with probability p_t . The value of the i -th item of type t is a random variable X_i if the user

(t)
(t)
prefers type t and 0 otherwise (so its expected value is $p_t E[X_i]$). The recommender knows only how X_i are distributed, not their realizations. We refer to p_t as the likelihood of a type and the random variable

(t)
 X_i as a conditional item value (the value of an item conditional on the user preferring the item's type). Let $S_{n,k}$ be the set of n items that maximizes the expected total value of the k items with the highest realized values. We call $S_{n,k}$ the optimal set of n recommendations with respect to this objective. (We omit the dependency of $S_{n,k}$ on the other model parameters, as these will be clear from context.) $S_{n,k}$ arises naturally from an assumption that the user can only use k items, so that they derive value from the k highest value recommended items. So $S_{n,1}$ results from an assumption that the user uses only one item, while $S_{n,n}$ results from an assumption that the user uses every recommended item. $S_{n,n}$ maximizes the expected total value of all of the recommended items, so by linearity of expectation, $S_{n,n}$ contains the n items with the highest individual expected values.

[Next verbatim source excerpt]

Consider the setting in which $X_i \sim D$, i.e., conditional item values are drawn from a shared distribution. This implies that the values of items behave similarly across types and within types, and the platform cannot easily distinguish between the items in a type. From the perspective of a hiring platform, there may be many candidates with similar backgrounds (e.g., education or work history), none of whom can be distinguished from another by the platform. Conditional on a recruiter preferring this background, candidate values can be modeled as coming from a shared distribution.

We show that for a fixed k , as n grows large, the diversity of $S_{n,k}$ theoretically varies between equal representation, proportional representation, and near-complete homogeneity depending on the tail-behavior of D . In particular, distributions that are bounded or have exponential tails induce at least proportional representation. Meanwhile, $S_{n,n}$ is completely homogeneous.

[Next verbatim source excerpt]

Theorem 1. Suppose X_i

$\sim D$ where D has finite mean. Then the following statements hold.

Settled source reading All-coordinate share limits: possibly zero type masses $\rightarrow$ strictly positive mass for every named type.

Expanded PaperInterface specification

```
∀ {T : ℕ} [NeZero T] (preferenceLaw : PRPKG24AccuracyDiversity.SourcePreferenceLaw T) {k : ℕ} {alpha : ℝ},
  1 < alpha →
  0 < k →
  (∀ (t : PRPKG24AccuracyDiversity.ItemType T), 0 < (preferenceLaw t).toReal) →
  (seq :
    PRPKG24AccuracyDiversity.OptimalAllocationSequence fun (x : ℕ) =>
      PRPKG24AccuracyDiversity.paretolidOrderStatisticConsumptionModel
        (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) k alpha),
  (∀ (a : PRPKG24AccuracyDiversity.CountAllocation T),
    (PRPKG24AccuracyDiversity.paretolidOrderStatisticConsumptionModel
      (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) k alpha).objective
      a =
    AppliedModelingLib.pmfExp preferenceLaw fun (t : PRPKG24AccuracyDiversity.ItemType T) =>
      AppliedModelingLib.Probability.expectedSampleTopKSum
        (PRPKG24AccuracyDiversity.paretolidSampleMeasure alpha (a.count t)) k) ∧
  (∀ (t : PRPKG24AccuracyDiversity.ItemType T),
    Filter.Tendsto (fun (N : ℕ) => (seq.allocation N).representation t) Filter.atTop
      (nhds
        ((preferenceLaw t).toReal ^ (alpha / (alpha - 1))) /
        (Σ i : PRPKG24AccuracyDiversity.ItemType T, (preferenceLaw i).toReal ^ (alpha / (alpha - 1)))))
```

Retained governing declarations

- [AppliedModelingLib.Allocation](#)
- [AppliedModelingLib.Probability.expectedSampleTopKSum](#)
- [AppliedModelingLib.pmfExp](#)
- [PRPKG24AccuracyDiversity.ConsumptionModel.objective](#)
- [PRPKG24AccuracyDiversity.CountAllocation](#)

- [PRPKG24AccuracyDiversity.CountAllocation.representation](#)
- [PRPKG24AccuracyDiversity.ItemType](#)
- [PRPKG24AccuracyDiversity.OptimalAllocationSequence](#)
- [PRPKG24AccuracyDiversity.SourcePreferenceLaw](#)
- [PRPKG24AccuracyDiversity.paretolidOrderStatisticConsumptionModel](#)
- [PRPKG24AccuracyDiversity.paretolidSampleMeasure](#)

Lean-elaborated claim roles

```

1. parameter:  $\mathbb{N}$ 
2. assumption: NeZero T
3. parameter: PRPKG24AccuracyDiversity.SourcePreferenceLaw T
4. parameter:  $\mathbb{N}$ 
5. parameter:  $\mathbb{R}$ 
6. assumption:  $1 < \alpha$ 
7. assumption:  $0 < k$ 
8. assumption:  $\forall (t : \text{PRPKG24AccuracyDiversity.ItemType } T), 0 < (\text{preferenceLaw } t).\text{toReal}$ 
9. parameter: PRPKG24AccuracyDiversity.OptimalAllocationSequence fun (x :  $\mathbb{N}$ ) =>
  PRPKG24AccuracyDiversity.paretolidOrderStatisticConsumptionModel
  (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) k alpha
10. conclusion: ( $\forall (a : \text{PRPKG24AccuracyDiversity.CountAllocation } T),$ 
  (PRPKG24AccuracyDiversity.paretolidOrderStatisticConsumptionModel
  (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) k alpha).objective
  a =
  AppliedModelingLib.pmfExp preferenceLaw fun (t : PRPKG24AccuracyDiversity.ItemType T) =>
  AppliedModelingLib.Probability.expectedSampleTopKSum
  (PRPKG24AccuracyDiversity.paretolidSampleMeasure alpha (a.count t)) k)  $\wedge$ 
 $\forall (t : \text{PRPKG24AccuracyDiversity.ItemType } T),$ 
  Filter.Tendsto (fun (N :  $\mathbb{N}$ ) => (seq.allocation N).representation t) Filter.atTop
  (nhds
  ((preferenceLaw t).toReal  $^{\alpha / (\alpha - 1)}$ ) /
   $\sum i : \text{PRPKG24AccuracyDiversity.ItemType } T, (\text{preferenceLaw } i).\text{toReal}  $^{\alpha / (\alpha - 1)}$ )))$ 

```

Lean proof endpoint (built separately)

PRPKG24AccuracyDiversity.PaperInterface.theorem1_iv_formulaSpec_proof

Recorded source-to-Spec screening Verdict: matches

Reason: Matches: the target retains the Pareto $\alpha > 1$ fixed-positive-k iid model, makes its expectation bridge explicit, and has the source exponent $\alpha/(\alpha-1)$ in the normalized share profile.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 5

Source locator: cited publication, lines 438-441

Verbatim source input

(v) $S_{n,n}$ contains only items of type $t = \arg \max_{t \in [m]} p_t$.
As Table 2 illustrates, the theorem shows how for fixed k , the diversity of optimal solutions depends on the tail behavior of D . In fact, we can obtain γ -homogeneity for any γ :
(t) iid

[Next verbatim source excerpt]

There are m types of items indexed by $[m] = \{1, 2, \dots, m\}$. A user prefers exactly one type of item, preferring (t)
type $t \in [m]$ with probability p_t . The value of the i -th item of type t is a random variable X_i if the user (t)
(t)
prefers type t and 0 otherwise (so its expected value is $p_t E[X_i]$). The recommender knows only how X_i are distributed, not their realizations. We refer to p_t as the likelihood of a type and the random variable (t)
 X_i as a conditional item value (the value of an item conditional on the user preferring the item's type). Let $S_{n,k}$ be the set of n items that maximizes the expected total value of the k items with the highest realized values. We call $S_{n,k}$ the optimal set of n recommendations with respect to this objective. (We omit the dependency of $S_{n,k}$ on the other model parameters, as these will be clear from context.) $S_{n,k}$ arises naturally from an assumption that the user can only use k items, so that they derive value from the k highest value recommended items. So $S_{n,1}$ results from an assumption that the user uses only one item, while $S_{n,n}$ results from an assumption that the user uses every recommended item. $S_{n,n}$ maximizes the expected total value of all of the recommended items, so by linearity of expectation, $S_{n,n}$ contains the n items with the highest individual expected values.

[Next verbatim source excerpt]

Theorem 1. Suppose X_i

$\sim D$ where D has finite mean. Then the following statements hold.

Formalized review target The archival source above is not asserted equivalent to this formalized target.

For a finite type space with a preferred-type PMF, one common iid conditional-value probability law D whose identity is integrable and whose mean is
 $\hookrightarrow$ nonnegative, and a selected type maximizing the PMF atom mass, the source all-consumed objective is the stated selected-type and finite-iid-sample
 $\hookrightarrow$ expectation and the all-on allocation at that type is an optimal selected allocation at every total.

Archival source anchor: cited publication, lines 437-438

Expanded PaperInterface specification

```
∀ {T : ℕ} (preferenceLaw : PRPKG24AccuracyDiversity.SourcePreferenceLaw T) (D : MeasureTheory.Measure ℝ)
[MeasureTheory.IsProbabilityMeasure D],
MeasureTheory.Integrable (fun (x : ℝ) => x) D →
0 ≤ ∫ (x : ℝ), x ∂D →
  ∀ (N : ℕ) (best : PRPKG24AccuracyDiversity.ItemType T),
  (∀ (t : PRPKG24AccuracyDiversity.ItemType T), (preferenceLaw t).toReal ≤ (preferenceLaw best).toReal) →
  (PRPKG24AccuracyDiversity.iidAllConsumedSourceModel preferenceLaw D).IsOptimalAtTotal N
  (PRPKG24AccuracyDiversity.allOnTypeAllocation N best)
```

Retained governing declarations

- [PRPKG24AccuracyDiversity.ConsumptionModel.IsOptimalAtTotal](#)
- [PRPKG24AccuracyDiversity.ItemType](#)
- [PRPKG24AccuracyDiversity.SourcePreferenceLaw](#)
- [PRPKG24AccuracyDiversity.allOnTypeAllocation](#)
- [PRPKG24AccuracyDiversity.iidAllConsumedSourceModel](#)

Lean-elaborated claim roles

1. parameter: $\mathbb{N}$
2. parameter: $\text{PRPKG24AccuracyDiversity.SourcePreferenceLaw } T$
3. parameter: $\text{MeasureTheory.Measure } \mathbb{R}$
4. assumption: $\text{MeasureTheory.IsProbabilityMeasure } D$
5. assumption: $\text{MeasureTheory.Integrable } (\text{fun } (x : \mathbb{R}) \Rightarrow x) D$
6. assumption: $0 \leq \int (x : \mathbb{R}), x \partial D$
7. parameter: $\mathbb{N}$
8. parameter: $\text{PRPKG24AccuracyDiversity.ItemType } T$
9. assumption: $\forall (t : \text{PRPKG24AccuracyDiversity.ItemType } T), (\text{preferenceLaw } t).\text{toReal} \leq (\text{preferenceLaw best}).\text{toReal}$
10. conclusion: $(\text{PRPKG24AccuracyDiversity.iidAllConsumedSourceModel } \text{preferenceLaw } D).\text{IsOptimalAtTotal } N$
($\text{PRPKG24AccuracyDiversity.allOnTypeAllocation } N \text{ best}$)

Lean proof endpoint (built separately)

$\text{PRPKG24AccuracyDiversity.PaperInterface.theorem1_v_selected_maximizer_conclusionSpec_proof}$

Recorded source-to-Spec screening **Verdict:** matches_approved_corrected_target

Reason: Approved corrected target: under the approved nonnegative common-mean condition, Lean establishes an all-on allocation at a likelihood maximizer as optimal and does not make a false universal-optimizer claim.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 6

Source locator: cited publication, lines 443-457

Verbatim source input

Corollary 1. For any $\gamma \geq 0$, there exists D such that when X_i homogeneous.

$\sim D$ and k is fixed, $\{S_{n,k}\}_{n=1}^\infty$ is γ -

Intuitively, heavy-tailed distributions (part (iv)) induce less diverse recommendations since the marginal returns of recommending more items from the same type remains high: drawing more samples from a heavytailed distribution produces ever-increasing $\hookrightarrow$ item values. This contrasts with bounded distributions like the uniform distribution (part (ii)), where once an item has close to the maximum value, additional draws of that type will not further improve the utility significantly. A result for finite n and larger k . One limitation of our main results is that they are asymptotic ($n \rightarrow \infty$) and are restricted to fixed consumption constraints k . Stronger results can be obtained by considering specific distributions. For example, the result below characterizes for any n, k the representation of each type when conditional item values are uniformly distributed on $[0, 1]$.
(t) iid

[Next verbatim source excerpt]

There are m types of items indexed by $[m] = \{1, 2, \dots, m\}$. A user prefers exactly one type of item, preferring (t) type $t \in [m]$ with probability p_t . The value of the i -th item of type t is a random variable X_i if the user (t) (t) prefers type t and 0 otherwise (so its expected value is $p_t E[X_i]$). The recommender knows only how X_i are distributed, not their realizations. We refer to p_t as the likelihood of a type and the random variable (t) X_i as a conditional item value (the value of an item conditional on the user preferring the item's type). Let $S_{n,k}$ be the set of n items that maximizes the expected total value of the k items with the highest realized values. We call $S_{n,k}$ the optimal set of n recommendations with respect to this objective. (We omit the dependency of $S_{n,k}$ on the other model parameters, as these will be clear from context.) $S_{n,k}$ arises naturally from an assumption that the user can only use k items, so that they derive value from the k highest value recommended items. So $S_{n,1}$ results from an assumption that the user uses only one item, while $S_{n,n}$ results from an assumption that the user uses every recommended item. $S_{n,n}$ maximizes the expected total value of all of the recommended items, so by linearity of expectation, $S_{n,n}$ contains the n items with the highest individual expected values.

Settled source reading All-coordinate share limits: possibly zero type masses $\rightarrow$ strictly positive mass for every named type.

Expanded PaperInterface specification

```
∀ {T : ℕ} [NeZero T] {k : ℕ} (likelihood : PRPKG24AccuracyDiversity.ItemType T → ℝ) (gamma : ℝ),
  0 < k →
  0 ≤ gamma →
  (∀ (t : PRPKG24AccuracyDiversity.ItemType T), 0 < likelihood t) →
  ∃ (M : PRPKG24AccuracyDiversity.ConsumptionModel T),
    PRPKG24AccuracyDiversity.Corollary1SourceIdFamily likelihood gamma k M ∧
    ∀ (seq : PRPKG24AccuracyDiversity.OptimalAllocationSequence fun (x : ℕ) => M)
      (t : PRPKG24AccuracyDiversity.ItemType T),
      Filter.Tendsto (fun (N : ℕ) => (seq.allocation N).representation t) Filter.atTop
      (nhds (likelihood t ^ gamma / ∑ i : PRPKG24AccuracyDiversity.ItemType T, likelihood i ^ gamma))
```

Retained governing declarations

- [PRPKG24AccuracyDiversity.ConsumptionModel](#)
- [PRPKG24AccuracyDiversity.Corollary1SourceIdFamily](#)
- [PRPKG24AccuracyDiversity.CountAllocation.representation](#)
- [PRPKG24AccuracyDiversity.ItemType](#)
- [PRPKG24AccuracyDiversity.OptimalAllocationSequence](#)

Lean-elaborated claim roles

1. parameter: $\mathbb{N}$
2. assumption: $\text{NeZero } T$
3. parameter: $\mathbb{N}$
4. parameter: $\text{PRPKG24AccuracyDiversity.ItemType } T \rightarrow \mathbb{R}$
5. parameter: $\mathbb{R}$
6. assumption: $0 < k$
7. assumption: $0 \leq \text{gamma}$
8. assumption: $\forall (t : \text{PRPKG24AccuracyDiversity.ItemType } T), 0 < \text{likelihood } t$
9. conclusion: $\exists (M : \text{PRPKG24AccuracyDiversity.ConsumptionModel } T),$
 $\text{PRPKG24AccuracyDiversity.Corollary1SourceIdFamily likelihood gamma k M } \wedge$
 $\forall (\text{seq} : \text{PRPKG24AccuracyDiversity.OptimalAllocationSequence fun (x : } \mathbb{N}) \Rightarrow M)$
 $(t : \text{PRPKG24AccuracyDiversity.ItemType } T),$
 $\text{Filter.Tendsto (fun (N : } \mathbb{N}) \Rightarrow (\text{seq.allocation } N).\text{representation } t) \text{ Filter.atTop}$
 $(\text{nhds (likelihood } t \wedge \text{gamma} / \sum i : \text{PRPKG24AccuracyDiversity.ItemType } T, \text{likelihood } i \wedge \text{gamma}))$

Lean proof endpoint (built separately)

PRPKG24AccuracyDiversity.PaperInterface.corollary1Spec_proof

Recorded source-to-Spec screening Verdict: matches

Reason: Matches: the source's existence claim for every nonnegative gamma and fixed k is realized by a common-iid Corollary1SourceIidFamily witness and its gamma-power limiting shares; positive likelihood is the visible all-coordinate convention.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 7

Source locator: cited publication, lines 459-516

Verbatim source input

Proposition 2. When X_i

$\sim U([0, 1])$,

$$\frac{\sqrt{p_t}}{r_t(S_{n,k})} = \frac{p_m}{\sqrt{m+1}}$$

(8)

$\leq$

$\frac{1}{n}$

$\frac{1}{n}$

$\frac{1}{n}$

(8)

[form-feed in source extraction]

$\frac{1}{p}$

for all $k \leq \frac{p_m}{p_i} n - m - 1$.

Therefore, for any n , $S_{n,k}$ is approximately $\frac{1}{21}$ -homogeneous. In addition, for any k that is smaller than a constant fraction of n , the diversity $S_{n,k}$ does not depend on k . Thus, even for small sets of recommendations and large consumption constraints, diversity is optimal in this setting. We further give simulated results for small n and large k in Appendix B corroborating our theoretical results.

3.2

Heterogeneous Bernoulli conditional item values

(t)

(t)

Now consider when X_i are independent random variables drawn from $\text{Ber}(q_i)$, reflecting a model in which items have binary values (i.e., a user is either satisfied or not satisfied by an item). In this section, we allow

q_i to differ across i and t , implying that the recommender has knowledge about which items are more likely

to be successful conditional on a user's preferred type. Specifically, in Section 3.2.1 we allow q_i to vary

across i and in Section 3.2.2 we allow q_i to vary across t .

Our results will focus on $S_{n,1}$ and $S_{n,n}$, which both have natural interpretations in this setting:

- $S_{n,1}$ maximizes the probability that the user will be satisfied by at least one recommended item.
- $S_{n,n}$ maximizes the the expected number of recommended items the user will be satisfied by, which is equivalent to the standard metric of accuracy.

(t)

Before proceeding, we note that the basic case $q_i = \frac{1}{m+1}$.

$= q$ for all i , t is handled as a direct corollary of Theo(t) iid

[Next verbatim source excerpt]

There are m types of items indexed by $[m] = \{1, 2, \dots, m\}$. A user prefers exactly one type of item, preferring

type $t \in [m]$ with probability p_t . The value of the i -th item of type t is a random variable X_i if the user

prefers type t and 0 otherwise (so its expected value is $p_t E[X_i]$). The recommender knows only how X_i are distributed, not their realizations. We refer to p_t as the likelihood of a type and the random variable

X_i as a conditional item value (the value of an item conditional on the user preferring the item's type).

Let $S_{n,k}$ be the set of n items that maximizes the expected total value of the k items with the highest realized values. We call $S_{n,k}$ the optimal set of n recommendations with respect to this objective. (We omit the dependency of $S_{n,k}$ on the other model parameters, as these will be clear from context.)

$S_{n,k}$ arises naturally from an assumption that the user can only use k items, so that they derive value from the k highest value recommended items. So $S_{n,1}$ results from an assumption that the user uses only one item, while $S_{n,n}$ results from an assumption that the user uses every recommended item. $S_{n,n}$ maximizes the expected total value of all of the recommended items, so by linearity of expectation, $S_{n,n}$ contains the n items with the highest individual expected values.

Formalized review target The archival source above is not asserted equivalent to this formalized target.

For the iid Uniform([0,1]) source model with positive type likelihoods and every displayed-admissible positive top- k sequence, every fixed-total optimal allocation has representation error at most $(2m+1)/n$ from the square-root likelihood profile, and every optimal allocation sequence converges to that profile.

Archival source anchor: cited publication, lines 459-485

Expanded PaperInterface specification

```

∀ {T : ℕ} [inst : NeZero T] (likelihood : PRPKG24AccuracyDiversity.ItemType T → ℝ) (kseq : ℕ → ℕ),
(∀ (t : PRPKG24AccuracyDiversity.ItemType T), 0 < likelihood t) →
(∀ (N : ℕ), 0 < N → 0 < kseq N) →
(∀ (N : ℕ), 0 < N → ↑(kseq N) + 1 ≤ ↑N * PRPKG24AccuracyDiversity.uniformSqrtMinShare likelihood - ↑T) →
  (seq :
    PRPKG24AccuracyDiversity.OptimalAllocationSequence fun (N : ℕ) =>
      PRPKG24AccuracyDiversity.uniformTopKConsumptionModel likelihood (kseq N)),
(∀ (N : ℕ),
  0 < N →
    ∀ (a : PRPKG24AccuracyDiversity.CountAllocation T),
      (PRPKG24AccuracyDiversity.uniformTopKConsumptionModel likelihood (kseq N)).IsOptimalAtTotal N a →
        (PRPKG24AccuracyDiversity.PaperInterface.proposition2SqrtProfile likelihood).Approx a
        ((2 * ↑(Fintype.card (PRPKG24AccuracyDiversity.ItemType T)) + 1) / ↑N)) ∧
    seq.toAllocationSequence.ConvergesToProfile
      (PRPKG24AccuracyDiversity.PaperInterface.proposition2SqrtProfile likelihood)

```

Retained governing declarations

- [PRPKG24AccuracyDiversity.AllocationSequence.ConvergesToProfile](#)
- [PRPKG24AccuracyDiversity.ConsumptionModel.IsOptimalAtTotal](#)
- [PRPKG24AccuracyDiversity.CountAllocation](#)
- [PRPKG24AccuracyDiversity.GammaHomogeneityProfile.Approx](#)
- [PRPKG24AccuracyDiversity.ItemType](#)
- [PRPKG24AccuracyDiversity.OptimalAllocationSequence](#)
- [PRPKG24AccuracyDiversity.OptimalAllocationSequence.toAllocationSequence](#)
- [PRPKG24AccuracyDiversity.PaperInterface.proposition2SqrtProfile](#)
- [PRPKG24AccuracyDiversity.uniformSqrtMinShare](#)
- [PRPKG24AccuracyDiversity.uniformTopKConsumptionModel](#)

Lean-elaborated claim roles

```

1. parameter: ℕ
2. assumption: NeZero T
3. parameter: PRPKG24AccuracyDiversity.ItemType T → ℝ
4. parameter: ℕ → ℕ
5. assumption: ∀ (t : PRPKG24AccuracyDiversity.ItemType T), 0 < likelihood t
6. assumption: ∀ (N : ℕ), 0 < N → 0 < kseq N
7. assumption: ∀ (N : ℕ), 0 < N → ↑(kseq N) + 1 ≤ ↑N * PRPKG24AccuracyDiversity.uniformSqrtMinShare likelihood - ↑T
8. parameter: PRPKG24AccuracyDiversity.OptimalAllocationSequence fun (N : ℕ) =>
  PRPKG24AccuracyDiversity.uniformTopKConsumptionModel likelihood (kseq N)
9. conclusion: (∀ (N : ℕ),
  0 < N →
    ∀ (a : PRPKG24AccuracyDiversity.CountAllocation T),
      (PRPKG24AccuracyDiversity.uniformTopKConsumptionModel likelihood (kseq N)).IsOptimalAtTotal N a →
        (PRPKG24AccuracyDiversity.PaperInterface.proposition2SqrtProfile likelihood).Approx a
        ((2 * ↑(Fintype.card (PRPKG24AccuracyDiversity.ItemType T)) + 1) / ↑N)) ∧
    seq.toAllocationSequence.ConvergesToProfile
      (PRPKG24AccuracyDiversity.PaperInterface.proposition2SqrtProfile likelihood)

```

Lean proof endpoint (built separately)

PRPKG24AccuracyDiversity.PaperInterface.proposition2_corrected_finite_and_sequence_sourceSpec_proof

Recorded source-to-Spec screening Verdict: matches approved corrected target

Reason: Approved corrected target: Lean has the finite $(2*m+1)/n$ error bound and the same square-root profile for every fixed-total optimal sequence in the source's top-k regime.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 8

Source locator: cited publication, lines 518-535

Verbatim source input

Corollary 3 (Conditional item values are i.i.d. Bernoulli). When X_i 0-homogeneous.

$\sim \text{Ber}(q)$ for $q > 0$, then $S_{n,1}$ is

Therefore, if the success probability is the same for all items, optimal solutions are 0-homogeneous (each item is equally represented) for large n , even as the likelihoods p_t vary across type.
3.2.1

Decaying success probabilities

We now consider a setting in which among items of the same type, the recommender knows that some items have higher success probability. This maps onto settings where the recommender knows which items of a type are most likely to be satisfactory, e.g., some action movies are more commonly liked than are others. Thus, we assume that the recommender has access to items with decaying success probabilities.

(t) iid

(t)

[Next verbatim source excerpt]

There are m types of items indexed by $[m] = \{1, 2, \dots, m\}$. A user prefers exactly one type of item, preferring

(t)
type $t \in [m]$ with probability p_t . The value of the i -th item of type t is a random variable X_i if the user

(t)
(t)
prefers type t and 0 otherwise (so its expected value is $p_t E[X_i]$). The recommender knows only how X_i are distributed, not their realizations. We refer to p_t as the likelihood of a type and the random variable

(t)
 X_i as a conditional item value (the value of an item conditional on the user preferring the item's type). Let $S_{n,k}$ be the set of n items that maximizes the expected total value of the k items with the highest realized values. We call $S_{n,k}$ the optimal set of n recommendations with respect to this objective. (We omit the dependency of $S_{n,k}$ on the other model parameters, as these will be clear from context.) $S_{n,k}$ arises naturally from an assumption that the user can only use k items, so that they derive value from the k highest value recommended items. So $S_{n,1}$ results from an assumption that the user uses only one item, while $S_{n,n}$ results from an assumption that the user uses every recommended item. $S_{n,n}$ maximizes the expected total value of all of the recommended items, so by linearity of expectation, $S_{n,n}$ contains the n items with the highest individual expected values.

Formalized review target The archival source above is not asserted equivalent to this formalized target.

For a nonempty finite type space with a preferred-type PMF having every atom $p_t > 0$ and, conditional on each selected type, iid Bernoulli(q) values for
→ one common q in $(0,1)$, the source top-one objective is the stated selected-type and finite-iid-Bernoulli two-stage expectation and every fixed-total
→ optimal allocation sequence has limiting uniform type shares.

Archival source anchor: cited publication, lines 518-521

Expanded PaperInterface specification

```
∀ {T : ℕ} {NeZero T} (B : PRPKG24AccuracyDiversity.BernoulliSatisfactionModel T)
  (preferenceLaw : PRPKG24AccuracyDiversity.SourcePreferenceLaw T),
  (∀ (t : PRPKG24AccuracyDiversity.ItemType T), B.likelihood t = (preferenceLaw t).toReal) →
  (∀ (t : PRPKG24AccuracyDiversity.ItemType T), 0 < B.successProb t) →
  (∀ (t : PRPKG24AccuracyDiversity.ItemType T), B.successProb t < 1) →
  (∀ (t : PRPKG24AccuracyDiversity.ItemType T), 0 < (preferenceLaw t).toReal) →
  (∀ (i j : PRPKG24AccuracyDiversity.ItemType T), B.successProb i = B.successProb j) →
  PRPKG24AccuracyDiversity.ConsumptionModel.AsymptoticHomogeneity (fun (x : ℕ) => B.toConsumptionModel)
  (PRPKG24AccuracyDiversity.uniformProfile T)
```

Retained governing declarations

- [PRPKG24AccuracyDiversity.BernoulliSatisfactionModel](#)
- [PRPKG24AccuracyDiversity.BernoulliSatisfactionModel.toConsumptionModel](#)
- [PRPKG24AccuracyDiversity.ConsumptionModel.AsymptoticHomogeneity](#)
- [PRPKG24AccuracyDiversity.ItemType](#)
- [PRPKG24AccuracyDiversity.SourcePreferenceLaw](#)
- [PRPKG24AccuracyDiversity.uniformProfile](#)

Lean-elaborated claim roles

```
1. parameter: ℕ
2. assumption: NeZero T
3. parameter: PRPKG24AccuracyDiversity.BernoulliSatisfactionModel T
4. parameter: PRPKG24AccuracyDiversity.SourcePreferenceLaw T
5. assumption: ∀ (t : PRPKG24AccuracyDiversity.ItemType T), B.likelihood t = (preferenceLaw t).toReal
6. assumption: ∀ (t : PRPKG24AccuracyDiversity.ItemType T), 0 < B.successProb t
7. assumption: ∀ (t : PRPKG24AccuracyDiversity.ItemType T), B.successProb t < 1
8. assumption: ∀ (t : PRPKG24AccuracyDiversity.ItemType T), 0 < (preferenceLaw t).toReal
9. assumption: ∀ (i j : PRPKG24AccuracyDiversity.ItemType T), B.successProb i = B.successProb j
10. conclusion: PRPKG24AccuracyDiversity.ConsumptionModel.AsymptoticHomogeneity (fun (x : ℕ) => B.toConsumptionModel)
    (PRPKG24AccuracyDiversity.uniformProfile T)
```

Lean proof endpoint (built separately)

PRPKG24AccuracyDiversity.PaperInterface.corollary3_iid_bernoulli_conclusionSpec_proof

Recorded source-to-Spec screening Verdict: matches_approved_corrected target

Reason: Approved corrected target: q in $(0,1)$ and strictly positive preferred-type PMF support repair the source endpoint/all-type ambiguity; Lean uses the selected-type finite iid Bernoulli model and uniform limiting shares.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 9

Source locator: cited publication, lines 540-543

Verbatim source input

(i) $\{S_{n,1}\}_{\infty}$
 $n=1$ is 0-homogeneous for $\alpha < 1$.

1

[Next verbatim source excerpt]

There are m types of items indexed by $[m] = \{1, 2, \dots, m\}$. A user prefers exactly one type of item, preferring
(t)
type $t \in [m]$ with probability p_t . The value of the i -th item of type t is a random variable X_i if the user
(t)
(t)
prefers type t and 0 otherwise (so its expected value is $p_t E[X_i]$). The recommender knows only how X_i
are distributed, not their realizations. We refer to p_t as the likelihood of a type and the random variable
(t)
 X_i as a conditional item value (the value of an item conditional on the user preferring the item's type).
Let $S_{n,k}$ be the set of n items that maximizes the expected total value of the k items with the highest
realized values. We call $S_{n,k}$ the optimal set of n recommendations with respect to this objective. (We omit
the dependency of $S_{n,k}$ on the other model parameters, as these will be clear from context.)
 $S_{n,k}$ arises naturally from an assumption that the user can only use k items, so that they derive value
from the k highest value recommended items. So $S_{n,1}$ results from an assumption that the user uses only
one item, while $S_{n,n}$ results from an assumption that the user uses every recommended item.
 $S_{n,n}$ maximizes the expected total value of all of the recommended items, so by linearity of expectation,
 $S_{n,n}$ contains the n items with the highest individual expected values.

[Next verbatim source excerpt]

Theorem 2 (Decaying success probabilities). Suppose that $X_i \sim \text{Ber}(q_i)$ are i.i.d. Bernoulli random
(t)
variables such that $q_i = c(i + d)^{-\alpha}$ for all $i \geq 1$ and some $\alpha, c, d \geq 0$. Then the following statements hold.

Formalized review target The archival source above is not asserted equivalent to this formalized target.

For a nonempty finite type space with a preferred-type PMF having every atom $p_t > 0$ and, conditional on the selected type, independent rank-varying
↪ Bernoulli values satisfying $P[X_i(t)=1] = c*((i+1)+d)^{-\alpha}$, $0 \leq \alpha < 1$, $c > 0$, $d \geq 0$, and $c*(1+d)^{-\alpha} < 1$, the source top-one
↪ objective is the stated two-stage PMF experiment and every optimal top-one allocation sequence has limiting uniform type shares.

Archival source anchor: cited publication, lines 537-541

Expanded PaperInterface specification

```
∀ {T : ℕ} {NeZero T} (preferenceLaw : PRPKG24AccuracyDiversity.SourcePreferenceLaw T) (alpha c d : ℝ)
(halpha_nonneg : 0 ≤ alpha),
alpha < 1 →
  ∀ (hc_pos : 0 < c) (hd_nonneg : 0 ≤ d)
  (hfirst_lt_one : PRPKG24AccuracyDiversity.decayingBernoulliSuccess c d alpha 0 < 1),
  (∀ (t : PRPKG24AccuracyDiversity.ItemType T), 0 < (preferenceLaw t).toReal) →
    ∀
      (seq :
        PRPKG24AccuracyDiversity.OptimalAllocationSequence fun (x : ℕ) =>
          PRPKG24AccuracyDiversity.decayingBernoulliTopOneConsumptionModel
            (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) c d alpha),
      (∀ (a : PRPKG24AccuracyDiversity.CountAllocation T),
        (PRPKG24AccuracyDiversity.decayingBernoulliTopOneConsumptionModel
          (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) c d alpha).objective
          a =
            AppliedModelingLib.pmfExp preferenceLaw fun (t : PRPKG24AccuracyDiversity.ItemType T) =>
              AppliedModelingLib.pmfExp
                (PRPKG24AccuracyDiversity.decayingBernoulliFiniteLaw c d alpha (a.count t)
                  (PRPKG24AccuracyDiversity.PaperInterface.theorem2_i_corrected_pmf_source_model_endpointSpec._proof_1
                    c hc_pos)
                  hd_nonneg halpha_nonneg
                  (PRPKG24AccuracyDiversity.PaperInterface.theorem2_i_corrected_pmf_source_model_endpointSpec._proof_2
                    alpha c d hfirst_lt_one))
                PRPKG24AccuracyDiversity.rankBernoulliFiniteTopOneSampleValue) a
      (∀ (t : PRPKG24AccuracyDiversity.ItemType T),
        Filter.Tendsto (fun (N : ℕ) => (seq.allocation N).representation t) Filter.atTop (nhds (1 / ↑T)))
```

Retained governing declarations

- [AppliedModelingLib.Allocation](#)
- [AppliedModelingLib.pmfExp](#)
- [PRPKG24AccuracyDiversity.ConsumptionModel.objective](#)
- [PRPKG24AccuracyDiversity.CountAllocation](#)
- [PRPKG24AccuracyDiversity.CountAllocation.representation](#)
- [PRPKG24AccuracyDiversity.ItemType](#)

- [PRPKG24AccuracyDiversity.OptimalAllocationSequence](#)
- [PRPKG24AccuracyDiversity.SourcePreferenceLaw](#)
- [PRPKG24AccuracyDiversity.decayingBernoulliFiniteLaw](#)
- [PRPKG24AccuracyDiversity.decayingBernoulliSuccess](#)
- [PRPKG24AccuracyDiversity.decayingBernoulliTopOneConsumptionModel](#)
- [PRPKG24AccuracyDiversity.rankBernoulliFiniteTopOneSampleValue](#)

Lean-elaborated claim roles

```

1. parameter: ℕ
2. assumption: NeZero T
3. parameter: PRPKG24AccuracyDiversity.SourcePreferenceLaw T
4. parameter: ℝ
5. parameter: ℝ
6. parameter: ℝ
7. assumption: 0 ≤ alpha
8. assumption: alpha < 1
9. assumption: 0 < c
10. assumption: 0 ≤ d
11. assumption: PRPKG24AccuracyDiversity.decayingBernoulliSuccess c d alpha 0 < 1
12. assumption: ∀ (t : PRPKG24AccuracyDiversity.ItemType T), 0 < (preferenceLaw t).toReal
13. parameter: PRPKG24AccuracyDiversity.OptimalAllocationSequence fun (x : ℕ) =>
  PRPKG24AccuracyDiversity.decayingBernoulliTopOneConsumptionModel
  (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) c d alpha
14. conclusion: (∀ (a : PRPKG24AccuracyDiversity.CountAllocation T),
  (PRPKG24AccuracyDiversity.decayingBernoulliTopOneConsumptionModel
    (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) c d alpha).objective
    a =
  AppliedModelingLib.pmfExp preferenceLaw fun (t : PRPKG24AccuracyDiversity.ItemType T) =>
    AppliedModelingLib.pmfExp
      (PRPKG24AccuracyDiversity.decayingBernoulliFiniteLaw c d alpha (a.count t)
        (PRPKG24AccuracyDiversity.PaperInterface.theorem2_i_corrected_pmf_source_model_endpointSpec._proof_1 c
          hc_pos)
          hd_nonneg halpha_nonneg
          (PRPKG24AccuracyDiversity.PaperInterface.theorem2_i_corrected_pmf_source_model_endpointSpec._proof_2 alpha c
            d hfirst_lt_one))
        PRPKG24AccuracyDiversity.rankBernoulliFiniteTopOneSampleValue) a
  ∀ (t : PRPKG24AccuracyDiversity.ItemType T),
  Filter.Tendsto (fun (N : ℕ) => (seq.allocation N).representation t) Filter.atTop (nhds (1 / ↑T)))

```

Lean proof endpoint (built separately)

PRPKG24AccuracyDiversity.PaperInterface.theorem2_i_corrected_pmf_source_model_endpointSpec_proof

Recorded source-to-Spec screening Verdict: matches approved corrected target

Reason: Approved corrected target: Lean replaces the contradictory iid header by the specified selected-type PMF and independent rank-varying Bernoulli law $q_i = c \cdot ((i+1)+d)^{-\alpha}$, with $0 \leq \alpha < 1$, $c > 0$, $q_1 < 1$, and uniform shares.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 10

Source locator: cited publication, lines 544-546

Verbatim source input

(ii) $\{S_{n,1}\}_{n=1}^{\infty}$ is $1+c$ -homogeneous for $\alpha = 1$.

[Next verbatim source excerpt]

There are m types of items indexed by $[m] = \{1, 2, \dots, m\}$. A user prefers exactly one type of item, preferring type $t \in [m]$ with probability p_t . The value of the i -th item of type t is a random variable X_i if the user prefers type t and 0 otherwise (so its expected value is $p_t E[X_i]$). The recommender knows only how X_i are distributed, not their realizations. We refer to p_t as the likelihood of a type and the random variable X_i as a conditional item value (the value of an item conditional on the user preferring the item's type). Let $S_{n,k}$ be the set of n items that maximizes the expected total value of the k items with the highest realized values. We call $S_{n,k}$ the optimal set of n recommendations with respect to this objective. (We omit the dependency of $S_{n,k}$ on the other model parameters, as these will be clear from context.) $S_{n,k}$ arises naturally from an assumption that the user can only use k items, so that they derive value from the k highest value recommended items. So $S_{n,1}$ results from an assumption that the user uses only one item, while $S_{n,n}$ results from an assumption that the user uses every recommended item. $S_{n,n}$ maximizes the expected total value of all of the recommended items, so by linearity of expectation, $S_{n,n}$ contains the n items with the highest individual expected values.

[Next verbatim source excerpt]

Theorem 2 (Decaying success probabilities). Suppose that $X_i \sim \text{Ber}(q_i)$ are i.i.d. Bernoulli random variables such that $q_i = c(i + d) - \alpha$ for all $i \geq 1$ and some $\alpha, c, d \geq 0$. Then the following statements hold.

Formalized review target The archival source above is not asserted equivalent to this formalized target.

For a nonempty finite type space with a preferred-type PMF having every atom $p_t > 0$ and, conditional on the selected type, independent rank-varying Bernoulli values satisfying $P[X_i = 1] = c*((i+1)+d)^{-1}$, $c > 0$, $d \geq 0$, and $c*(1+d)^{-1} < 1$, the source top-one objective is the stated two-stage PMF experiment and every optimal top-one allocation sequence has shares converging to $p_t^{1/(1+c)} / \sum_i p_i^{1/(1+c)}$.

Archival source anchor: cited publication, lines 537-545

Expanded PaperInterface specification

```
∀ {T : ℕ} {NeZero T} (preferenceLaw : PRPKG24AccuracyDiversity.SourcePreferenceLaw T) (c d : ℝ) (hc_pos : 0 < c)
(hd_nonneg : 0 ≤ d) (hfirst_lt_one : PRPKG24AccuracyDiversity.decayingBernoulliSuccess c d 1 0 < 1),
(∀ (t : PRPKG24AccuracyDiversity.ItemType T), 0 < (preferenceLaw t).toReal) →
  (seq :
    PRPKG24AccuracyDiversity.OptimalAllocationSequence fun (x : ℕ) =>
      PRPKG24AccuracyDiversity.decayingBernoulliTopOneConsumptionModel
        (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) c d 1),
  (∀ (a : PRPKG24AccuracyDiversity.CountAllocation T),
    (PRPKG24AccuracyDiversity.decayingBernoulliTopOneConsumptionModel
      (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) c d 1).objective
      a =
    AppliedModelingLib.pmfExp preferenceLaw fun (t : PRPKG24AccuracyDiversity.ItemType T) =>
      AppliedModelingLib.pmfExp
        (PRPKG24AccuracyDiversity.decayingBernoulliFiniteLaw c d 1 (a.count t)
          (PRPKG24AccuracyDiversity.PaperInterface.theorem2_i_corrected_pmf_source_model_endpointSpec._proof_1 c
            hc_pos)
            hd_nonneg
            PRPKG24AccuracyDiversity.PaperInterface.theorem2_ii_corrected_pmf_source_model_endpointSpec._proof_1
            (PRPKG24AccuracyDiversity.PaperInterface.theorem2_ii_corrected_pmf_source_model_endpointSpec._proof_2
              c d hfirst_lt_one))
          PRPKG24AccuracyDiversity.rankBernoulliFiniteTopOneSampleValue) ∧
    ∀ (t : PRPKG24AccuracyDiversity.ItemType T),
      Filter.Tendsto (fun (N : ℕ) => (seq.allocation N).representation t) Filter.atTop
        (nhds
          ((preferenceLaw t).toReal ^ (1 / (1 + c))) /
            ∑ i : PRPKG24AccuracyDiversity.ItemType T, (preferenceLaw i).toReal ^ (1 / (1 + c))))
```

Retained governing declarations

- [AppliedModelingLib.Allocation](#)
- [AppliedModelingLib.pmfExp](#)
- [PRPKG24AccuracyDiversity.ConsumptionModel.objective](#)
- [PRPKG24AccuracyDiversity.CountAllocation](#)
- [PRPKG24AccuracyDiversity.CountAllocation.representation](#)
- [PRPKG24AccuracyDiversity.ItemType](#)

- [PRPKG24AccuracyDiversity.OptimalAllocationSequence](#)
- [PRPKG24AccuracyDiversity.SourcePreferenceLaw](#)
- [PRPKG24AccuracyDiversity.decayingBernoulliFiniteLaw](#)
- [PRPKG24AccuracyDiversity.decayingBernoulliSuccess](#)
- [PRPKG24AccuracyDiversity.decayingBernoulliTopOneConsumptionModel](#)
- [PRPKG24AccuracyDiversity.rankBernoulliFiniteTopOneSampleValue](#)

Lean-elaborated claim roles

```

1. parameter: ℕ
2. assumption: NeZero T
3. parameter: PRPKG24AccuracyDiversity.SourcePreferenceLaw T
4. parameter: ℝ
5. parameter: ℝ
6. assumption: 0 < c
7. assumption: 0 ≤ d
8. assumption: PRPKG24AccuracyDiversity.decayingBernoulliSuccess c d 1 0 < 1
9. assumption: ∀ (t : PRPKG24AccuracyDiversity.ItemType T), 0 < (preferenceLaw t).toReal
10. parameter: PRPKG24AccuracyDiversity.OptimalAllocationSequence fun (x : ℕ) =>
    PRPKG24AccuracyDiversity.decayingBernoulliTopOneConsumptionModel
    (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) c d 1
11. conclusion: (∀ (a : PRPKG24AccuracyDiversity.CountAllocation T),
    (PRPKG24AccuracyDiversity.decayingBernoulliTopOneConsumptionModel
    (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) c d 1).objective
    a =
    AppliedModelingLib.pmfExp preferenceLaw fun (t : PRPKG24AccuracyDiversity.ItemType T) =>
    AppliedModelingLib.pmfExp
    (PRPKG24AccuracyDiversity.decayingBernoulliFiniteLaw c d 1 (a.count t)
    (PRPKG24AccuracyDiversity.PaperInterface.theorem2_i_corrected_pmf_source_model_endpointSpec._proof_1 c
    hc_pos)
    hd_nonneg
    PRPKG24AccuracyDiversity.PaperInterface.theorem2_ii_corrected_pmf_source_model_endpointSpec._proof_1
    (PRPKG24AccuracyDiversity.PaperInterface.theorem2_ii_corrected_pmf_source_model_endpointSpec._proof_2 c d
    hfirst_lt_one))
    PRPKG24AccuracyDiversity.rankBernoulliFiniteTopOneSampleValue) ∧
    ∀ (t : PRPKG24AccuracyDiversity.ItemType T),
    Filter.Tendsto (fun (N : ℕ) => (seq.allocation N).representation t) Filter.atTop
    (nhds
    ((preferenceLaw t).toReal ^ (1 / (1 + c)) /
    ∑ i : PRPKG24AccuracyDiversity.ItemType T, (preferenceLaw i).toReal ^ (1 / (1 + c))))

```

Lean proof endpoint (built separately)

PRPKG24AccuracyDiversity.PaperInterface.theorem2_ii_corrected_pmf_source_model_endpointSpec_proof

Recorded source-to-Spec screening Verdict: matches_approved_corrected_target

Reason: Approved corrected target: the same two-stage independent rank-varying Bernoulli bridge is explicit at $\alpha=1$, with $c>0$, $d\geq 0$, $q_1<1$, positive PMF support, and the approved $p^{1/(1+c)}$ shares.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 11

Source locator: cited publication, lines 547-551

Verbatim source input

(iii) $\{S_{n,1}\}_{n=1}^{\infty}$
 $n=1$ is α -homogeneous for $\alpha > 1$.

Additionally,
1

[Next verbatim source excerpt]

There are m types of items indexed by $[m] = \{1, 2, \dots, m\}$. A user prefers exactly one type of item, preferring
(t)
type $t \in [m]$ with probability p_t . The value of the i -th item of type t is a random variable X_i if the user
(t)
(t)
prefers type t and 0 otherwise (so its expected value is $p_t E[X_i]$). The recommender knows only how X_i
(t)
are distributed, not their realizations. We refer to p_t as the likelihood of a type and the random variable
(t)
 X_i as a conditional item value (the value of an item conditional on the user preferring the item's type).
Let $S_{n,k}$ be the set of n items that maximizes the expected total value of the k items with the highest
realized values. We call $S_{n,k}$ the optimal set of n recommendations with respect to this objective. (We omit
the dependency of $S_{n,k}$ on the other model parameters, as these will be clear from context.)
 $S_{n,k}$ arises naturally from an assumption that the user can only use k items, so that they derive value
from the k highest value recommended items. So $S_{n,1}$ results from an assumption that the user uses only
one item, while $S_{n,n}$ results from an assumption that the user uses every recommended item.
 $S_{n,n}$ maximizes the expected total value of all of the recommended items, so by linearity of expectation,
 $S_{n,n}$ contains the n items with the highest individual expected values.

[Next verbatim source excerpt]

Theorem 2 (Decaying success probabilities). Suppose that $X_i \sim \text{Ber}(q_i)$ are i.i.d. Bernoulli random
(t)
variables such that $q_i = c(i+d)^{-\alpha}$ for all $i \geq 1$ and some $\alpha, c, d \geq 0$. Then the following statements hold.

Formalized review target The archival source above is not asserted equivalent to this formalized target.

For a nonempty finite type space with a preferred-type PMF having every atom $p_t > 0$ and, conditional on the selected type, independent rank-varying
↪ Bernoulli values satisfying $P[X_i^{(t)}=1] = c*((i+1)+d)^{-\alpha}$, $\alpha > 1$, $c > 0$, $d \geq 0$, and $c*(1+d)^{-\alpha} < 1$, the source top-one objective is
↪ the stated two-stage PMF experiment and every optimal top-one allocation sequence has shares converging to $p_t^{1/\alpha} / \sum_i p_i^{1/\alpha}$.

Archival source anchor: cited publication, lines 537-548

Expanded PaperInterface specification

```
∀ {T : ℕ} [NeZero T] (preferenceLaw : PRPKG24AccuracyDiversity.SourcePreferenceLaw T) (alpha c d : ℝ)
(halpha_gt_one : 1 < alpha) (hc_pos : 0 < c) (hd_nonneg : 0 ≤ d)
(hfirst_lt_one : PRPKG24AccuracyDiversity.decayingBernoulliSuccess c d alpha 0 < 1),
(∀ (t : PRPKG24AccuracyDiversity.ItemType T), 0 < (preferenceLaw t).toReal) →
  (seq :
    PRPKG24AccuracyDiversity.OptimalAllocationSequence fun (x : ℕ) =>
      PRPKG24AccuracyDiversity.decayingBernoulliTopOneConsumptionModel
        (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) c d alpha),
  (∀ (a : PRPKG24AccuracyDiversity.CountAllocation T),
    (PRPKG24AccuracyDiversity.decayingBernoulliTopOneConsumptionModel
      (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) c d alpha).objective
      a =
    AppliedModelingLib.pmfExp preferenceLaw fun (t : PRPKG24AccuracyDiversity.ItemType T) =>
      AppliedModelingLib.pmfExp
        (PRPKG24AccuracyDiversity.decayingBernoulliFiniteLaw c d alpha (a.count t)
          (PRPKG24AccuracyDiversity.PaperInterface.theorem2_i_corrected_pmf_source_model_endpointSpec_proof_1 c
            hc_pos)
            hd_nonneg
            (PRPKG24AccuracyDiversity.PaperInterface.theorem2_iii_corrected_pmf_source_model_endpointSpec_proof_1
              alpha halpha_gt_one)
            (PRPKG24AccuracyDiversity.PaperInterface.theorem2_i_corrected_pmf_source_model_endpointSpec_proof_2
              alpha c d hfirst_lt_one))
          PRPKG24AccuracyDiversity.rankBernoulliFiniteTopOneSampleValue) ∧
    ∀ (t : PRPKG24AccuracyDiversity.ItemType T),
    Filter.Tendsto (fun (N : ℕ) => (seq.allocation N).representation t) Filter.atTop
      (nhds
        ((preferenceLaw t).toReal ^ (1 / alpha) /
          ∑ i : PRPKG24AccuracyDiversity.ItemType T, (preferenceLaw i).toReal ^ (1 / alpha))))
```

Retained governing declarations

- [AppliedModelingLib.Allocation](#)
- [AppliedModelingLib.pmfExp](#)
- [PRPKG24AccuracyDiversity.ConsumptionModel.objective](#)
- [PRPKG24AccuracyDiversity.CountAllocation](#)

- [PRPKG24AccuracyDiversity.CountAllocation.representation](#)
- [PRPKG24AccuracyDiversity.ItemType](#)
- [PRPKG24AccuracyDiversity.OptimalAllocationSequence](#)
- [PRPKG24AccuracyDiversity.SourcePreferenceLaw](#)
- [PRPKG24AccuracyDiversity.decayingBernoulliFiniteLaw](#)
- [PRPKG24AccuracyDiversity.decayingBernoulliSuccess](#)
- [PRPKG24AccuracyDiversity.decayingBernoulliTopOneConsumptionModel](#)
- [PRPKG24AccuracyDiversity.rankBernoulliFiniteTopOneSampleValue](#)

Lean-elaborated claim roles

```

1. parameter: ℕ
2. assumption: NeZero T
3. parameter: PRPKG24AccuracyDiversity.SourcePreferenceLaw T
4. parameter: ℝ
5. parameter: ℝ
6. parameter: ℝ
7. assumption: 1 < alpha
8. assumption: 0 < c
9. assumption: 0 ≤ d
10. assumption: PRPKG24AccuracyDiversity.decayingBernoulliSuccess c d alpha 0 < 1
11. assumption: ∀ (t : PRPKG24AccuracyDiversity.ItemType T), 0 < (preferenceLaw t).toReal
12. parameter: PRPKG24AccuracyDiversity.OptimalAllocationSequence fun (x : ℕ) =>
  PRPKG24AccuracyDiversity.decayingBernoulliTopOneConsumptionModel
  (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) c d alpha
13. conclusion: (∀ (a : PRPKG24AccuracyDiversity.CountAllocation T),
  (PRPKG24AccuracyDiversity.decayingBernoulliTopOneConsumptionModel
    (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) c d alpha).objective
    a =
  AppliedModelingLib.pmfExp preferenceLaw fun (t : PRPKG24AccuracyDiversity.ItemType T) =>
    AppliedModelingLib.pmfExp
      (PRPKG24AccuracyDiversity.decayingBernoulliFiniteLaw c d alpha (a.count t)
        (PRPKG24AccuracyDiversity.PaperInterface.theorem2_i_corrected_pmf_source_model_endpointSpec._proof_1 c
          hc_pos)
          hd_nonneg
          (PRPKG24AccuracyDiversity.PaperInterface.theorem2_iii_corrected_pmf_source_model_endpointSpec._proof_1 alpha
            halpha_gt_one)
            (PRPKG24AccuracyDiversity.PaperInterface.theorem2_j_corrected_pmf_source_model_endpointSpec._proof_2 alpha c
              d hfirst_lt_one))
          PRPKG24AccuracyDiversity.rankBernoulliFiniteTopOneSampleValue) a
  ∀ (t : PRPKG24AccuracyDiversity.ItemType T),
  Filter.Tendsto (fun (N : ℕ) => (seq.allocation N).representation t) Filter.atTop
    (nhds
      ((preferenceLaw t).toReal ^ (1 / alpha) /
        ∑ i : PRPKG24AccuracyDiversity.ItemType T, (preferenceLaw i).toReal ^ (1 / alpha))))

```

Lean proof endpoint (built separately)

PRPKG24AccuracyDiversity.PaperInterface.theorem2_iii_corrected_pmf_source_model_endpointSpec_proof

Recorded source-to-Spec screening Verdict: matches_approved_corrected_target

Reason: Approved corrected target: Lean realizes the source ranks as $i+1$ in the independent Bernoulli law and, for $\alpha > 1$ with $c > 0$ and $q_1 < 1$, states the approved $p^{1/\alpha}$ limit and objective bridge.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 12

Source locator: cited publication, lines 552-564

Verbatim source input

(iv) $\{S_{n,n}\}_{n=1}^{\infty}$ is α -homogeneous for $\alpha \geq 0$.

When the success probabilities of items have moderate decay ($\alpha < 1$), then 0-homogeneity is maintained in the case $k = 1$ (note that $\alpha = 0$ recovers $\hookrightarrow$ Corollary 3). Moreover, for all rates of decay, optimal recommendations reflect at least proportional diversity for large n and $k = 1$. Theorem 2 also reveals surprising non-monotonic behavior. In particular, there is a discontinuity at $\alpha = 1$, where homogeneity suddenly increases, but then decreases as α continues to increase. At $\alpha = 1$, the optimal amount of diversity when $\alpha = 1$ can range between 0 and 1 depending on c . When $k = n$, a larger rate of decay induces more diverse recommendations. Intuitively, when there is a larger rate of decay, the recommender has fewer high-quality options of a given type and is more incentivized to recommend high-quality options of other types. Note that for moderate rates of decay ($\alpha < 1$), $S_{n,n}$ remains less than proportionally diverse for large n , unlike $S_{n,1}$.

7

[Next verbatim source excerpt]

There are m types of items indexed by $[m] = \{1, 2, \dots, m\}$. A user prefers exactly one type of item, preferring (t) type $t \in [m]$ with probability p_t . The value of the i -th item of type t is a random variable X_i if the user (t) prefers type t and 0 otherwise (so its expected value is $p_t E[X_i]$). The recommender knows only how X_i are distributed, not their realizations. We refer to p_t as the likelihood of a type and the random variable (t) X_i as a conditional item value (the value of an item conditional on the user preferring the item's type). Let $S_{n,k}$ be the set of n items that maximizes the expected total value of the k items with the highest realized values. We call $S_{n,k}$ the optimal set of n recommendations with respect to this objective. (We omit the dependency of $S_{n,k}$ on the other model parameters, as these will be clear from context.) $S_{n,k}$ arises naturally from an assumption that the user can only use k items, so that they derive value from the k highest value recommended items. So $S_{n,1}$ results from an assumption that the user uses only one item, while $S_{n,n}$ results from an assumption that the user uses every recommended item. $S_{n,n}$ maximizes the expected total value of all of the recommended items, so by linearity of expectation, $S_{n,n}$ contains the n items with the highest individual expected values.

[Next verbatim source excerpt]

Theorem 2 (Decaying success probabilities). Suppose that $X_i \sim \text{Ber}(q_i)$ are i.i.d. Bernoulli random (t) variables such that $q_i = c(i + d)^{-\alpha}$ for all $i \geq 1$ and some $\alpha, c, d \geq 0$. Then the following statements hold.

Formalized review target The archival source above is not asserted equivalent to this formalized target.

For a nonempty finite type space with a preferred-type PMF having every atom $p_t > 0$ and, conditional on the selected type, independent rank-varying $\hookrightarrow$ Bernoulli values satisfying $P[X_i(t)=1] = c*((i+1)+d)^{-\alpha}$, $\alpha \geq 0$, $c \geq 0$, $d \geq 0$, and $c*(1+d)^{-\alpha} \leq 1$: the source all-consumed $\hookrightarrow$ objective is the stated two-stage PMF experiment; if $\alpha > 0$ and $c > 0$, every optimal all-consumed allocation sequence has shares converging to $\hookrightarrow$ $p_t^{1/(1+\alpha)} / \sum_i p_i^{1/(1+\alpha)}$; if $\alpha = 0$, the corrected selected-optimizer convention yields an all-on allocation at a likelihood-maximizing $\hookrightarrow$ type.

Archival source anchor: cited publication, lines 537-553

Expanded PaperInterface specification

```
∀ {T : ℕ} [NeZero T] (preferenceLaw : PRPKG24AccuracyDiversity.SourcePreferenceLaw T) (alpha c d : ℝ)
(halpha_nonneg : 0 ≤ alpha) (hc_nonneg : 0 ≤ c) (hd_nonneg : 0 ≤ d)
(hfirst_le_one : PRPKG24AccuracyDiversity.decayingBernoulliSuccess c d alpha 0 ≤ 1),
(∀ (t : PRPKG24AccuracyDiversity.ItemType T), 0 < (preferenceLaw t).toReal) →
(0 < alpha →
  ∀ (hc_pos : 0 < c)
    (seq :
      PRPKG24AccuracyDiversity.OptimalAllocationSequence fun (x : ℕ) =>
        PRPKG24AccuracyDiversity.decayingBernoulliAllConsumedConsumptionModel
          (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) c d alpha),
    (∀ (a : PRPKG24AccuracyDiversity.CountAllocation T),
      (PRPKG24AccuracyDiversity.decayingBernoulliAllConsumedConsumptionModel
        (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) c d alpha).objective
        a =
        AppliedModelingLib.pmfExp preferenceLaw fun (t : PRPKG24AccuracyDiversity.ItemType T) =>
          AppliedModelingLib.pmfExp
            (PRPKG24AccuracyDiversity.decayingBernoulliFiniteLaw c d alpha (a.count t)
              (PRPKG24AccuracyDiversity.PaperInterface.theorem2_i_corrected_pmf_source_model_endpointSpec._proof_1
                c hc_pos)
              hd_nonneg halpha_nonneg hfirst_le_one)
            PRPKG24AccuracyDiversity.rankBernoulliFiniteAllConsumedSampleValue) ∧
      ∀ (t : PRPKG24AccuracyDiversity.ItemType T),
        Filter.Tendsto (fun (N : ℕ) => (seq.allocation N).representation t) Filter.atTop
          (nhds
            ((preferenceLaw t).toReal ^ (1 / alpha) /
              ∑ i : PRPKG24AccuracyDiversity.ItemType T, (preferenceLaw i).toReal ^ (1 / alpha)))) ∧
      (alpha = 0 →
        ∀ (N : ℕ) (best : PRPKG24AccuracyDiversity.ItemType T),
          (∀ (t : PRPKG24AccuracyDiversity.ItemType T), (preferenceLaw t).toReal ≤ (preferenceLaw best).toReal) →
            (∀ (a : PRPKG24AccuracyDiversity.CountAllocation T),
              (PRPKG24AccuracyDiversity.decayingBernoulliAllConsumedConsumptionModel
```



```

    (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) c d alpha).objective
  a =
  AppliedModelingLib.pmfExp preferenceLaw fun (t : PRPKG24AccuracyDiversity.ItemType T) =>
  AppliedModelingLib.pmfExp
    (PRPKG24AccuracyDiversity.decayingBernoulliFiniteLaw c d alpha (a.count t) hc_nonneg hd_nonneg
      halpha_nonneg hfirst_le_one)
  PRPKG24AccuracyDiversity.rankBernoulliFiniteAllConsumedSampleValue) ^
  (PRPKG24AccuracyDiversity.decayingBernoulliAllConsumedConsumptionModel
    (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) c d
      alpha).IsOptimalAtTotal
  N (PRPKG24AccuracyDiversity.allOnTypeAllocation N best))

```

Retained governing declarations

- [AppliedModelingLib.Allocation](#)
- [AppliedModelingLib.pmfExp](#)
- [PRPKG24AccuracyDiversity.ConsumptionModel.IsOptimalAtTotal](#)
- [PRPKG24AccuracyDiversity.ConsumptionModel.objective](#)
- [PRPKG24AccuracyDiversity.CountAllocation](#)
- [PRPKG24AccuracyDiversity.CountAllocation.representation](#)
- [PRPKG24AccuracyDiversity.ItemType](#)
- [PRPKG24AccuracyDiversity.OptimalAllocationSequence](#)
- [PRPKG24AccuracyDiversity.SourcePreferenceLaw](#)
- [PRPKG24AccuracyDiversity.allOnTypeAllocation](#)
- [PRPKG24AccuracyDiversity.decayingBernoulliAllConsumedConsumptionModel](#)
- [PRPKG24AccuracyDiversity.decayingBernoulliFiniteLaw](#)
- [PRPKG24AccuracyDiversity.decayingBernoulliSuccess](#)
- [PRPKG24AccuracyDiversity.rankBernoulliFiniteAllConsumedSampleValue](#)

Lean-elaborated claim roles

```

1. parameter: ℕ
2. assumption: NeZero T
3. parameter: PRPKG24AccuracyDiversity.SourcePreferenceLaw T
4. parameter: ℝ
5. parameter: ℝ
6. parameter: ℝ
7. assumption: 0 ≤ alpha
8. assumption: 0 ≤ c
9. assumption: 0 ≤ d
10. assumption: PRPKG24AccuracyDiversity.decayingBernoulliSuccess c d alpha 0 ≤ 1
11. assumption: ∀ (t : PRPKG24AccuracyDiversity.ItemType T), 0 < (preferenceLaw t).toReal
12. conclusion: (0 < alpha →
  ∀ (hc_pos : 0 < c)
    (seq :
      PRPKG24AccuracyDiversity.OptimalAllocationSequence fun (x : ℕ) =>
      PRPKG24AccuracyDiversity.decayingBernoulliAllConsumedConsumptionModel
        (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) c d alpha),
    (∀ (a : PRPKG24AccuracyDiversity.CountAllocation T),
      (PRPKG24AccuracyDiversity.decayingBernoulliAllConsumedConsumptionModel
        (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) c d alpha).objective
        a =
        AppliedModelingLib.pmfExp preferenceLaw fun (t : PRPKG24AccuracyDiversity.ItemType T) =>
        AppliedModelingLib.pmfExp
          (PRPKG24AccuracyDiversity.decayingBernoulliFiniteLaw c d alpha (a.count t)
            (PRPKG24AccuracyDiversity.PaperInterface.theorem2_i_corrected_pmf_source_model_endpointSpec._proof_1 c
              hc_pos)
            hd_nonneg halpha_nonneg hfirst_le_one)
          PRPKG24AccuracyDiversity.rankBernoulliFiniteAllConsumedSampleValue) ^
        ∀ (t : PRPKG24AccuracyDiversity.ItemType T),
        Filter.Tendsto (fun (N : ℕ) => (seq.allocation N).representation t) Filter.atTop
          (nhds
            ((preferenceLaw t).toReal ^ (1 / alpha) /
              ∑ i : PRPKG24AccuracyDiversity.ItemType T, (preferenceLaw i).toReal ^ (1 / alpha)))) ^
        (alpha = 0 →
          ∀ (N : ℕ) (best : PRPKG24AccuracyDiversity.ItemType T),
          (∀ (t : PRPKG24AccuracyDiversity.ItemType T), (preferenceLaw t).toReal ≤ (preferenceLaw best).toReal) →
          (∀ (a : PRPKG24AccuracyDiversity.CountAllocation T),
            (PRPKG24AccuracyDiversity.decayingBernoulliAllConsumedConsumptionModel
              (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) c d alpha).objective
              a =
              AppliedModelingLib.pmfExp preferenceLaw fun (t : PRPKG24AccuracyDiversity.ItemType T) =>
              AppliedModelingLib.pmfExp
                (PRPKG24AccuracyDiversity.decayingBernoulliFiniteLaw c d alpha (a.count t) hc_nonneg hd_nonneg
                  halpha_nonneg hfirst_le_one)
                PRPKG24AccuracyDiversity.rankBernoulliFiniteAllConsumedSampleValue) ^
            (PRPKG24AccuracyDiversity.decayingBernoulliAllConsumedConsumptionModel
              (fun (t : PRPKG24AccuracyDiversity.ItemType T) => (preferenceLaw t).toReal) c d alpha).IsOptimalAtTotal
              N (PRPKG24AccuracyDiversity.allOnTypeAllocation N best)))

```

Lean proof endpoint (built separately)

PRPKG24AccuracyDiversity.PaperInterface.theorem2_iv_corrected_pmf_source_model_endpointSpec_proof

Recorded source-to-Spec screening Verdict: matches_approved_corrected_target

Reason: Approved corrected target: Lean gives the all-consumed two-stage PMF bridge, separates $\alpha > 0 / c > 0$ $p^{(1/\alpha)}$ shares from $\alpha = 0$, and uses the approved likelihood-argmax all-on selected optimizer without asserting $1/0$.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 13

Source locator: cited publication, lines 605-662

Verbatim source input

Theorem 3 (Varying success probability across types). Suppose that for each fixed t , X_i i.i.d. Bernoulli random variables. Then

$\lim_{n \rightarrow \infty} \mathbb{E}(\sum_{t=1}^m S_{n,t}) \propto$

$\frac{1}{\log \frac{1}{1-q_t}}$

$\sim \text{Ber}(q_t)$ are

(9)

while $S_{n,t}$ contains only items of type $t = \arg \max_{t \in [m]} p_t q_t$.

The surprising high-level takeaway from Theorem 3 is that, for large n , a smaller success probability q_t results in more representation of type t . The less likely an item of a given type is satisfactory, the more that type is recommended. Moreover, note that the amount of representation in this setting is independent of the popularities $p_1, \dots, p_m$.

This paradox is illustrated in grocery stores, where more space is devoted to ice cream than milk, despite milk being much more popular than ice cream. Here, p_1 (the popularity of milk) is higher than p_2 (the popularity of ice cream). However, q_1 (the likelihood a given milk product satisfies a shopper looking for milk) is also higher than q_2 (the likelihood a given ice cream product satisfies a shopper looking for ice cream), since people tend to have more specific tastes for ice cream. Thus, since q_2 is smaller than q_1 and the grocery store should “recommend” many more ice creams than milks, explaining why more space is devoted to ice cream. Intuitively, while more shoppers want milk, these consumers can be satisfied with a small selection of milk; thus, it is more beneficial to devote more space to ice cream, for which shoppers have more specific tastes.

4

Sketch of Proofs

We now sketch the proof of Theorem 1(ii). (The general strategy applies to the rest of Theorem 1, as well as Theorem 2 and Theorem 3.) First, recall the general setup in Theorem 1, where the conditional item

values X_i are drawn i.i.d. from a shared distribution D . In other words, a user prefers type $t \in [m]$ with probability p_t such that the user prefers exactly one type of item, and conditioned on the user preferring type

[form-feed in source extraction]

t , the value of an item of that type is drawn i.i.d. from D . We are interested in analyzing, depending on D , the composition of $S_{n,k}$, the set of n items maximizing the expected sum of the k highest value items in the

set. If $S_{n,k}$ contains at items of type t , we need to analyze

at

$\lim_{n \rightarrow \infty} \mathbb{E}(\sum_{t=1}^m S_{n,t}) =$

$\lim_{n \rightarrow \infty} \mathbb{E}(\sum_{t=1}^m S_{n,t}) =$

(10)

We first provide an expression for the expected sum of the k highest value items in a set S with at items of type t . The following definition will be useful.

[Next verbatim source excerpt]

There are m types of items indexed by $[m] = \{1, 2, \dots, m\}$. A user prefers exactly one type of item, preferring

type $t \in [m]$ with probability p_t . The value of the i -th item of type t is a random variable X_i if the user

prefers type t and 0 otherwise (so its expected value is $p_t \mathbb{E}[X_i]$). The recommender knows only how X_i are distributed, not their realizations. We refer to p_t as the likelihood of a type and the random variable

X_i as a conditional item value (the value of an item conditional on the user preferring the item's type).

Let $S_{n,k}$ be the set of n items that maximizes the expected total value of the k items with the highest realized values. We call $S_{n,k}$ the optimal set of n recommendations with respect to this objective. (We omit the dependency of $S_{n,k}$ on the other model parameters, as these will be clear from context.)

$S_{n,k}$ arises naturally from an assumption that the user can only use k items, so that they derive value from the k highest value recommended items. So $S_{n,1}$ results from an assumption that the user uses only one item, while $S_{n,n}$ results from an assumption that the user uses every recommended item.

$S_{n,n}$ maximizes the expected total value of all of the recommended items, so by linearity of expectation, $S_{n,n}$ contains the n items with the highest individual expected values.

[Next verbatim source excerpt]

Theorem 3 (Varying success probability across types). Suppose that for each fixed t , X_i i.i.d. Bernoulli random variables. Then

[Next verbatim source excerpt]

$\sim \text{Ber}(q_t)$ are

Formalized review target The archival source above is not asserted equivalent to this formalized target.

For a nonempty finite type space with a preferred-type PMF having every atom $p_t > 0$ and, conditional on each selected type t , iid Bernoulli values with q_t in $(0,1)$, the source top-one objective is the stated selected-type and finite-iid-Bernoulli two-stage expectation and every fixed-total optimal allocation sequence has limiting type share $(1 / \log(1 / (1 - q_t))) / \sum_i (1 / \log(1 / (1 - q_i)))$ for every t .

Archival source anchor: cited publication, lines 605-619

Expanded PaperInterface specification

```

∀ {T : ℕ} [NeZero T] (B : PRPKG24AccuracyDiversity.BernoulliSatisfactionModel T)
  (preferenceLaw : PRPKG24AccuracyDiversity.SourcePreferenceLaw T),
  (∀ (t : PRPKG24AccuracyDiversity.ItemType T), B.likelihood t = (preferenceLaw t).toReal) →
  (∀ (t : PRPKG24AccuracyDiversity.ItemType T), 0 < B.successProb t) →
  (∀ (t : PRPKG24AccuracyDiversity.ItemType T), B.successProb t < 1) →
  (∀ (t : PRPKG24AccuracyDiversity.ItemType T), 0 < (preferenceLaw t).toReal) →
  ∀ (seq : PRPKG24AccuracyDiversity.OptimalAllocationSequence fun (x : ℕ) => B.toConsumptionModel
    (t : PRPKG24AccuracyDiversity.ItemType T),
    Filter.Tendsto (fun (N : ℕ) => (seq.allocation N).representation t) Filter.atTop
    (nhds
      (PRPKG24AccuracyDiversity.theorem3LogShareWeight B t /
        ∑ i : PRPKG24AccuracyDiversity.ItemType T, PRPKG24AccuracyDiversity.theorem3LogShareWeight B i))

```

Retained governing declarations

- [PRPKG24AccuracyDiversity.BernoulliSatisfactionModel](#)
- [PRPKG24AccuracyDiversity.BernoulliSatisfactionModel.toConsumptionModel](#)
- [PRPKG24AccuracyDiversity.CountAllocation.representation](#)
- [PRPKG24AccuracyDiversity.ItemType](#)
- [PRPKG24AccuracyDiversity.OptimalAllocationSequence](#)
- [PRPKG24AccuracyDiversity.SourcePreferenceLaw](#)
- [PRPKG24AccuracyDiversity.theorem3LogShareWeight](#)

Lean-elaborated claim roles

1. parameter: $\mathbb{N}$
2. assumption: NeZero T
3. parameter: PRPKG24AccuracyDiversity.BernoulliSatisfactionModel T
4. parameter: PRPKG24AccuracyDiversity.SourcePreferenceLaw T
5. assumption: $\forall (t : \text{PRPKG24AccuracyDiversity.ItemType } T), B.\text{likelihood } t = (\text{preferenceLaw } t).\text{toReal}$
6. assumption: $\forall (t : \text{PRPKG24AccuracyDiversity.ItemType } T), 0 < B.\text{successProb } t$
7. assumption: $\forall (t : \text{PRPKG24AccuracyDiversity.ItemType } T), B.\text{successProb } t < 1$
8. assumption: $\forall (t : \text{PRPKG24AccuracyDiversity.ItemType } T), 0 < (\text{preferenceLaw } t).\text{toReal}$
9. parameter: PRPKG24AccuracyDiversity.OptimalAllocationSequence fun (x : ℕ) => B.toConsumptionModel
10. parameter: PRPKG24AccuracyDiversity.ItemType T
11. conclusion: Filter.Tendsto (fun (N : ℕ) => (seq.allocation N).representation t) Filter.atTop (nhds (PRPKG24AccuracyDiversity.theorem3LogShareWeight B t / ∑ i : PRPKG24AccuracyDiversity.ItemType T, PRPKG24AccuracyDiversity.theorem3LogShareWeight B i))

Lean proof endpoint (built separately)

PRPKG24AccuracyDiversity.PaperInterface.theorem3_log_share_conclusionSpec_proof

Recorded source-to-Spec screening Verdict: matches_approved_corrected_target

Reason: Approved corrected target: Lean retains the inverse-log share formula and makes the selected-type finite iid Bernoulli experiment, q_t in $(0,1)$, and strictly positive PMF support explicit for every type.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 14

Source locator: cited publication, lines 619

Verbatim source input

while $S_{n,n}$ contains only items of type $t = \arg \max_{t \in [m]} p_t q_t$.

[Next verbatim source excerpt]

There are m types of items indexed by $[m] = \{1, 2, \dots, m\}$. A user prefers exactly one type of item, preferring (t)
type $t \in [m]$ with probability p_t . The value of the i -th item of type t is a random variable X_i if the user (t)
(t)
prefers type t and 0 otherwise (so its expected value is $p_t E[X_i]$). The recommender knows only how X_i are distributed, not their realizations. We refer to p_t as the likelihood of a type and the random variable (t)
 X_i as a conditional item value (the value of an item conditional on the user preferring the item's type). Let $S_{n,k}$ be the set of n items that maximizes the expected total value of the k items with the highest realized values. We call $S_{n,k}$ the optimal set of n recommendations with respect to this objective. (We omit the dependency of $S_{n,k}$ on the other model parameters, as these will be clear from context.) $S_{n,k}$ arises naturally from an assumption that the user can only use k items, so that they derive value from the k highest value recommended items. So $S_{n,1}$ results from an assumption that the user uses only one item, while $S_{n,n}$ results from an assumption that the user uses every recommended item. $S_{n,n}$ maximizes the expected total value of all of the recommended items, so by linearity of expectation, $S_{n,n}$ contains the n items with the highest individual expected values.

[Next verbatim source excerpt]

Theorem 3 (Varying success probability across types). Suppose that for each fixed t , X_i i.i.d. Bernoulli random variables. Then

[Next verbatim source excerpt]

$\sim \text{Ber}(q_t)$ are

Expanded PaperInterface specification

```
∀ {T : ℕ} (B : PRPKG24AccuracyDiversity.BernoulliSatisfactionModel T)
  (preferenceLaw : PRPKG24AccuracyDiversity.SourcePreferenceLaw T),
  (∀ (t : PRPKG24AccuracyDiversity.ItemType T), B.likelihood t = (preferenceLaw t).toReal) →
  PRPKG24AccuracyDiversity.assumption_bernoulli_success_probability_range B →
  ∀ (N : ℕ) (best : PRPKG24AccuracyDiversity.ItemType T),
  (∀ (t : PRPKG24AccuracyDiversity.ItemType T),
    B.likelihood t * B.successProb t ≤ B.likelihood best * B.successProb best) →
  (PRPKG24AccuracyDiversity.bernoulliAllConsumedModel B).IsOptimalAtTotal N
  (PRPKG24AccuracyDiversity.allOnTypeAllocation N best)
```

Retained governing declarations

- [PRPKG24AccuracyDiversity.BernoulliSatisfactionModel](#)
- [PRPKG24AccuracyDiversity.ConsumptionModel.IsOptimalAtTotal](#)
- [PRPKG24AccuracyDiversity.ItemType](#)
- [PRPKG24AccuracyDiversity.SourcePreferenceLaw](#)
- [PRPKG24AccuracyDiversity.allOnTypeAllocation](#)
- [PRPKG24AccuracyDiversity.assumption_bernoulli_success_probability_range](#)
- [PRPKG24AccuracyDiversity.bernoulliAllConsumedModel](#)

Lean-elaborated claim roles

1. parameter: $\mathbb{N}$
2. parameter: $\text{PRPKG24AccuracyDiversity.BernoulliSatisfactionModel } T$
3. parameter: $\text{PRPKG24AccuracyDiversity.SourcePreferenceLaw } T$
4. assumption: $\forall (t : \text{PRPKG24AccuracyDiversity.ItemType } T), B.\text{likelihood } t = (\text{preferenceLaw } t).\text{toReal}$
5. assumption: $\text{PRPKG24AccuracyDiversity.assumption_bernoulli_success_probability_range } B$
6. parameter: $\mathbb{N}$
7. parameter: $\text{PRPKG24AccuracyDiversity.ItemType } T$
8. assumption: $\forall (t : \text{PRPKG24AccuracyDiversity.ItemType } T), B.\text{likelihood } t * B.\text{successProb } t \leq B.\text{likelihood best} * B.\text{successProb best}$
9. conclusion: $(\text{PRPKG24AccuracyDiversity.bernoulliAllConsumedModel } B).\text{IsOptimalAtTotal } N$
($\text{PRPKG24AccuracyDiversity.allOnTypeAllocation } N \text{ best}$)

Lean proof endpoint (built separately)

$\text{PRPKG24AccuracyDiversity.PaperInterface.theorem3_all_consumed_argmax_conclusionSpec_proof}$

Recorded source-to-Spec screening Verdict: matches

Reason: Matches: the source all-consumed clause selects a type maximizing $p_t q_t$; Lean proves that the corresponding all-on allocation is optimal under the Bernoulli probability range, without an unwarranted every-optimizer claim.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 15

Source locator: cited publication, lines 703-729

Verbatim source input

Lemma 1. If D has support bounded from above by M with pdf f_D such that $\lim_{x \rightarrow M} (Mf - x) = 1$ for some

$\beta, c > 0$, then $\lim_{a \rightarrow \infty} M k - h(a) = 1$.

Combining Lemma 1 with Lemma D.1(ii), with $\sigma = -\beta 1$, we show that for D as in Theorem 1(ii),

(n)

a

p $\beta + 1$

$\lim_{n \rightarrow \infty} t = P t$

β .

m

$\beta + 1$

i=1 π_i

5

(11)

Settled source reading Theorem 1(ii): an upper-bounded conditional value law $\rightarrow$ an upper-bounded law with non-negative values almost surely. The current allocation proof does not establish the translated negative-value domain.

Expanded PaperInterface specification

```
∀ {beta c M L : ℝ} {k : ℕ} (baseMeasure : MeasureTheory.Measure ℝ) [MeasureTheory.IsProbabilityMeasure baseMeasure],
  (∀m (y : ℝ) ∂baseMeasure, L ≤ y ∧ y ≤ M) →
  0 < M - L →
  ∀ (f : ℝ → ℝ),
  (baseMeasure = MeasureTheory.volume.withDensity fun (y : ℝ) => ENNReal.ofReal (f y)) →
  (∀ (y : ℝ), 0 ≤ f y) →
  Measurable f →
  MeasureTheory.Integrable f MeasureTheory.volume →
  0 < beta →
  0 < c →
  Filter.Tendsto (fun (u : ℝ) => f (M - u) / (c * u ^ (beta - 1))) (nhdsWithin 0 (Set.Ioi 0))
  (nhds 1) →
  0 < k →
  AppliedModelingLib.Math.AsymptoticEquivalent
  (fun (a : ℕ) =>
    ↑k * M -
    PRPKG24AccuracyDiversity.orderStatisticTopKSumFromMean
    (PRPKG24AccuracyDiversity.expectedOrderStatisticMeanSeq fun (a : ℕ) =>
      MeasureTheory.Measure.pi fun (x : Fin a) => baseMeasure)
    k a)
  fun (a : ℕ) =>
  (Σ q : PRPKG24AccuracyDiversity.BoundedLemmaD2Index k,
    PRPKG24AccuracyDiversity.boundedLemmaD2LimitCoeff beta c ↑q.snd) *
    PRPKG24AccuracyDiversity.boundedTailScale beta a
```

Retained governing declarations

- [AppliedModelingLib.Math.AsymptoticEquivalent](#)
- [PRPKG24AccuracyDiversity.BoundedLemmaD2Index](#)
- [PRPKG24AccuracyDiversity.boundedLemmaD2LimitCoeff](#)
- [PRPKG24AccuracyDiversity.boundedTailScale](#)
- [PRPKG24AccuracyDiversity.expectedOrderStatisticMeanSeq](#)
- [PRPKG24AccuracyDiversity.orderStatisticTopKSumFromMean](#)

Lean-elaborated claim roles

1. parameter: $\mathbb{R}$
2. parameter: $\mathbb{R}$
3. parameter: $\mathbb{R}$
4. parameter: $\mathbb{R}$
5. parameter: $\mathbb{N}$

```

6. parameter: MeasureTheory.Measure ℝ
7. assumption: MeasureTheory.IsProbabilityMeasure baseMeasure
8. assumption:  $\forall^m (y : \mathbb{R}) \partial \text{baseMeasure}, L \leq y \wedge y \leq M$ 
9. assumption:  $0 < M - L$ 
10. parameter:  $\mathbb{R} \rightarrow \mathbb{R}$ 
11. assumption: baseMeasure = MeasureTheory.volume.withDensity fun (y : ℝ) => ENNReal.ofReal (f y)
12. assumption:  $\forall (y : \mathbb{R}), 0 \leq f y$ 
13. assumption: Measurable f
14. assumption: MeasureTheory.Integrable f MeasureTheory.volume
15. assumption:  $0 < \beta$ 
16. assumption:  $0 < c$ 
17. assumption: Filter.Tendsto (fun (u : ℝ) => f (M - u) / (c * u ^ (beta - 1))) (nhdsWithin 0 (Set.Ioi 0)) (nhds 1)
18. assumption:  $0 < k$ 
19. conclusion: AppliedModelingLib.Math.AsymptoticEquivalent
  (fun (a : ℕ) =>
    ↑ k * M -
    PRPKG24AccuracyDiversity.orderStatisticTopKSumFromMean
    (PRPKG24AccuracyDiversity.expectedOrderStatisticMeanSeq fun (a : ℕ) =>
      MeasureTheory.Measure.pi fun (x : Fin a) => baseMeasure)
    k a)
  fun (a : ℕ) =>
    (Σ q : PRPKG24AccuracyDiversity.BoundedLemmaD2Index k,
      PRPKG24AccuracyDiversity.boundedLemmaD2LimitCoeff beta c ↑ q.snd) *
    PRPKG24AccuracyDiversity.boundedTailScale beta a

```

Lean proof endpoint (built separately)

PRPKG24AccuracyDiversity.PaperInterface.lemma1_bounded_topk_loss_asymptotic_of_pdf_sourceSpec_proof

Recorded source-to-Spec screening Verdict: matches

Reason: Matches: Lean makes the bounded PDF/support, positive tail parameters, and fixed positive k explicit and states the source top-k loss equivalent with the displayed fixed-rank coefficient and $a^{(-1/\beta)}$ scale.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 16

Source locator: cited publication, lines 1122-1145

Verbatim source input

Proposition 4. Consider a non-constant function $p(u, v) : S_d \times S_d \rightarrow (0, 1]$, interpreted as the probability that item v does not satisfy a user with preference u , such that $p(u, v) = q(\|u - v\|)$ can be expressed as a function of the cosine distance between u and v . Then let μ be a probability measure with support S_d , representing the distribution of user preferences. Then for a probability measure α on S_d , define

[U+0014 control character] Z

[U+0015 control character] Z

Z

1

log

$\mu(u) \exp n$

$\alpha(v) \log p(u, v) dv du.$

(17)

$\Gamma(\alpha) = \lim$

$n \rightarrow \infty n$

S_d

S_d

Then

$\Gamma(\pi) \in \inf \Gamma(\alpha),$

(18)

α

where π is the uniform probability measure over S_d .

Formalized review target The archival source above is not asserted equivalent to this formalized target.

On the source unit-sphere model, with a full-support preference probability measure and a continuous nonconstant radial kernel p with $0 < p \leq 1$ on $\rightarrow$ realized distances, the uniform probability profile π satisfies $\Gamma(\pi) \leq \Gamma(\alpha)$ for every probability profile α .

Archival source anchor: cited publication, lines 1122-1145

Expanded PaperInterface specification

```
∀ {E : Type u_1} [inst : NormedAddCommGroup E] [inst_1 : InnerProductSpace ℝ E] [inst_2 : MeasurableSpace E]
[inst_3 : BorelSpace E] [OpensMeasurableSpace (PRPKG24AccuracyDiversity.Proposition4Sphere.UnitSphere E)]
[inst_5 : FiniteDimensional ℝ E] [inst_6 : Nontrivial E]
(preferenceMeasure : MeasureTheory.Measure (PRPKG24AccuracyDiversity.Proposition4Sphere.UnitSphere E))
[MeasureTheory.IsProbabilityMeasure preferenceMeasure] (p : ℝ → ℝ),
preferenceMeasure.IsOpenPosMeasure →
Continuous p →
(∃ r ∈ Set.Icc 0 2, ∃ s ∈ Set.Icc 0 2, p r ≠ p s) →
(∀ r ∈ Set.Icc 0 2, 0 < p r ∧ p r ≤ 1) →
∀ (alpha : MeasureTheory.ProbabilityMeasure (PRPKG24AccuracyDiversity.Proposition4Sphere.UnitSphere E)),
PRPKG24AccuracyDiversity.Proposition4Sphere.logRadialDistanceProfileSupValue p
PRPKG24AccuracyDiversity.Proposition4Sphere.sphereVolumeUniformProbabilityMeasure ≤
PRPKG24AccuracyDiversity.Proposition4Sphere.logRadialDistanceProfileSupValue p alpha
```

Retained governing declarations

- [PRPKG24AccuracyDiversity.Proposition4Sphere.UnitSphere](#)
- [PRPKG24AccuracyDiversity.Proposition4Sphere.logRadialDistanceProfileSupValue](#)
- [PRPKG24AccuracyDiversity.Proposition4Sphere.sphereVolumeUniformProbabilityMeasure](#)

Lean-elaborated claim roles

1. parameter: Type u_1
2. parameter: NormedAddCommGroup E
3. parameter: InnerProductSpace ℝ E
4. parameter: MeasurableSpace E
5. assumption: BorelSpace E
6. assumption: OpensMeasurableSpace (PRPKG24AccuracyDiversity.Proposition4Sphere.UnitSphere E)
7. assumption: FiniteDimensional ℝ E
8. assumption: Nontrivial E
9. parameter: MeasureTheory.Measure (PRPKG24AccuracyDiversity.Proposition4Sphere.UnitSphere E)
10. assumption: MeasureTheory.IsProbabilityMeasure preferenceMeasure
11. parameter: ℝ → ℝ
12. assumption: preferenceMeasure.IsOpenPosMeasure
13. assumption: Continuous p
14. assumption: ∃ r ∈ Set.Icc 0 2, ∃ s ∈ Set.Icc 0 2, p r ≠ p s
15. assumption: ∀ r ∈ Set.Icc 0 2, 0 < p r ∧ p r ≤ 1
16. parameter: MeasureTheory.ProbabilityMeasure (PRPKG24AccuracyDiversity.Proposition4Sphere.UnitSphere E)
17. conclusion: PRPKG24AccuracyDiversity.Proposition4Sphere.logRadialDistanceProfileSupValue p
PRPKG24AccuracyDiversity.Proposition4Sphere.sphereVolumeUniformProbabilityMeasure ≤
PRPKG24AccuracyDiversity.Proposition4Sphere.logRadialDistanceProfileSupValue p alpha

Lean proof endpoint (built separately)

PRPKG24AccuracyDiversity.PaperInterface.proposition4Spec_proof

Recorded source-to-Spec screening Verdict: matches_approved_corrected_target

Reason: Approved corrected target: the intended $\Gamma(\pi) \leq \Gamma(\alpha)$ comparison is represented by the log-radial supremum inequality, with the approved continuous radial kernel and full-support preference law.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 17

Source locator: cited publication, lines 1298-1322

Verbatim source input

(i) Suppose there exist constants $A, B > 0$ and $\sigma < 0$ such that

lim

$a \rightarrow \infty$

$$\frac{\log(A - h(a))}{Ba^\sigma} = 1.$$

Then
(n)

lim r_t

$n \rightarrow \infty$

=

$\frac{1}{m}$

(29)

(30)

[Next verbatim source excerpt]

Lemma D.1. Consider an integer m and $p_1, p_2, \dots, p_m \geq 0$. Let $h : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{R}$ be monotonically increasing.

(n)

(n)

For each positive integer n , choose $(a_1, \dots, a_m)$ such that

(n)

$$(a_1, \dots, a_m) \in \mathcal{X}$$

arg max

$\sum_{t=1}^m$

$p_t h(a_t),$

(27)

$(a_1, \dots, a_m) \in \mathcal{A}_n, t=1$

and define
(n)

r_t

(n)

a_t
 $\cdot$
 n

:=

(28)

Then the following statements hold.

Formalized review target The archival source above is not asserted equivalent to this formalized target.

For a nonempty finite type space with every weight $p_t > 0$, let $a^{(n)}$ be an exact fixed-total integer maximizer of $\sum_t p_t h(a_t)$. If h is monotone, $B < 0$, $\sigma > 0$, $A - h(a)$ is eventually positive, and $\log(A - h(a)) / (B * a^\sigma)$ tends to 1 as a tends to infinity, then $a_t^{(n)} / n$ tends to $1 / m$ for every t .

Archival source anchor: cited publication, lines 1298-1322

Expanded PaperInterface specification

```
∀ {m : ℕ} [NeZero m] {A B sigma : ℝ} {h : ℕ → ℝ} (p : PRPKG24AccuracyDiversity.ItemType m → ℝ)
  (seq :
    AppliedModelingLib.Allocation.OptimalSequence (fun (x : ℕ) => p)
      fun (x : ℕ) (x_1 : PRPKG24AccuracyDiversity.ItemType m) => h),
  (∀ (i : PRPKG24AccuracyDiversity.ItemType m), 0 < p i) →
  Monotone h →
  B < 0 →
  0 < sigma →
  (∀ (n : ℕ) in Filter.atTop, 0 < PRPKG24AccuracyDiversity.AppendixD1Generic1.saturationGap A h n) →
```

```

Filter.Tendsto
  (fun (n : ℕ) =>
    Real.log (PRPKG24AccuracyDiversity.AppendixD1GenericI.saturationGap A h n) / (B * ↑n ^ sigma))
  Filter.atTop (nhds 1) →
seq.toSequence.ConvergesToProfile fun (x : PRPKG24AccuracyDiversity.ItemType m) => 1 / ↑m

```

Retained governing declarations

- [AppliedModelingLib.Allocation.OptimalSequence](#)
- [AppliedModelingLib.Allocation.OptimalSequence.toSequence](#)
- [AppliedModelingLib.Allocation.Sequence.ConvergesToProfile](#)
- [PRPKG24AccuracyDiversity.AppendixD1GenericI.saturationGap](#)
- [PRPKG24AccuracyDiversity.ItemType](#)

Lean-elaborated claim roles

1. parameter: $\mathbb{N}$
2. assumption: NeZero m
3. parameter: $\mathbb{R}$
4. parameter: $\mathbb{R}$
5. parameter: $\mathbb{R}$
6. parameter: $\mathbb{N} \rightarrow \mathbb{R}$
7. parameter: PRPKG24AccuracyDiversity.ItemType m $\rightarrow \mathbb{R}$
8. parameter: AppliedModelingLib.Allocation.OptimalSequence (fun (x : ℕ) => p)
fun (x : ℕ) (x_1 : PRPKG24AccuracyDiversity.ItemType m) => h
9. assumption: $\forall (i : \text{PRPKG24AccuracyDiversity.ItemType } m), 0 < p \ i$
10. assumption: Monotone h
11. assumption: $B < 0$
12. assumption: $0 < \text{sigma}$
13. assumption: $\forall^* (n : \mathbb{N}) \text{ in } \text{Filter.atTop}, 0 < \text{PRPKG24AccuracyDiversity.AppendixD1GenericI.saturationGap } A \ h \ n$
14. assumption: Filter.Tendsto
(fun (n : ℕ) => Real.log (PRPKG24AccuracyDiversity.AppendixD1GenericI.saturationGap A h n) / (B * ↑n ^ sigma))
Filter.atTop (nhds 1)
15. conclusion: seq.toSequence.ConvergesToProfile fun (x : PRPKG24AccuracyDiversity.ItemType m) => 1 / ↑m

Lean proof endpoint (built separately)

PRPKG24AccuracyDiversity.PaperInterface.lemmaD1_i_corrected_uniform_optimizer_shares_sourceSpec_proof

Recorded source-to-Spec screening Verdict: matches_approved_corrected_target

Reason: Approved corrected target: Lean has $B < 0$, $\text{sigma} > 0$, eventual positive saturation gap, and the corrected log-gap limit, yielding the retained uniform-share conclusion.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 18

Source locator: cited publication, lines 1324-1350

Verbatim source input

(ii) Suppose there exist constants $A, B > 0$ and $\sigma < 0$ such that

$$\lim_{a \rightarrow \infty}$$

$$\frac{A - h(a)}{Ba^\sigma}$$

Then

$$(31)$$

$$\frac{1}{1 - \sigma}$$

$$\lim_{n \rightarrow \infty} \frac{p(n)}{m} = P$$

$$\lim_{n \rightarrow \infty} \frac{1}{m}$$

$$\lim_{i \rightarrow \infty} \frac{1}{p_i}$$

$$(32)$$

[Next verbatim source excerpt]

Lemma D.1. Consider an integer m and $p_1, p_2, \dots, p_m \geq 0$. Let $h : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{R}$ be monotonically increasing.

For each positive integer n , choose $(a_1, \dots, a_m)$ such that

$$(a_1, \dots, a_m) \in \mathbb{N}^m$$

arg max

$$\sum_{t=1}^m$$

$$p_t h(a_t),$$

$$(27)$$

$$(a_1, \dots, a_m) \in \mathbb{N}^m$$

and define

$$r_t$$

$$(n)$$

$$a_t$$

$$\cdot$$

$$:=$$

$$(28)$$

Then the following statements hold.

Formalized review target The archival source above is not asserted equivalent to this formalized target.

For a nonempty finite type space with every weight $p_t > 0$, let $a^{(n)}$ be an exact fixed-total integer maximizer of $\sum_t p_t h(a_t)$. If h is monotone, $B > 0$, $\sigma < 0$, and $(A - h(a)) / (B * a^\sigma)$ tends to 1 as a tends to infinity, then $a^{(n)} / n$ tends to $p^{(1 / (1 - \sigma))} / \sum_i p_i^{(1 / (1 - \sigma))}$ for every t .

Archival source anchor: cited publication, lines 1324-1350

Expanded PaperInterface specification

$\forall \{m : \mathbb{N}\} [\text{NeZero } m] \{A \ B \ \sigma : \mathbb{R}\} \{h : \mathbb{N} \rightarrow \mathbb{R}\} (p : \text{PRPKG24AccuracyDiversity.ItemType } m \rightarrow \mathbb{R})$
(seq :
 AppliedModelingLib.Allocation.OptimalSequence (fun (x : $\mathbb{N}$) => p)
 fun (x : $\mathbb{N}$) (x_1 : PRPKG24AccuracyDiversity.ItemType m) => h),
($\forall (i : \text{PRPKG24AccuracyDiversity.ItemType } m), 0 < p_i \rightarrow$
 Monotone h $\rightarrow$
 $0 < B \rightarrow$

```

sigma < 0 →
  Filter.Tendsto
    (fun (n : ℕ) => PRPKG24AccuracyDiversity.AppendixD1GenericII.saturationGap A h n / (B * ↑n ^ sigma))
    Filter.atTop (nhds 1) →
      seq.toSequence.ConvergesToProfile (PRPKG24AccuracyDiversity.AppendixD1GenericIIFull.targetShare p sigma)

```

Retained governing declarations

- [AppliedModelingLib.Allocation.OptimalSequence](#)
- [AppliedModelingLib.Allocation.OptimalSequence.toSequence](#)
- [AppliedModelingLib.Allocation.Sequence.ConvergesToProfile](#)
- [PRPKG24AccuracyDiversity.AppendixD1GenericII.saturationGap](#)
- [PRPKG24AccuracyDiversity.AppendixD1GenericIIFull.targetShare](#)
- [PRPKG24AccuracyDiversity.ItemType](#)

Lean-elaborated claim roles

1. parameter: $\mathbb{N}$
2. assumption: $\text{NeZero } m$
3. parameter: $\mathbb{R}$
4. parameter: $\mathbb{R}$
5. parameter: $\mathbb{R}$
6. parameter: $\mathbb{N} \rightarrow \mathbb{R}$
7. parameter: $\text{PRPKG24AccuracyDiversity.ItemType } m \rightarrow \mathbb{R}$
8. parameter: $\text{AppliedModelingLib.Allocation.OptimalSequence } (\text{fun } (x : \mathbb{N}) \Rightarrow p)$
 $\text{fun } (x : \mathbb{N}) (x_1 : \text{PRPKG24AccuracyDiversity.ItemType } m) \Rightarrow h$
9. assumption: $\forall (i : \text{PRPKG24AccuracyDiversity.ItemType } m), 0 < p \ i$
10. assumption: $\text{Monotone } h$
11. assumption: $0 < B$
12. assumption: $\text{sigma} < 0$
13. assumption: $\text{Filter.Tendsto } (\text{fun } (n : \mathbb{N}) \Rightarrow \text{PRPKG24AccuracyDiversity.AppendixD1GenericII.saturationGap } A \ h \ n / (B * \uparrow n ^ \text{sigma}))$
 $\text{Filter.atTop } (\text{nhds } 1)$
14. conclusion: $\text{seq.toSequence.ConvergesToProfile } (\text{PRPKG24AccuracyDiversity.AppendixD1GenericIIFull.targetShare } p \ \text{sigma})$

Lean proof endpoint (built separately)

`PRPKG24AccuracyDiversity.PaperInterface.lemmaD1_ii_corrected_powerTail_optimizer_shares_sourceSpec_proof`

Recorded source-to-Spec screening Verdict: matches approved corrected target

Reason: Approved corrected target: all-positive weights, monotonicity, $B > 0$, $\text{sigma} < 0$, and the saturation-gap power limit give exactly the normalized $p^{1/(1-\text{sigma})}$ profile.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 19

Source locator: cited publication, lines 1352-1369

Verbatim source input

(iii) Suppose h is strictly concave, and that there exist constants $B, C > 0$ such that $\lim_{a \rightarrow \infty} h(a) - B \log a - C = 0$.

(33)

(n)

(34)

$a \rightarrow \infty$

Then

$\lim_{n \rightarrow \infty} r_t$

$n \rightarrow \infty$

$= p_t$.

[Next verbatim source excerpt]

Lemma D.1. Consider an integer m and $p_1, p_2, \dots, p_m \geq 0$. Let $h : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{R}$ be monotonically increasing.

(n)

(n)

For each positive integer n , choose $(a_1, \dots, a_m)$ such that

(n)

$(a_1, \dots, a_m) \in$

$\arg \max$

$\sum_{t=1}^m$

$p_t h(a_t),$

(27)

(27)

$(a_1, \dots, a_m) \in \mathcal{A}_n$ $t=1$

and define

(n)

r_t

(n)

a_t

$\cdot$

$:=$

(28)

Then the following statements hold.

[Next verbatim source excerpt]

Let $\mathbb{Z}_{\geq 0}$ denote the set of non-negative integers and $\mathcal{A}_n \subset \mathbb{Z}^m$

≥ 0 denote the set of m -tuples whose elements

sum to n . We will say that a function $h : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{R}$ is strictly concave if $h(a+1) - h(a) < h(a) - h(a-1)$

for all a .

Formalized review target The archival source above is not asserted equivalent to this formalized target.

For a nonempty finite type space with every weight $p_t > 0$ and $\sum_t p_t = 1$, let $a^{(n)}$ be an exact fixed-total integer maximizer of $\sum_t p_t h(a_t)$. If h is monotone and strictly discretely concave, $B > 0$, and $h(a) - B \log(a) - C$ tends to 0 as a tends to infinity, then $a^{(n)} / n$ tends to p_t for every t .

Archival source anchor: cited publication, lines 1352-1369

Expanded PaperInterface specification

```
∀ {m : ℕ} [NeZero m] {B C : ℝ} {h : ℕ → ℝ} (p : PRPKG24AccuracyDiversity.ItemType m → ℝ)
  (seq :
    AppliedModelingLib.Allocation.OptimalSequence (fun (x : ℕ) => p)
      fun (x : ℕ) (x_1 : PRPKG24AccuracyDiversity.ItemType m) => h),
  (∀ (i : PRPKG24AccuracyDiversity.ItemType m), 0 < p i) →
  Σ i : PRPKG24AccuracyDiversity.ItemType m, p i = 1 →
  Monotone h →
  PRPKG24AccuracyDiversity.AppendixD1GenericIII.StrictDiscreteConcave h →
  0 < B →
  Filter.Tendsto (PRPKG24AccuracyDiversity.AppendixD1GenericIII.logRemainder h B C) Filter.atTop (nhds 0) →
  seq.toSequence.ConvergesToProfile p
```

Retained governing declarations

- [AppliedModelingLib.Allocation.OptimalSequence](#)
- [AppliedModelingLib.Allocation.OptimalSequence.toSequence](#)
- [AppliedModelingLib.Allocation.Sequence.ConvergesToProfile](#)
- [PRPKG24AccuracyDiversity.AppendixD1GenericIII.StrictDiscreteConcave](#)
- [PRPKG24AccuracyDiversity.AppendixD1GenericIII.logRemainder](#)
- [PRPKG24AccuracyDiversity.ItemType](#)

Lean-elaborated claim roles

```
1. parameter:  $\mathbb{N}$ 
2. assumption:  $\text{NeZero } m$ 
3. parameter:  $\mathbb{R}$ 
4. parameter:  $\mathbb{R}$ 
5. parameter:  $\mathbb{N} \rightarrow \mathbb{R}$ 
6. parameter:  $\text{PRPKG24AccuracyDiversity.ItemType } m \rightarrow \mathbb{R}$ 
7. parameter:  $\text{AppliedModelingLib.Allocation.OptimalSequence } (\text{fun } (x : \mathbb{N}) \Rightarrow p)$ 
   fun (x :  $\mathbb{N}$ ) (x_1 :  $\text{PRPKG24AccuracyDiversity.ItemType } m$ ) => h
8. assumption:  $\forall (i : \text{PRPKG24AccuracyDiversity.ItemType } m), 0 < p \ i$ 
9. assumption:  $\sum i : \text{PRPKG24AccuracyDiversity.ItemType } m, p \ i = 1$ 
10. assumption:  $\text{Monotone } h$ 
11. assumption:  $\text{PRPKG24AccuracyDiversity.AppendixD1GenericIII.StrictDiscreteConcave } h$ 
12. assumption:  $0 < B$ 
13. assumption:  $\text{Filter.Tendsto } (\text{PRPKG24AccuracyDiversity.AppendixD1GenericIII.logRemainder } h \ B \ C) \ \text{Filter.atTop } (nhds \ 0)$ 
14. conclusion:  $\text{seq.toSequence.ConvergesToProfile } p$ 
```

Lean proof endpoint (built separately)

`PRPKG24AccuracyDiversity.PaperInterface.lemmaD1_iii_corrected_probability_optimizer_shares_sourceSpec_proof`

Recorded source-to-Spec screening Verdict: `matches_approved_corrected_target`

Reason: Approved corrected target: Lean supplies positive probability weights summing to one, monotonicity and strict discrete concavity, $B > 0$, and vanishing logarithmic remainder, with limiting profile p .

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 20

Source locator: cited publication, lines 1371-1399

Verbatim source input

(iv) Suppose h is strictly concave, and that there exist constants $B > 0$ and $0 < \sigma < 1$ such that

$$\lim_{a \rightarrow \infty} \frac{h(a)}{a} = B\sigma$$

$$h(a) = 1.$$

$$(35)$$

$$\frac{1}{1-\sigma}$$

$$\lim_{n \rightarrow \infty} \frac{p(n)}{m} = P$$

$$\frac{1}{1-\sigma}$$

$$\lim_{n \rightarrow \infty} \frac{p(n)}{m} = P$$

$$\frac{1}{1-\sigma}$$

$$\frac{p}{i=1}^m$$

$$\frac{1}{1-\sigma}$$

$$(36)$$

[Next verbatim source excerpt]

Lemma D.1. Consider an integer m and $p_1, p_2, \dots, p_m \geq 0$. Let $h : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{R}$ be monotonically increasing.

For each positive integer n , choose $(a_1, \dots, a_m)$ such that

$$(a_1, \dots, a_m) \in \arg \max_{(a_1, \dots, a_m) \in \mathbb{Z}_{\geq 0}^m} \sum_{i=1}^m p_i h(a_i)$$

$$\arg \max_{(a_1, \dots, a_m) \in \mathbb{Z}_{\geq 0}^m} \sum_{i=1}^m p_i h(a_i)$$

$$\sum_{i=1}^m p_i h(a_i)$$

$$p_i h(a_i),$$

$$(27)$$

$$(a_1, \dots, a_m) \in \mathbb{Z}_{\geq 0}^m$$

$$\text{and define}$$

$$r_t$$

$$(n)$$

$$a_t$$

$$\frac{1}{n}$$

$$:=$$

$$(28)$$

Then the following statements hold.

[Next verbatim source excerpt]

Let $\mathbb{Z}_{\geq 0}$ denote the set of non-negative integers and $\mathbb{A}_n \subset \mathbb{Z}_{\geq 0}^m$ denote the set of m -tuples whose elements sum to n . We will say that a function $h : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{R}$ is strictly concave if $h(a+1) - h(a) < h(a) - h(a-1)$ for all a .

Formalized review target The archival source above is not asserted equivalent to this formalized target.

For a nonempty finite type space with every weight $p_t > 0$, let $a^{(n)}$ be an exact fixed-total integer maximizer of $\sum_t p_t h(a_t)$. If h is strictly discretely concave, $B > 0$, $0 < \sigma < 1$, and $h(a) / (B * a^\sigma)$ tends to 1 as a tends to infinity, then $a^{(n)} / n$ tends to $p^{(1)} / (1 - \sigma) / \sum_i p_i^{(1)} / (1 - \sigma)$ for every t .

Archival source anchor: cited publication, lines 1371-1399

Expanded PaperInterface specification

```

∀ {m : ℕ} [inst : NeZero m] {B sigma : ℝ} {h : ℕ → ℝ} (p : Fin m → ℝ)
(seq : AppliedModelingLib.Allocation.OptimalSequence (fun (x : ℕ) => p) fun (x : ℕ) (x_1 : Fin m) => h),
(∀ (i : Fin m), 0 < p i) →
0 < B →
0 < sigma →
sigma < 1 →
PRPKG24AccuracyDiversity.AppendixD1GenericIV.StrictDiscreteConcave h →
Filter.Tendsto (PRPKG24AccuracyDiversity.AppendixD1GenericIV.powerTailQuotient h B sigma) Filter.atTop
(nhds 1) →
seq.toSequence.ConvergesToProfile (PRPKG24AccuracyDiversity.AppendixD1GenericIV.targetShare p sigma)

```

Retained governing declarations

- [AppliedModelingLib.Allocation.OptimalSequence](#)
- [AppliedModelingLib.Allocation.OptimalSequence.toSequence](#)
- [AppliedModelingLib.Allocation.Sequence.ConvergesToProfile](#)
- [PRPKG24AccuracyDiversity.AppendixD1GenericIV.StrictDiscreteConcave](#)
- [PRPKG24AccuracyDiversity.AppendixD1GenericIV.powerTailQuotient](#)
- [PRPKG24AccuracyDiversity.AppendixD1GenericIV.targetShare](#)

Lean-elaborated claim roles

```

1. parameter: ℕ
2. assumption: NeZero m
3. parameter: ℝ
4. parameter: ℝ
5. parameter: ℕ → ℝ
6. parameter: Fin m → ℝ
7. parameter: AppliedModelingLib.Allocation.OptimalSequence (fun (x : ℕ) => p) fun (x : ℕ) (x_1 : Fin m) => h
8. assumption: ∀ (i : Fin m), 0 < p i
9. assumption: 0 < B
10. assumption: 0 < sigma
11. assumption: sigma < 1
12. assumption: PRPKG24AccuracyDiversity.AppendixD1GenericIV.StrictDiscreteConcave h
13. assumption: Filter.Tendsto (PRPKG24AccuracyDiversity.AppendixD1GenericIV.powerTailQuotient h B sigma) Filter.atTop (nhds 1)
14. conclusion: seq.toSequence.ConvergesToProfile (PRPKG24AccuracyDiversity.AppendixD1GenericIV.targetShare p sigma)

```

Lean proof endpoint (built separately)

PRPKG24AccuracyDiversity.PaperInterface.lemmaD1_iv_corrected_powerTail_optimizer_shares_sourceSpec_proof

Recorded source-to-Spec screening Verdict: matches_approved_corrected_target

Reason: Approved corrected target: Lean records positive weights, monotonicity and strict discrete concavity, $B > 0$ and $0 < \sigma < 1$, and the power quotient leading to the approved power-share profile.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 21

Source locator: cited publication, lines 2264-2287

Verbatim source input

Lemma D.2. For $\beta > 0$,

$Z \propto [U+0012 \text{ control character}] [U+0013 \text{ control character}]$

$$\int_0^a GX(x)^j (1 - GX(x))^{a-j} dx \propto a - \beta.$$

(97)

$$\begin{aligned} & fX(x) \\ & gX(x) \\ & = \lim_{x \rightarrow M} - \\ & = 1 \\ & \lim_{x \rightarrow M} c(M - x)^{\beta-1} \\ & x \rightarrow 0+ \quad cx^{\beta-1} \end{aligned}$$

(98)

$$(1 - \epsilon)cx^{\beta-1} \leq gX(x) \leq (1 + \epsilon)cx^{\beta-1}$$

(99)

Settled source reading Theorem 1(ii): an upper-bounded conditional value law $\rightarrow$ an upper-bounded law with non-negative values almost surely. The current allocation proof does not establish the translated negative-value domain.

Expanded PaperInterface specification

```
∀ {beta c M L : ℝ} {j : ℕ} (baseMeasure : MeasureTheory.Measure ℝ) [MeasureTheory.IsProbabilityMeasure baseMeasure],
  (∀m (y : ℝ) ∂baseMeasure, L ≤ y ∧ y ≤ M) →
    0 < M - L →
      ∀ (f : ℝ → ℝ),
        (baseMeasure = MeasureTheory.volume.withDensity fun (y : ℝ) => ENNReal.ofReal (f y)) →
          (∀ (y : ℝ), 0 ≤ f y) →
            Measurable f →
              MeasureTheory.Integrable f MeasureTheory.volume →
                0 < beta →
                  0 < c →
                    Filter.Tendsto (fun (u : ℝ) => f (M - u) / (c * u ^ (beta - 1))) (nhdsWithin 0 (Set.Ioi 0))
                      (nhds 1) →
                        AppliedModelingLib.Math.AsymptoticEquivalent
                          (PRPKG24AccuracyDiversity.boundedLemmaD2IntegralTerm
                            (AppliedModelingLib.Probability.reflectedCDFMass baseMeasure M) j)
                            fun (a : ℕ) =>
                              PRPKG24AccuracyDiversity.boundedLemmaD2LimitCoeff beta c j *
                                PRPKG24AccuracyDiversity.boundedTailScale beta a
```

Retained governing declarations

- [AppliedModelingLib.Math.AsymptoticEquivalent](#)
- [AppliedModelingLib.Probability.reflectedCDFMass](#)
- [PRPKG24AccuracyDiversity.boundedLemmaD2IntegralTerm](#)
- [PRPKG24AccuracyDiversity.boundedLemmaD2LimitCoeff](#)
- [PRPKG24AccuracyDiversity.boundedTailScale](#)

Lean-elaborated claim roles

1. parameter: $\mathbb{R}$
2. parameter: $\mathbb{R}$
3. parameter: $\mathbb{R}$
4. parameter: $\mathbb{R}$
5. parameter: $\mathbb{N}$
6. parameter: $\text{MeasureTheory.Measure } \mathbb{R}$
7. assumption: $\text{MeasureTheory.IsProbabilityMeasure baseMeasure}$
8. assumption: $\forall^m (y : \mathbb{R}) \partial \text{baseMeasure}, L \leq y \wedge y \leq M$
9. assumption: $0 < M - L$
10. parameter: $\mathbb{R} \rightarrow \mathbb{R}$
11. assumption: $\text{baseMeasure} = \text{MeasureTheory.volume.withDensity fun (y : } \mathbb{R}) \Rightarrow \text{ENNReal.ofReal (f y)}$
12. assumption: $\forall (y : \mathbb{R}), 0 \leq f y$
13. assumption: Measurable f
14. assumption: $\text{MeasureTheory.Integrable f MeasureTheory.volume}$
15. assumption: $0 < \text{beta}$
16. assumption: $0 < c$
17. assumption: $\text{Filter.Tendsto (fun (u : } \mathbb{R}) \Rightarrow f (M - u) / (c * u ^ (\text{beta} - 1))) (\text{nhdsWithin } 0 (\text{Set.Ioi } 0)) (\text{nhds } 1)$

```

18. conclusion: AppliedModelingLib.Math.AsymptoticEquivalent
   (PRPKG24AccuracyDiversity.boundedLemmaD2IntegralTerm (AppliedModelingLib.Probability.reflectedCDFMass baseMeasure M)
    j)
   fun (a : ℕ) =>
   PRPKG24AccuracyDiversity.boundedLemmaD2LimitCoeff beta c j * PRPKG24AccuracyDiversity.boundedTailScale beta a

```

Lean proof endpoint (built separately)

PRPKG24AccuracyDiversity.PaperInterface.lemmaD2_bounded_fixed_rank_integral_asymptotic_of_pdf_sourceSpec_proof

Recorded source-to-Spec screening Verdict: matches

Reason: Matches: the target renders the source bounded-density and reflected-CDF fixed-rank integral under a literal PDF law, with its rank coefficient and $a^{(-1/\beta)}$ asymptotic scale.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 22

Source locator: cited publication, lines 2778-2785

Verbatim source input

Lemma D.3. For D an exponential distribution with rate parameter λ , so that $f_X(x) = \lambda e^{-\lambda x}$ for $\lambda > 0$,
 $\lim_{\mu D(a-i, a)} -\log a - B(j) = 0$

$a \rightarrow \infty$

(127)

for a constant $B(j) > 0$. Moreover, $\mu D(a-i, a)$ is strictly concave.

Formalized review target The archival source above is not asserted equivalent to this formalized target.

For an iid exponential law with rate $\lambda > 0$ and each fixed valid upper rank r , $\mu(q-r, q) = (H_q - H_r)/\lambda$ for $q > r$; after subtracting $(\log q)/\lambda$ it
 $\hookrightarrow$ tends to $(\text{EulerMascheroniConstant} - H_r)/\lambda$, and the fixed-rank forward marginal is eventually strictly diminishing on valid ranks.

Archival source anchor: cited publication, lines 2778-2785

Expanded PaperInterface specification

```
∀ (lambda : ℝ) (hlambda_pos : 0 < lambda) (r : ℕ),
  (∀ (q : ℕ),
    r < q →
      PRPKG24AccuracyDiversity.expectedOrderStatisticMeanSeq
        (fun (a : ℕ) =>
          (PRPKG24AccuracyDiversity.exponentialDistributionModel lambda hlambda_pos).iidProductMeasure a)
        (q - r) q =
          1 / lambda * (PRPKG24AccuracyDiversity.harmonicReal q - PRPKG24AccuracyDiversity.harmonicReal r)) ∧
    Filter.Tendsto
      (fun (q : ℕ) =>
        PRPKG24AccuracyDiversity.expectedOrderStatisticMeanSeq
          (fun (a : ℕ) =>
            (PRPKG24AccuracyDiversity.exponentialDistributionModel lambda hlambda_pos).iidProductMeasure a)
          (q - r) q -
            1 / lambda * Real.log ↑q)
        Filter.atTop (nhds (1 / lambda * (Real.eulerMascheroniConstant - PRPKG24AccuracyDiversity.harmonicReal r)))) ∧
    ∀r (q : ℕ) in Filter.atTop,
      PRPKG24AccuracyDiversity.expectedOrderStatisticMeanSeq
        (fun (a : ℕ) =>
          (PRPKG24AccuracyDiversity.exponentialDistributionModel lambda hlambda_pos).iidProductMeasure a)
        (q + 2 - r) (q + 2) -
          PRPKG24AccuracyDiversity.expectedOrderStatisticMeanSeq
            (fun (a : ℕ) =>
              (PRPKG24AccuracyDiversity.exponentialDistributionModel lambda hlambda_pos).iidProductMeasure a)
            (q + 1 - r) (q + 1) <
          PRPKG24AccuracyDiversity.expectedOrderStatisticMeanSeq
            (fun (a : ℕ) =>
              (PRPKG24AccuracyDiversity.exponentialDistributionModel lambda hlambda_pos).iidProductMeasure a)
            (q + 1 - r) (q + 1) -
          PRPKG24AccuracyDiversity.expectedOrderStatisticMeanSeq
            (fun (a : ℕ) =>
              (PRPKG24AccuracyDiversity.exponentialDistributionModel lambda hlambda_pos).iidProductMeasure a)
            (q - r) q
```

Retained governing declarations

- [AppliedModelingLib.Probability.Exponential.Model.iidProductMeasure](#)
- [PRPKG24AccuracyDiversity.expectedOrderStatisticMeanSeq](#)
- [PRPKG24AccuracyDiversity.exponentialDistributionModel](#)
- [PRPKG24AccuracyDiversity.harmonicReal](#)

Lean-elaborated claim roles

```
1. parameter: ℝ
2. assumption: 0 < lambda
3. parameter: ℕ
4. conclusion: (∀ (q : ℕ),
  r < q →
    PRPKG24AccuracyDiversity.expectedOrderStatisticMeanSeq
      (fun (a : ℕ) =>
        (PRPKG24AccuracyDiversity.exponentialDistributionModel lambda hlambda_pos).iidProductMeasure a)
      (q - r) q =
        1 / lambda * (PRPKG24AccuracyDiversity.harmonicReal q - PRPKG24AccuracyDiversity.harmonicReal r)) ∧
    Filter.Tendsto
      (fun (q : ℕ) =>
        PRPKG24AccuracyDiversity.expectedOrderStatisticMeanSeq
          (fun (a : ℕ) =>
            (PRPKG24AccuracyDiversity.exponentialDistributionModel lambda hlambda_pos).iidProductMeasure a)
```

```

(q - r) q -
1 / lambda * Real.log ↑q)
Filter.atTop (nhds (1 / lambda * (Real.eulerMascheroniConstant - PRPKG24AccuracyDiversity.harmonicReal r))) ∧
∀! (q : ℕ) in Filter.atTop,
PRPKG24AccuracyDiversity.expectedOrderStatisticMeanSeq
(fun (a : ℕ) =>
  (PRPKG24AccuracyDiversity.exponentialDistributionModel lambda hlambda_pos).iidProductMeasure a)
(q + 2 - r) (q + 2) -
PRPKG24AccuracyDiversity.expectedOrderStatisticMeanSeq
(fun (a : ℕ) =>
  (PRPKG24AccuracyDiversity.exponentialDistributionModel lambda hlambda_pos).iidProductMeasure a)
(q + 1 - r) (q + 1) <
PRPKG24AccuracyDiversity.expectedOrderStatisticMeanSeq
(fun (a : ℕ) =>
  (PRPKG24AccuracyDiversity.exponentialDistributionModel lambda hlambda_pos).iidProductMeasure a)
(q + 1 - r) (q + 1) -
PRPKG24AccuracyDiversity.expectedOrderStatisticMeanSeq
(fun (a : ℕ) =>
  (PRPKG24AccuracyDiversity.exponentialDistributionModel lambda hlambda_pos).iidProductMeasure a)
(q - r) q

```

Lean proof endpoint (built separately)

PRPKG24AccuracyDiversity.PaperInterface.lemmaD3_corrected_exponential_fixed_rank_sourceSpec_proof

Recorded source-to-Spec screening Verdict: matches approved_corrected target

Reason: Approved corrected target: Lean includes the exact $1/\lambda$ harmonic identity, (EulerGamma-H_r)/lambda logarithmic constant, valid fixed ranks, and eventual strict marginal decrease.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 23

Source locator: cited publication, lines 2844-2858

Verbatim source input

Lemma D.4. For D a Pareto distribution with pdf $f_X(x) = x^{-\alpha-1}$ for $\alpha > 1$,
lim

at $\rightarrow \infty$

$\mu_D(a - i, a)$
1

a^α

$= C$

(132)

for a constant $C > 0$. Moreover, $\mu_D(a - i, a)$ is strictly concave.

Formalized review target The archival source above is not asserted equivalent to this formalized target.

For the normalized iid Pareto law with density $\alpha x^{-(\alpha+1)}$ on its Pareto support, $\alpha > 1$, and each fixed valid upper rank, the order-statistic
↪ mean has the source power asymptotic with its corrected rank-dependent coefficient and its forward marginal is eventually strictly diminishing on
↪ valid ranks.

Archival source anchor: cited publication, lines 2844-2858

Expanded PaperInterface specification

```
∀ {alpha : ℝ},  
  1 < alpha →  
    ∀ (r : ℕ),  
      (AppliedModelingLib.Math.AsymptoticEquivalent  
        (fun (q : ℕ) =>  
          PRPKG24AccuracyDiversity.expectedOrderStatisticMeanSeq  
            (PRPKG24AccuracyDiversity.paretolidSampleMeasure alpha) (q - r) q)  
        fun (q : ℕ) => PRPKG24AccuracyDiversity.paretoRankValueCoeff alpha r * ↑q ^ (1 / alpha)) ∧  
      ∀* (q : ℕ) in Filter.atTop,  
        PRPKG24AccuracyDiversity.expectedOrderStatisticMeanSeq (PRPKG24AccuracyDiversity.paretolidSampleMeasure alpha)  
          (q + 2 - r) (q + 2) -  
        PRPKG24AccuracyDiversity.expectedOrderStatisticMeanSeq  
          (PRPKG24AccuracyDiversity.paretolidSampleMeasure alpha) (q + 1 - r) (q + 1) <  
        PRPKG24AccuracyDiversity.expectedOrderStatisticMeanSeq  
          (PRPKG24AccuracyDiversity.paretolidSampleMeasure alpha) (q + 1 - r) (q + 1) -  
        PRPKG24AccuracyDiversity.expectedOrderStatisticMeanSeq  
          (PRPKG24AccuracyDiversity.paretolidSampleMeasure alpha) (q - r) q
```

Retained governing declarations

- [AppliedModelingLib.Math.AsymptoticEquivalent](#)
- [PRPKG24AccuracyDiversity.expectedOrderStatisticMeanSeq](#)
- [PRPKG24AccuracyDiversity.paretolidSampleMeasure](#)
- [PRPKG24AccuracyDiversity.paretoRankValueCoeff](#)

Lean-elaborated claim roles

```
1. parameter: ℝ  
2. assumption: 1 < alpha  
3. parameter: ℕ  
4. conclusion: (AppliedModelingLib.Math.AsymptoticEquivalent  
  (fun (q : ℕ) =>  
    PRPKG24AccuracyDiversity.expectedOrderStatisticMeanSeq (PRPKG24AccuracyDiversity.paretolidSampleMeasure alpha)  
      (q - r) q)  
  fun (q : ℕ) => PRPKG24AccuracyDiversity.paretoRankValueCoeff alpha r * ↑q ^ (1 / alpha)) ∧  
  ∀* (q : ℕ) in Filter.atTop,  
    PRPKG24AccuracyDiversity.expectedOrderStatisticMeanSeq (PRPKG24AccuracyDiversity.paretolidSampleMeasure alpha)  
      (q + 2 - r) (q + 2) -  
    PRPKG24AccuracyDiversity.expectedOrderStatisticMeanSeq (PRPKG24AccuracyDiversity.paretolidSampleMeasure alpha)  
      (q + 1 - r) (q + 1) <  
    PRPKG24AccuracyDiversity.expectedOrderStatisticMeanSeq (PRPKG24AccuracyDiversity.paretolidSampleMeasure alpha)  
      (q + 1 - r) (q + 1) -  
    PRPKG24AccuracyDiversity.expectedOrderStatisticMeanSeq (PRPKG24AccuracyDiversity.paretolidSampleMeasure alpha)  
      (q - r) q
```

Lean proof endpoint (built separately)

PRPKG24AccuracyDiversity.PaperInterface.lemmaD4_corrected_pareto_fixed_rank_sourceSpec_proof

Recorded source-to-Spec screening Verdict: matches approved corrected target

Reason: Approved corrected target: Lean uses the normalized Pareto(α) law for $\alpha > 1$, the corrected rank coefficient and $q^{(1/\alpha)}$ asymptotic, and valid-rank eventual strict marginal decrease.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 24

Source locator: cited publication, lines 3819-3840

Verbatim source input

Lemma D.5. Let $g_1, g_2, \dots, g_m : [0, \infty)^m \rightarrow \mathbb{R}$ be strictly convex functions over the non-negative reals. Then

define

m

X

$gt(xt)$.

(190)

$g(x_1, \dots, x_m) :=$

$t=1$

Pm

$*$

$*$

$t=1 \quad xt = n$, if $(x_1, \dots, x_m)$ is the maximum of g over the non-negative

Then, under the constraint that

reals and $(a_1, \dots, a_m)$ is the maximum of g over the non-negative integers, then

$\lfloor x \rfloor - m < a \cdot t < \lfloor x \rfloor + m$

(191)

for all t .

Formalized review target The archival source above is not asserted equivalent to this formalized target.

For separable strictly concave summands on the nonnegative reals, given fixed-sum real and integer maximizers, every coordinate lies in the strict floor

$\hookrightarrow \text{window floor}(x_t) < a_t + m$ and $a_t < \text{floor}(x_t) + m$. The proof basis is derivative-free strict concavity.

Archival source anchor: cited publication, lines 3819-3840

Expanded PaperInterface specification

$\forall \{ \kappa : \text{Type } u_1 \} [\text{inst} : \text{Fintype } \kappa] (g : \kappa \rightarrow \mathbb{R} \rightarrow \mathbb{R}) (N : \mathbb{N}) (x : \kappa \rightarrow \mathbb{R}) (a : \kappa \rightarrow \mathbb{N}),$

$(\forall (i : \kappa), \text{StrictConcaveOn } \mathbb{R} (\text{Set.Ici } 0) (g \ i)) \rightarrow$

$(\forall (i : \kappa), 0 \leq x \ i) \rightarrow$

$\sum i : \kappa, x \ i = \uparrow N \rightarrow$

$(\forall (z : \kappa \rightarrow \mathbb{R}),$

$(\forall (i : \kappa), 0 \leq z \ i) \rightarrow$

$\sum i : \kappa, z \ i = \uparrow N \rightarrow$

$\text{PRPKG24AccuracyDiversity.GeneralRounding.objective } g \ z \leq$

$\text{PRPKG24AccuracyDiversity.GeneralRounding.objective } g \ x) \rightarrow$

$\sum i : \kappa, a \ i = N \rightarrow$

$(\forall (b : \kappa \rightarrow \mathbb{N}),$

$\sum i : \kappa, b \ i = N \rightarrow$

$(\text{PRPKG24AccuracyDiversity.GeneralRounding.objective } g \ \text{fun } (i : \kappa) => \uparrow (b \ i)) \leq$

$\text{PRPKG24AccuracyDiversity.GeneralRounding.objective } g \ \text{fun } (i : \kappa) => \uparrow (a \ i)) \rightarrow$

$\forall (t : \kappa), \lfloor x \rfloor_+ < a \ t + \text{Fintype.card } \kappa \wedge a \ t < \lfloor x \rfloor_+ + \text{Fintype.card } \kappa$

Retained governing declarations

- [PRPKG24AccuracyDiversity.GeneralRounding.objective](#)

Lean-elaborated claim roles

1. parameter: $\text{Type } u_1$
2. parameter: $\text{Fintype } \kappa$
3. parameter: $\kappa \rightarrow \mathbb{R} \rightarrow \mathbb{R}$
4. parameter: $\mathbb{N}$
5. parameter: $\kappa \rightarrow \mathbb{R}$
6. parameter: $\kappa \rightarrow \mathbb{N}$
7. assumption: $\forall (i : \kappa), \text{StrictConcaveOn } \mathbb{R} (\text{Set.Ici } 0) (g \ i)$
8. assumption: $\forall (i : \kappa), 0 \leq x \ i$
9. assumption: $\sum i : \kappa, x \ i = \uparrow N$
10. assumption: $\forall (z : \kappa \rightarrow \mathbb{R}),$
 $(\forall (i : \kappa), 0 \leq z \ i) \rightarrow$
 $\sum i : \kappa, z \ i = \uparrow N \rightarrow$
 $\text{PRPKG24AccuracyDiversity.GeneralRounding.objective } g \ z \leq \text{PRPKG24AccuracyDiversity.GeneralRounding.objective } g \ x$
11. assumption: $\sum i : \kappa, a \ i = N$
12. assumption: $\forall (b : \kappa \rightarrow \mathbb{N}),$
 $\sum i : \kappa, b \ i = N \rightarrow$
 $(\text{PRPKG24AccuracyDiversity.GeneralRounding.objective } g \ \text{fun } (i : \kappa) => \uparrow (b \ i)) \leq$
 $\text{PRPKG24AccuracyDiversity.GeneralRounding.objective } g \ \text{fun } (i : \kappa) => \uparrow (a \ i)$
13. parameter: κ
14. conclusion: $\lfloor x \rfloor_+ < a \ t + \text{Fintype.card } \kappa \wedge a \ t < \lfloor x \rfloor_+ + \text{Fintype.card } \kappa$

Lean proof endpoint (built separately)

PRPKG24AccuracyDiversity.PaperInterface.lemmaD5Spec_proof

Recorded source-to-Spec screening Verdict: matches_approved_corrected_target

Reason: Approved corrected target: strict concavity replaces the incompatible printed convex/maximization wording, and Lean states both fixed-sum real/integer maximizers and the approved strict floor window.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 25

Source locator: cited publication, lines 2071-2087

Verbatim source input

Proposition 5. For X_i

$\sim D$,

$\min\{k, a\}$

$h(a) =$

$\sum_{i=1}^X$

$\mu D(a - i + 1, a).$

(78)

Recall that $\mu D(i, a)$ is the expected value of the i -th order statistic of a random variables drawn i.i.d. from D .

Expanded PaperInterface specification

```
∀ (D : MeasureTheory.Measure ℝ) [MeasureTheory.IsProbabilityMeasure D],
  MeasureTheory.Integrable (fun (x : ℝ) => x) D →
  ∀ (k a : ℕ),
    AppliedModelingLib.Probability.orderStatisticTopKSumFromMean
      (PRPKG24AccuracyDiversity.definition3IidOrderStatisticMean D) k a =
      AppliedModelingLib.Probability.expectedSampleTopKSum (PRPKG24AccuracyDiversity.definition3IidSampleMeasure D a) k
```

Retained governing declarations

- [AppliedModelingLib.Probability.expectedSampleTopKSum](#)
- [AppliedModelingLib.Probability.orderStatisticTopKSumFromMean](#)
- [PRPKG24AccuracyDiversity.definition3IidOrderStatisticMean](#)
- [PRPKG24AccuracyDiversity.definition3IidSampleMeasure](#)

Lean-elaborated claim roles

1. parameter: `MeasureTheory.Measure ℝ`
2. assumption: `MeasureTheory.IsProbabilityMeasure D`
3. assumption: `MeasureTheory.Integrable (fun (x : ℝ) => x) D`
4. parameter: `ℕ`
5. parameter: `ℕ`
6. conclusion: `AppliedModelingLib.Probability.orderStatisticTopKSumFromMean (PRPKG24AccuracyDiversity.definition3IidOrderStatisticMean D) k a = AppliedModelingLib.Probability.expectedSampleTopKSum (PRPKG24AccuracyDiversity.definition3IidSampleMeasure D a) k`

Lean proof endpoint (built separately)

`PRPKG24AccuracyDiversity.PaperInterface.proposition5_iid_topk_identitySpec_proof`

Recorded source-to-Spec screening Verdict: matches

Reason: Matches: the target unfolds the source definition as the expected top-k sum from one common iid probability law; its probability and integrability premises make the source expectation well-defined.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation